# KỸ THUẬT SỐ

# Đánh giá

- **Điểm quá trình: 50%**
  - Chuyên cần
  - Kiểm tra
  - Tiểu luận
- **Điểm cuối kỳ: 50%**

# Tài liệu tham khảo


- Nguyễn Trường Duy, Giáo trình Kỹ thuật số, Đại học Quốc gia Tp. HCM, Năm 2019.
- Ronald J. Tocci - Neal S. Widmer – Gregory L. Moss, Digital Systems: Principles and Applications, 12th Ed. Prentice Hall, 2017.
- Thomas L. Floyd, Digital Fundamental, 11th Ed. Prentice Hall, 2015.

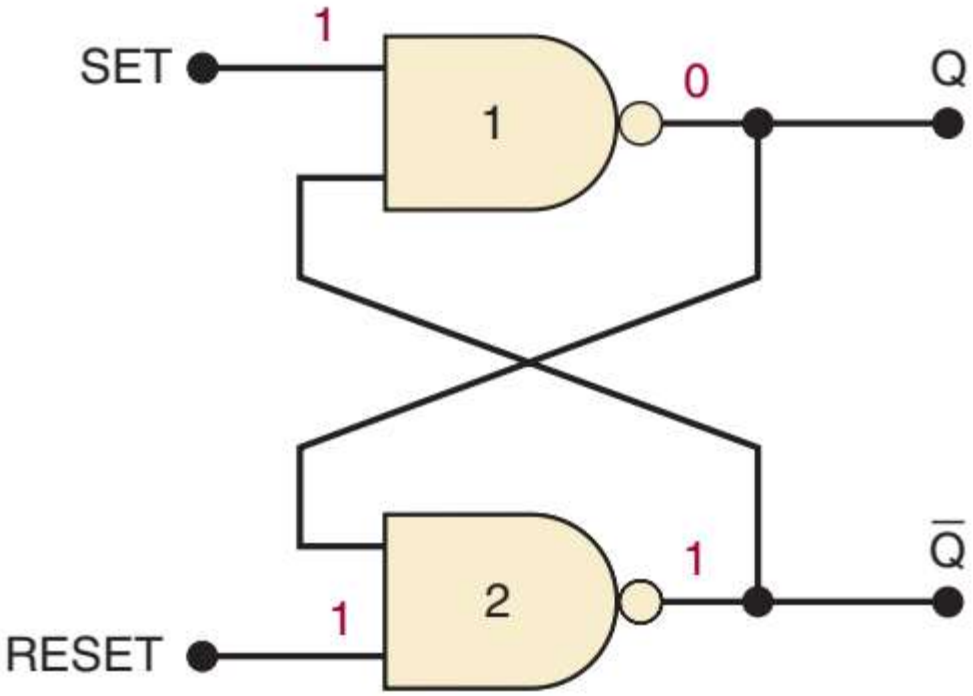
# Mạch tuần tự

# Nội dung

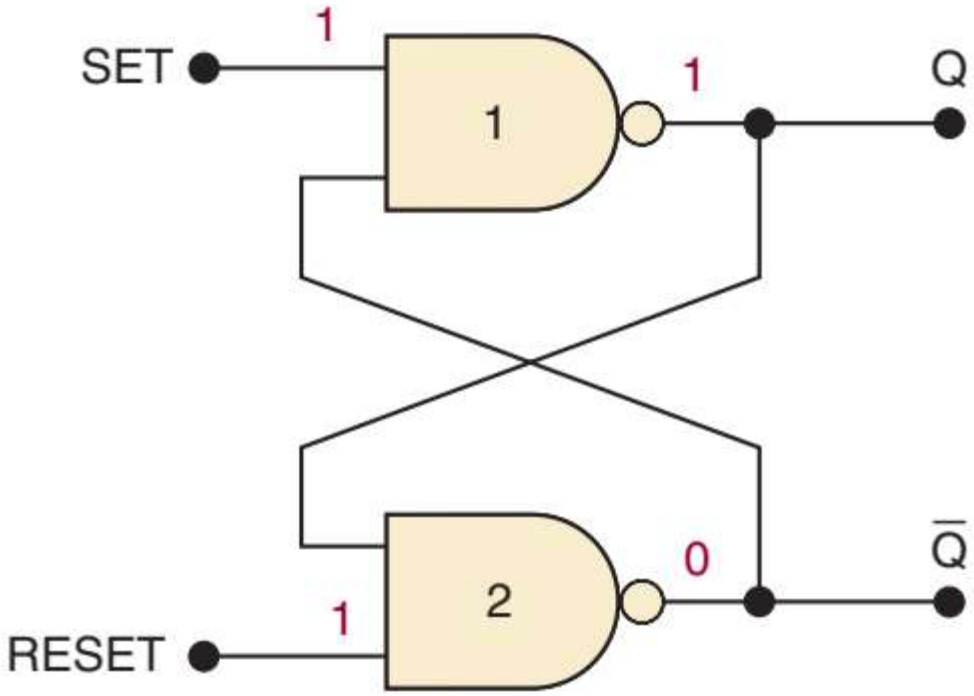
- Mạch chốt
- Clock
- Flip Flop
- Mạch đếm
- Thanh ghi

# Mạch chốt

- Mạch chốt cổng NAND



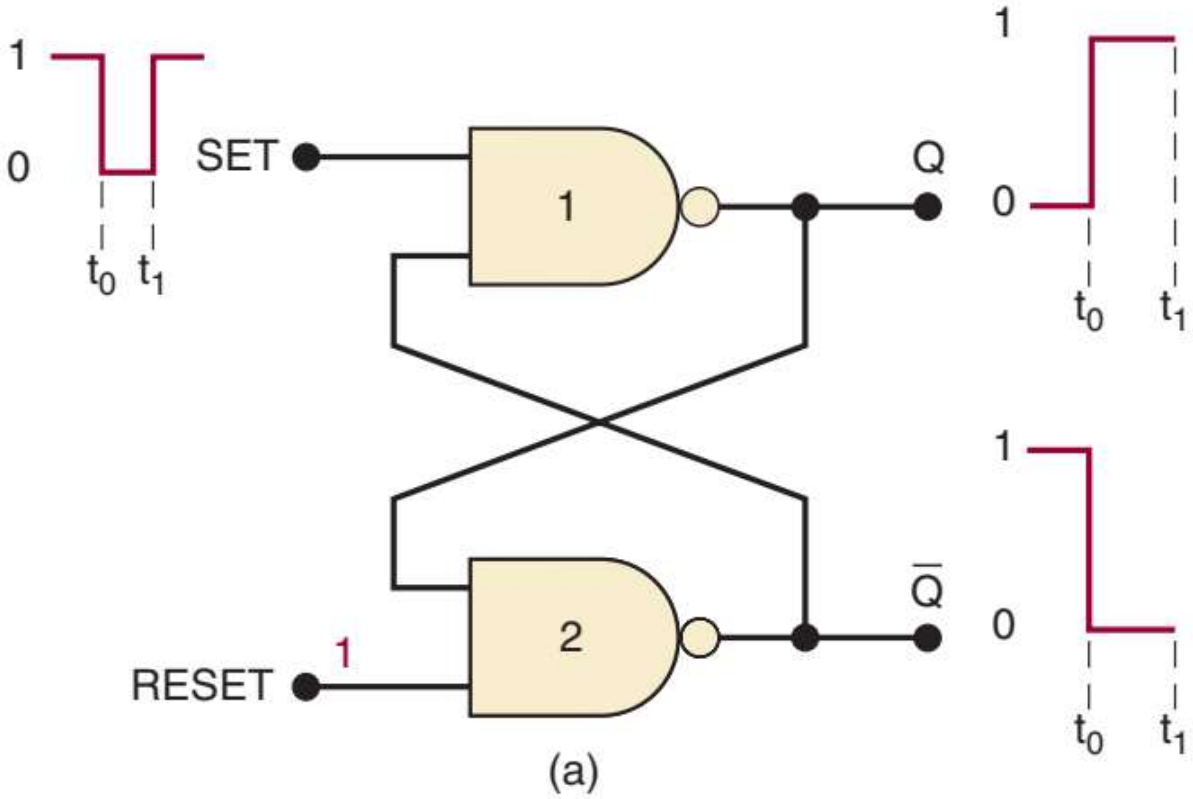

(a)

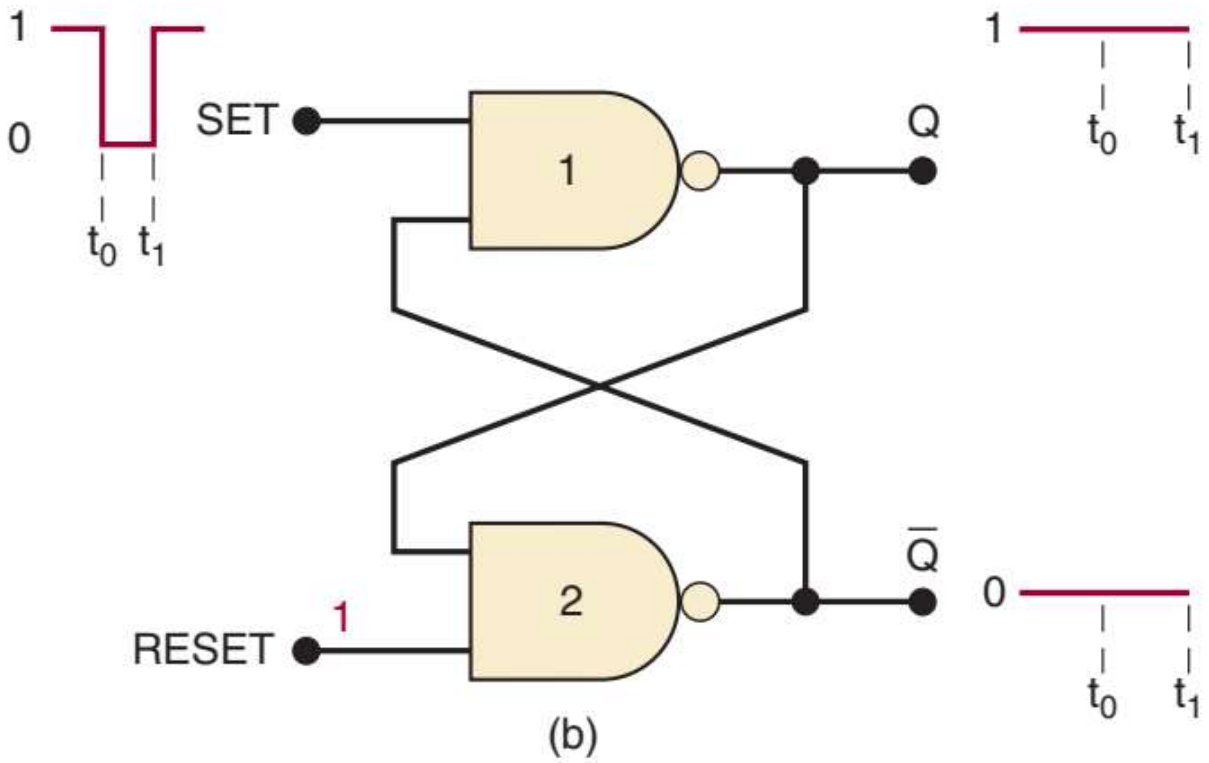



(b)

# Mạch chốt

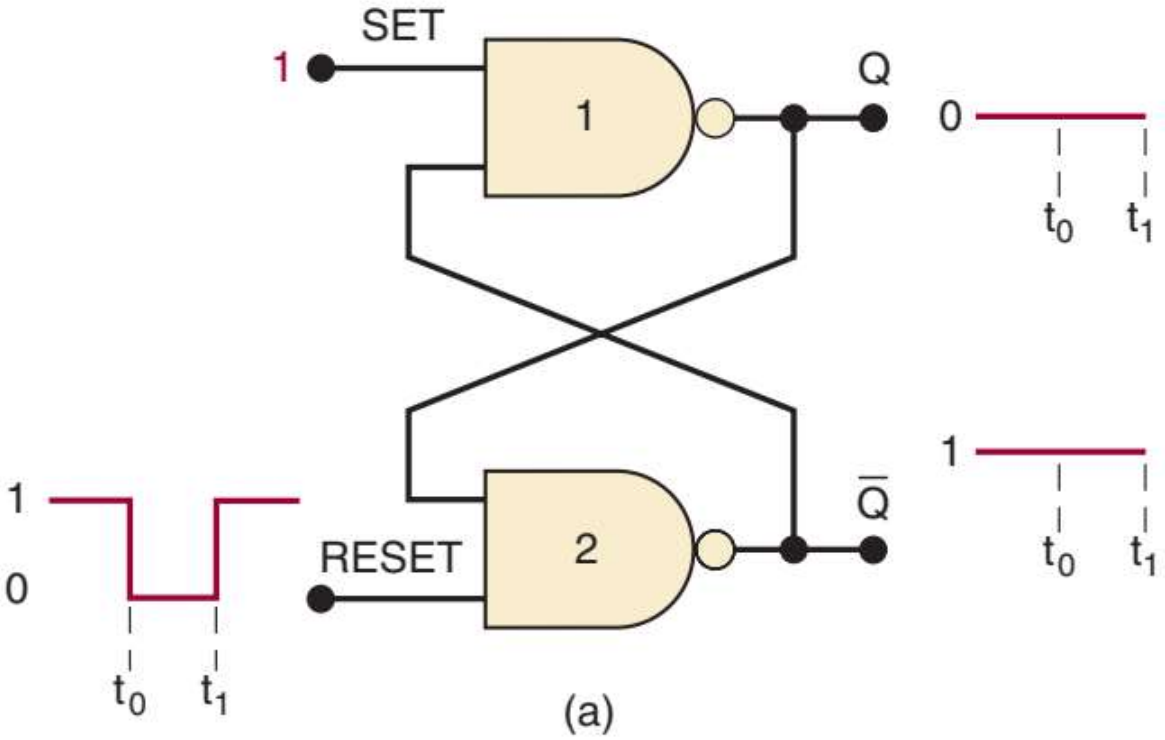
- Mạch chốt cổng NAND

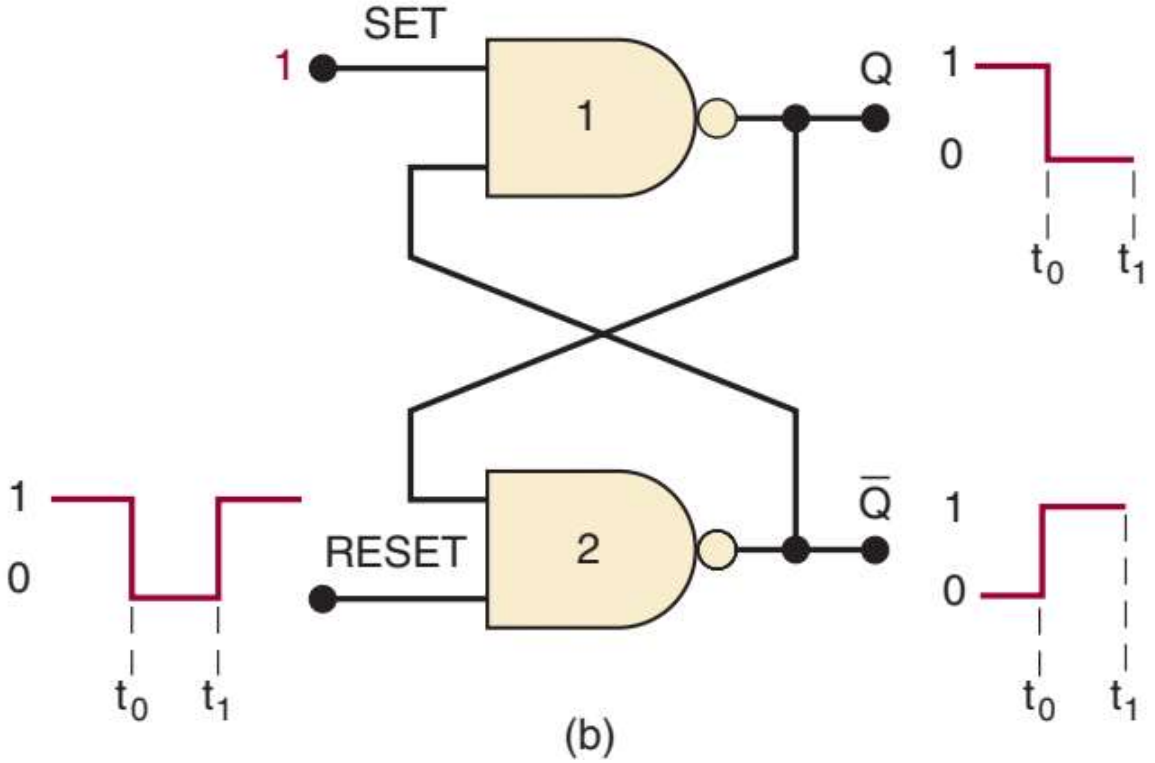


# Mạch chốt

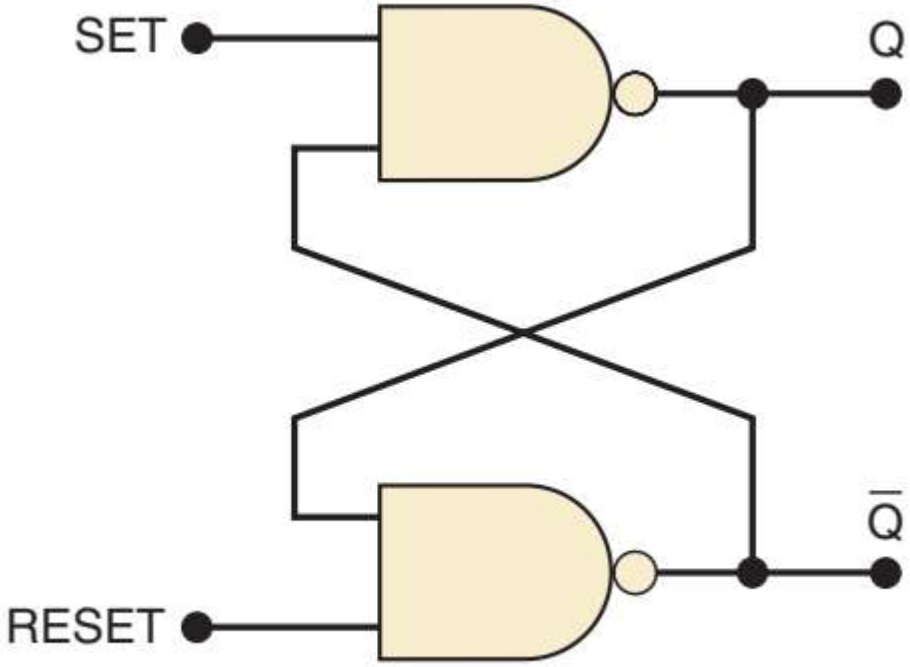
- Mạch chốt cổng NAND



# Mạch chốt

- Mạch chốt cổng NAND




(a)

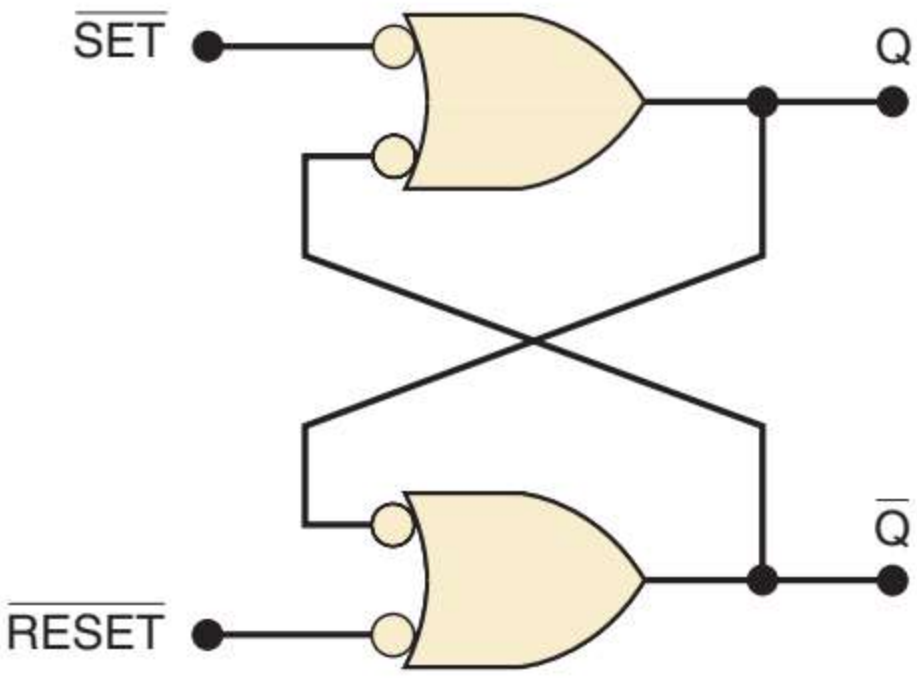
| Set | Reset | Output |
|---|---|---|
| 1 | 1 | No change |
| 0 | 1 | Q = 1 |
| 1 | 0 | Q = 0 |
| 0 | 0 | Invalid * |

*Produces $Q = \overline{Q} = 1$.

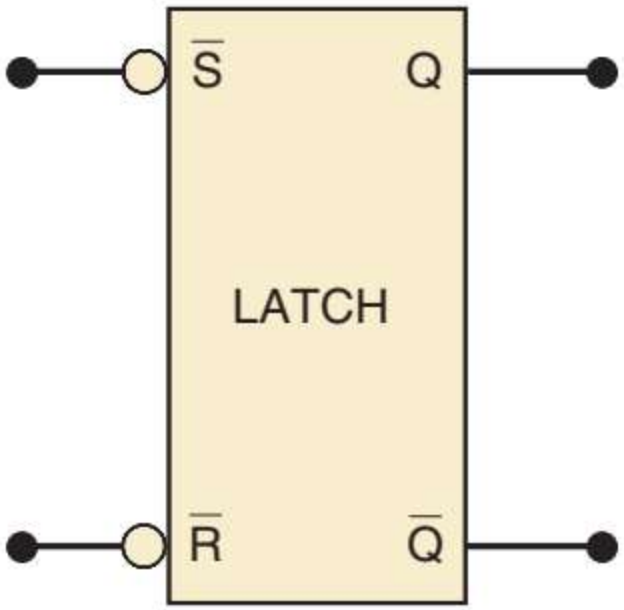
(b)

# Mạch chốt

- Mạch chốt cổng NAND



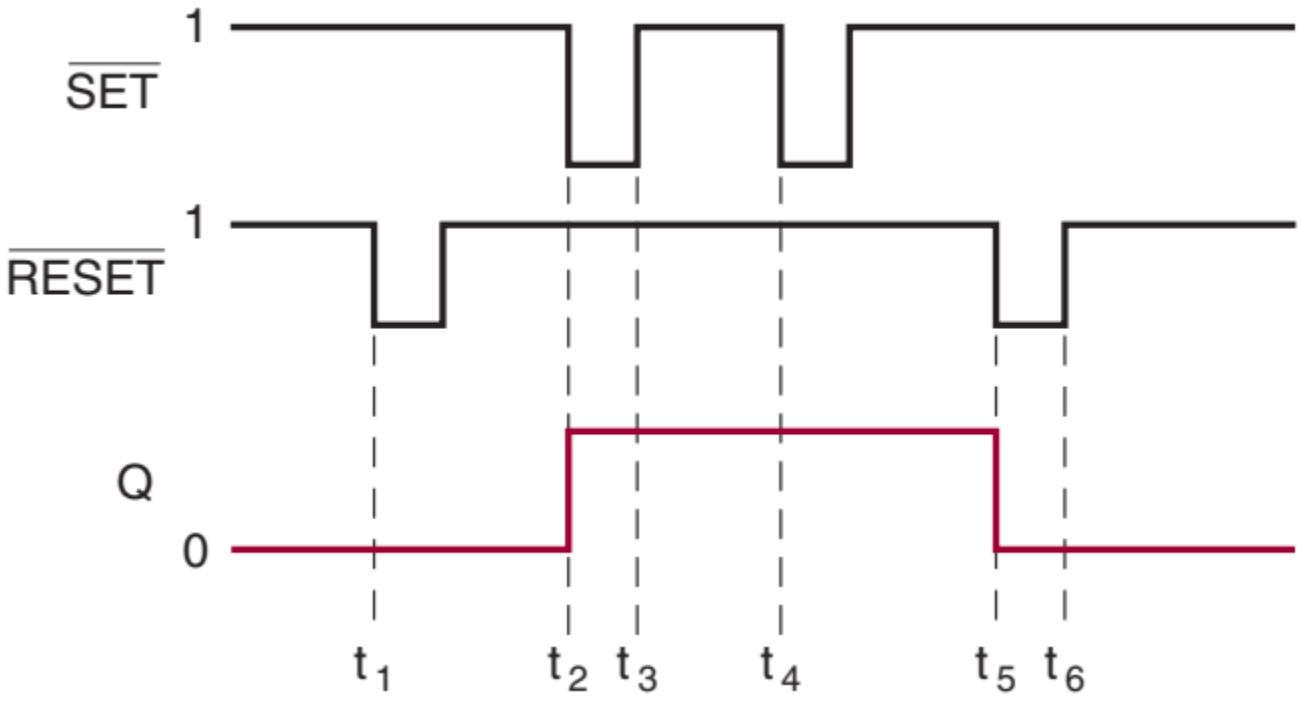

(a)




(b)

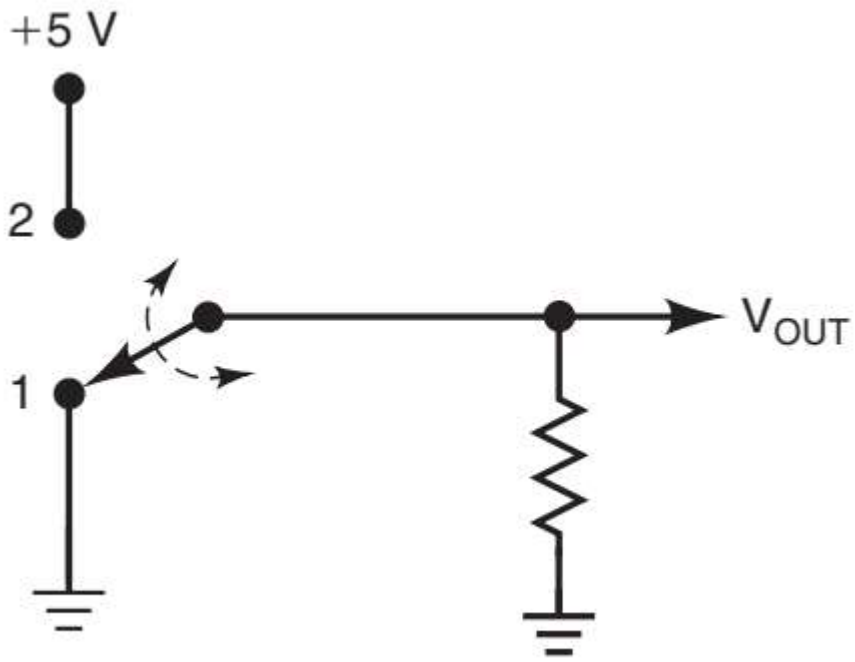
# Mạch chốt

- Mạch chốt cổng NAND

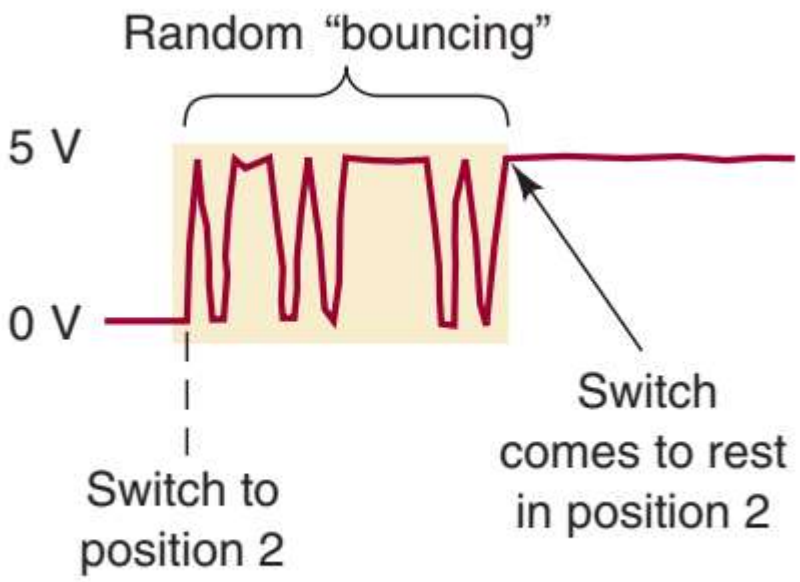


# Mạch chốt

- Mạch chốt cổng NAND

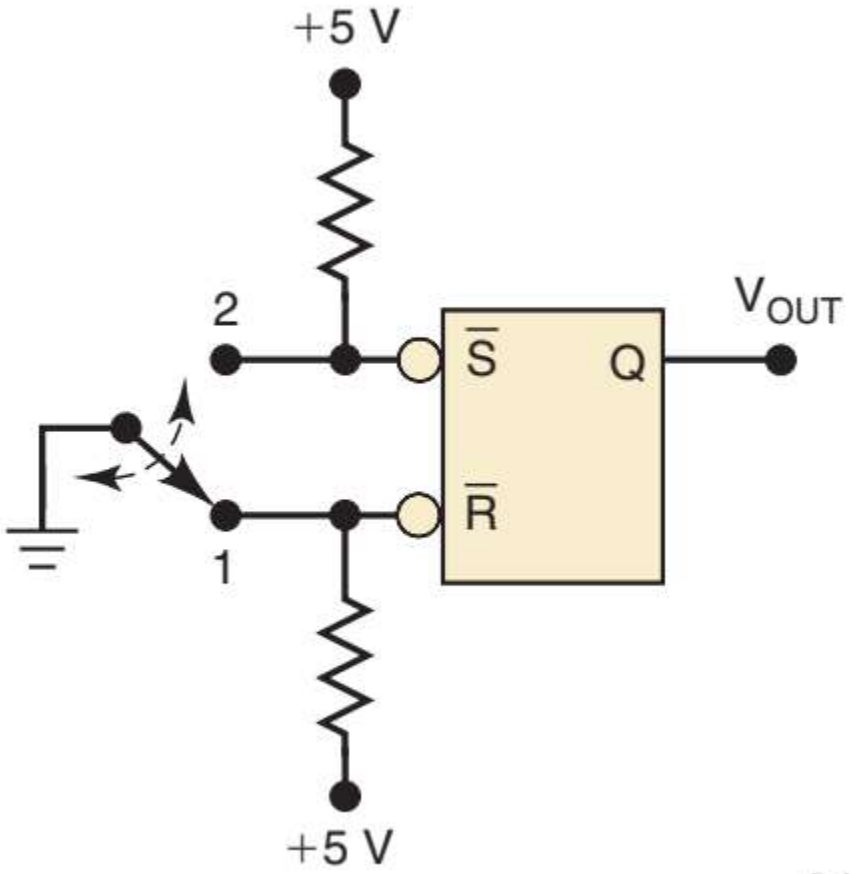


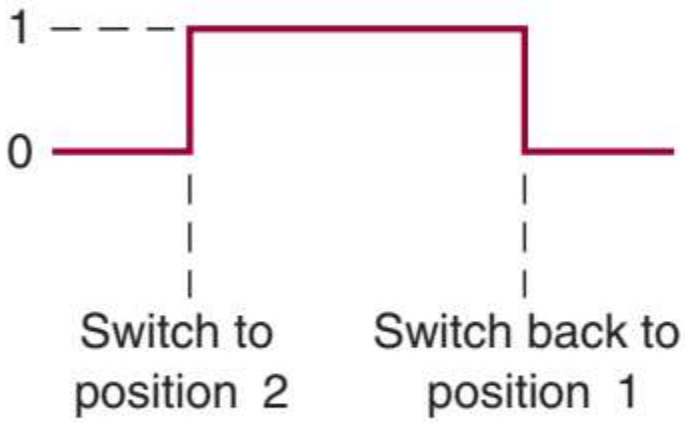

(a)



# Mạch chốt

- Mạch chốt cổng NAND

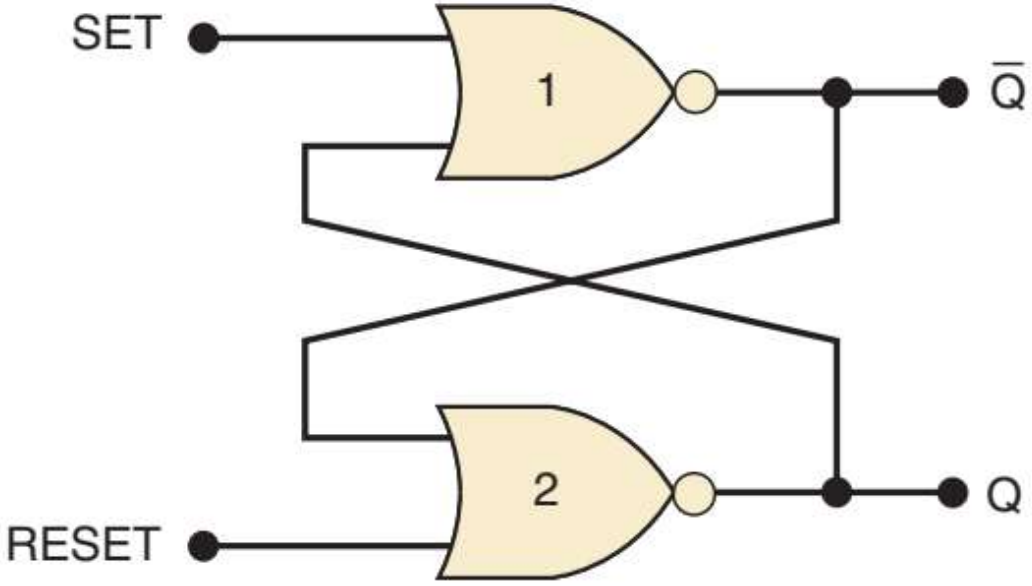



(b)

# Mạch chốt

- Mạch chốt cổng NOR

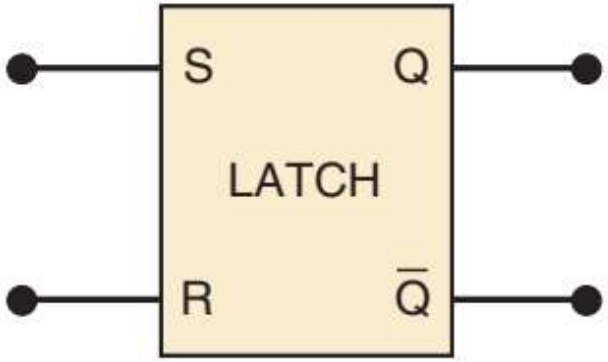



(a)

| Set | Reset | Output |
|---|---|---|
| 0 | 0 | No change |
| 1 | 0 | Q = 1 |
| 0 | 1 | Q = 0 |
| 1 | 1 | Invalid * |

*Produces Q = $\overline{Q}$ = 0.

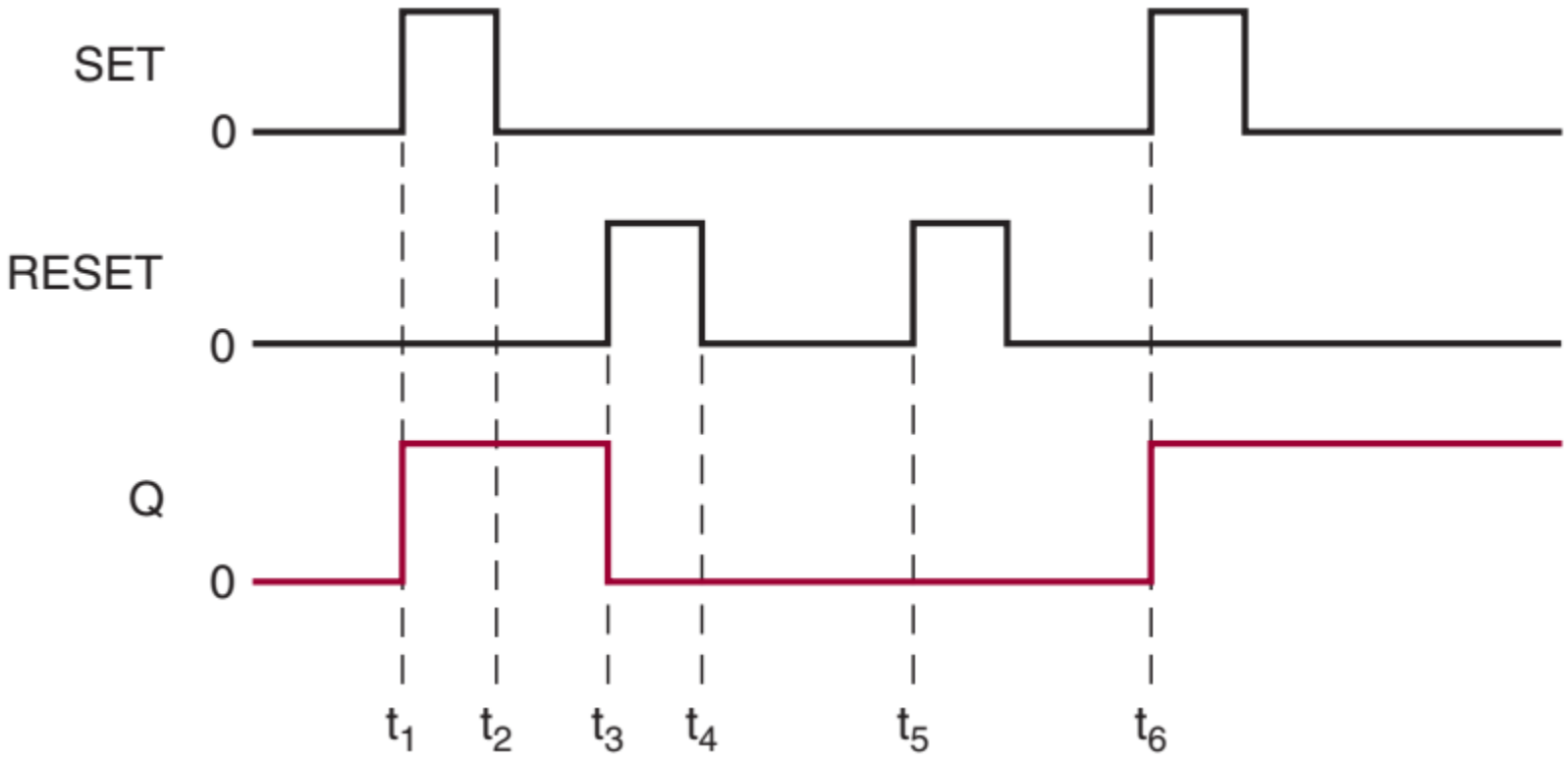
(b)




(c)

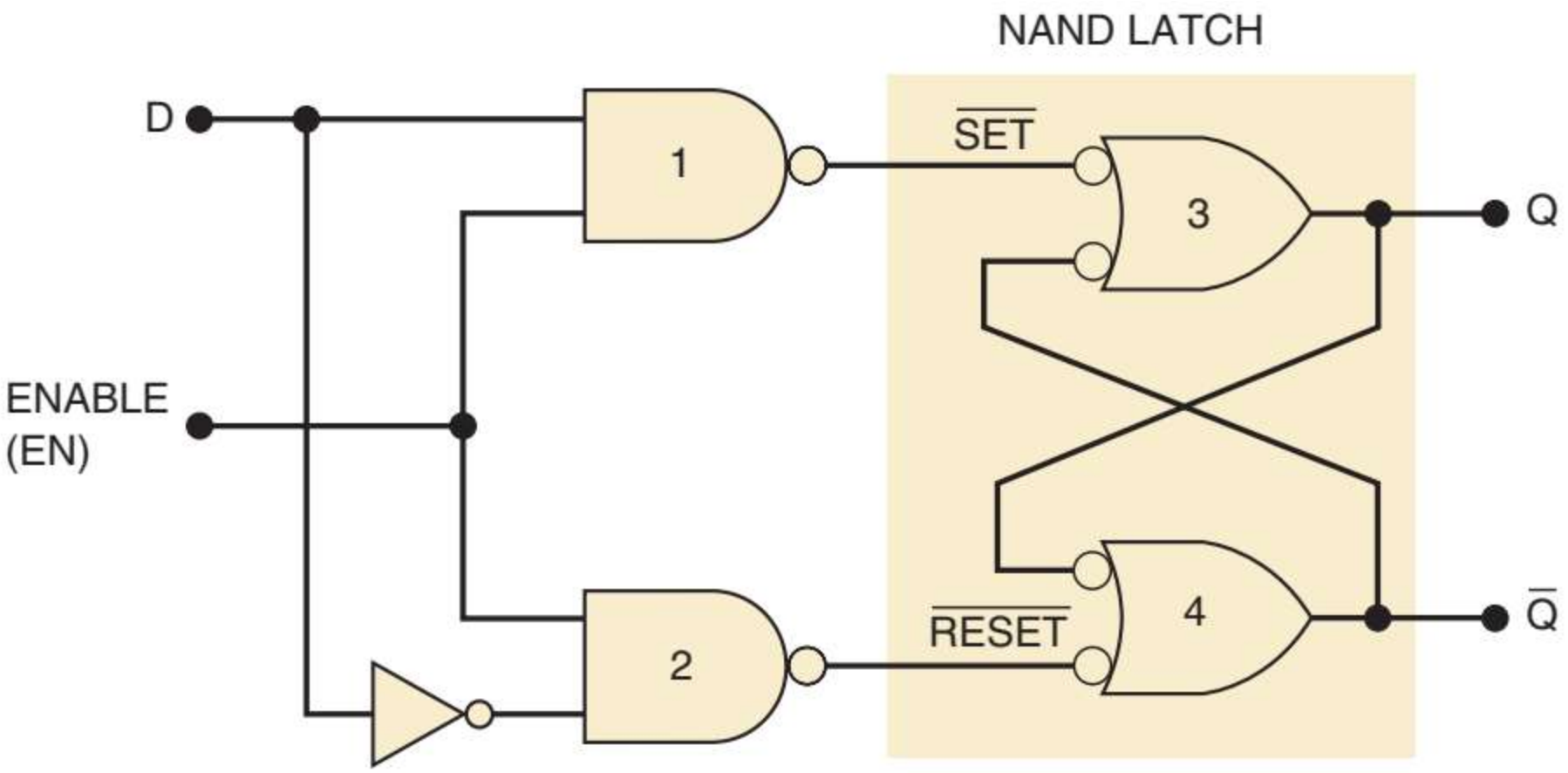
# Mạch chốt

- Mạch chốt cổng NOR



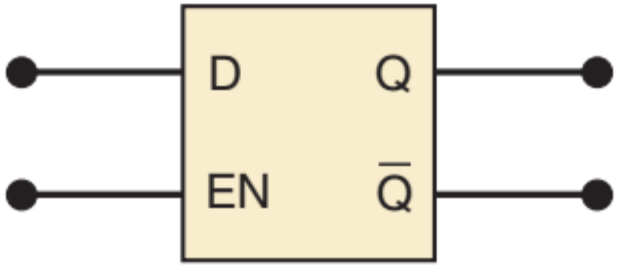
# Mạch chốt

- Mạch chốt cổng D



# Mạch chốt

- Mạch chốt cổng D

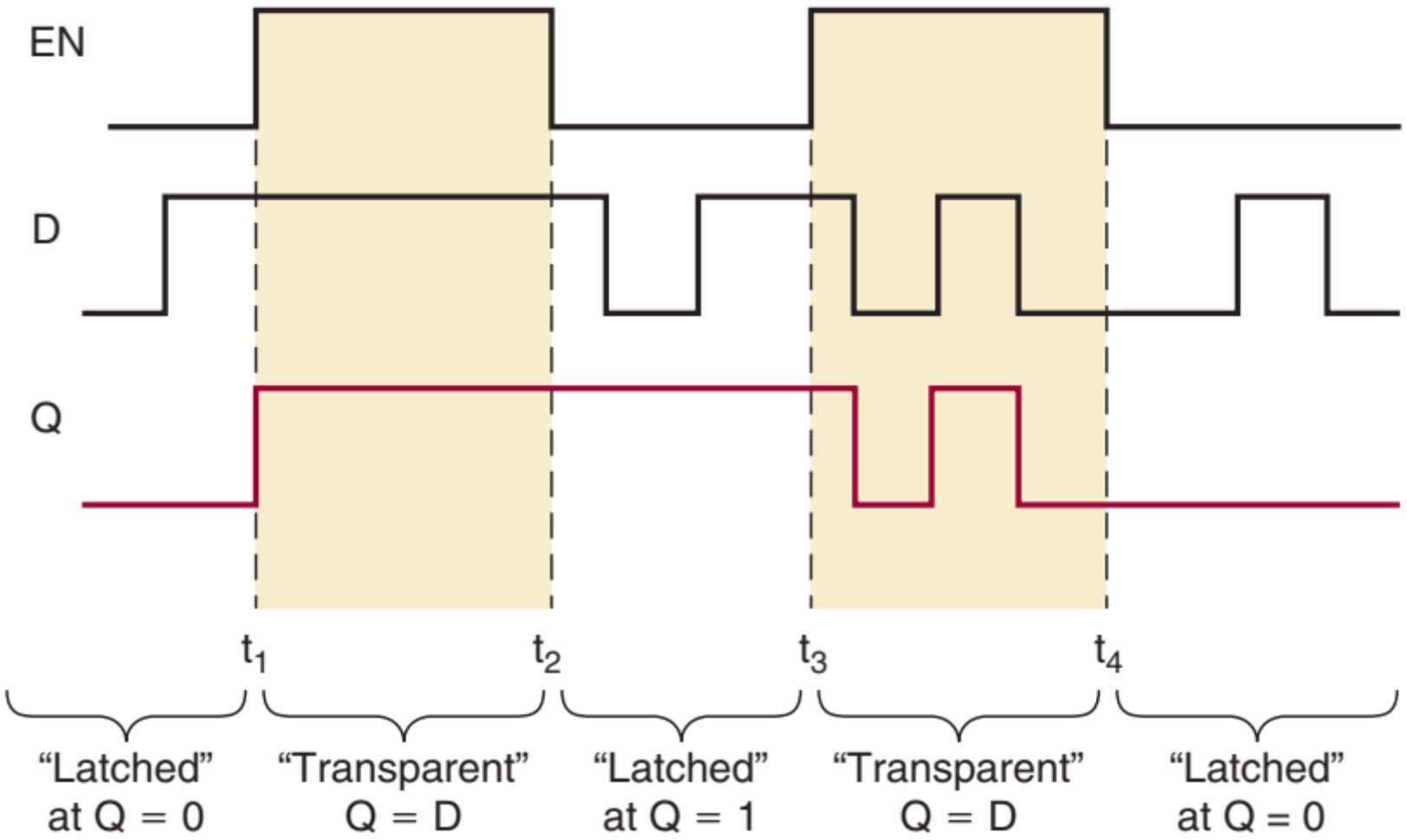



| Inputs | | Output |
|---|---|---|
| EN | D | Q |
| 0 | X | $Q_0$ (no change) |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

"X" indicates "don't care."
$Q_0$ is state Q just prior to EN going LOW.

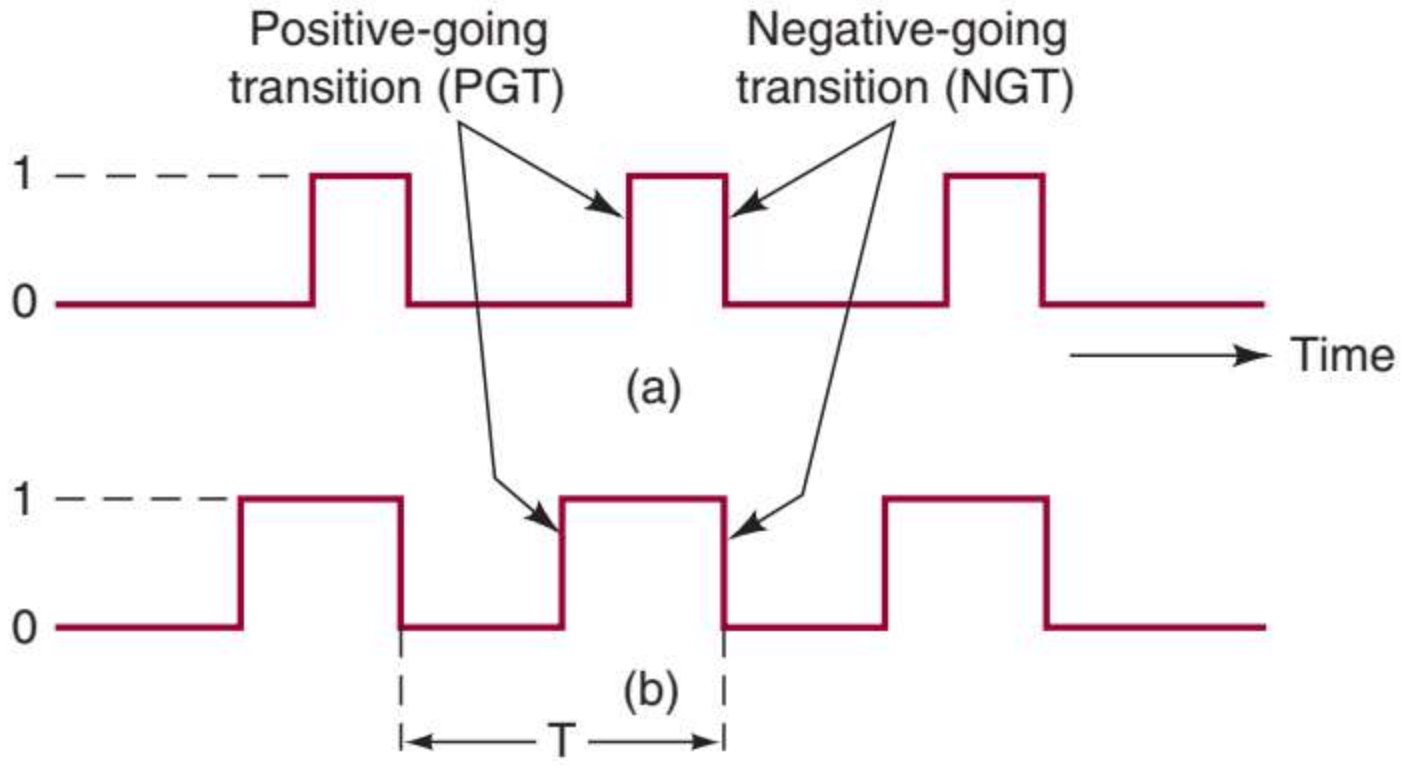
# Mạch chốt

- Mạch chốt cổng D



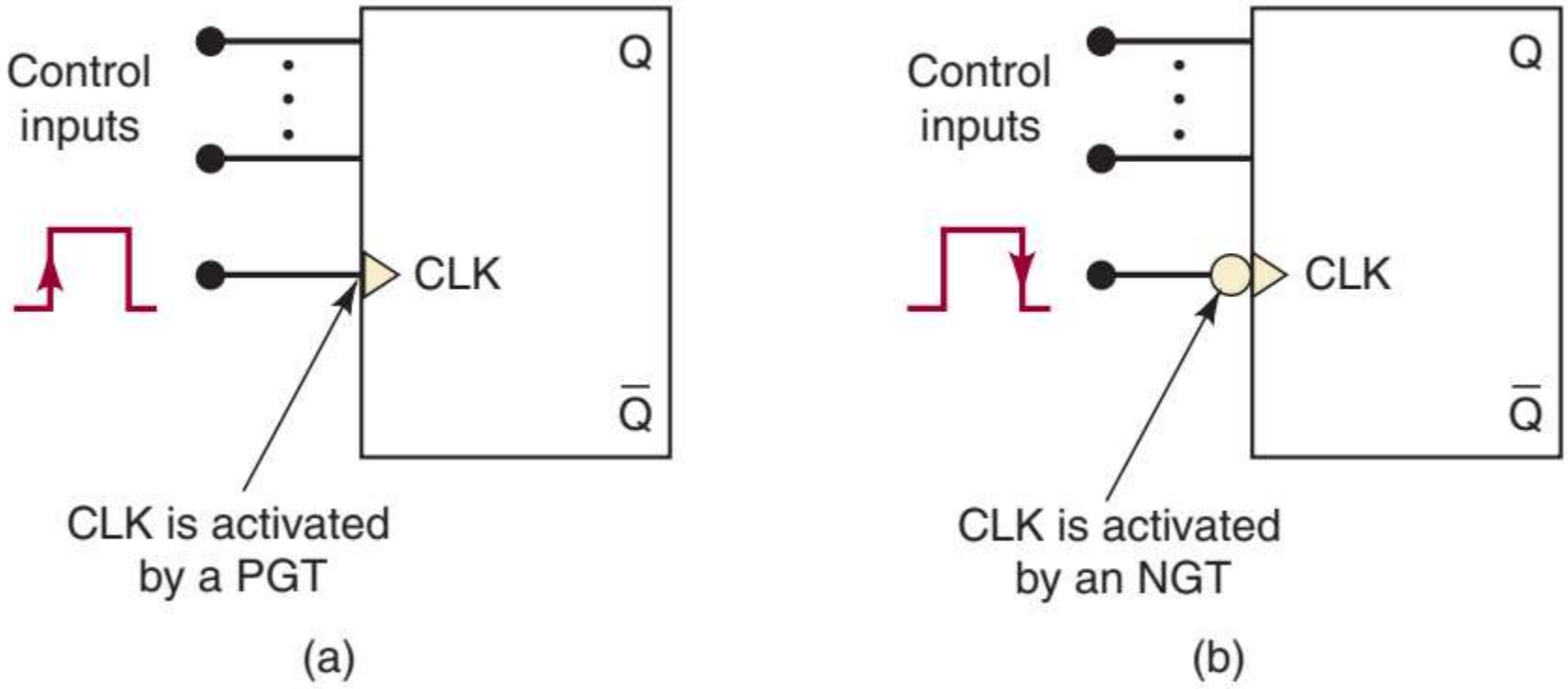
# Tín hiệu Clock

- Tín hiệu CLK



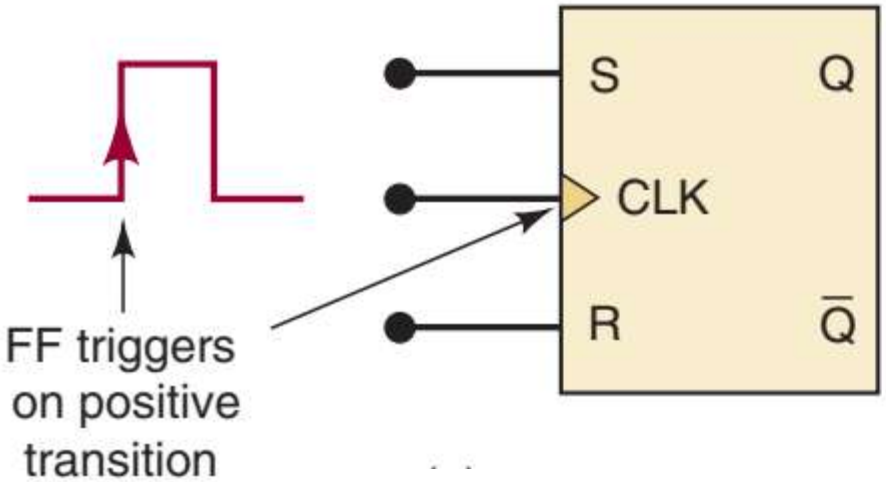
# Tín hiệu Clock

- Tín hiệu CLK và FF



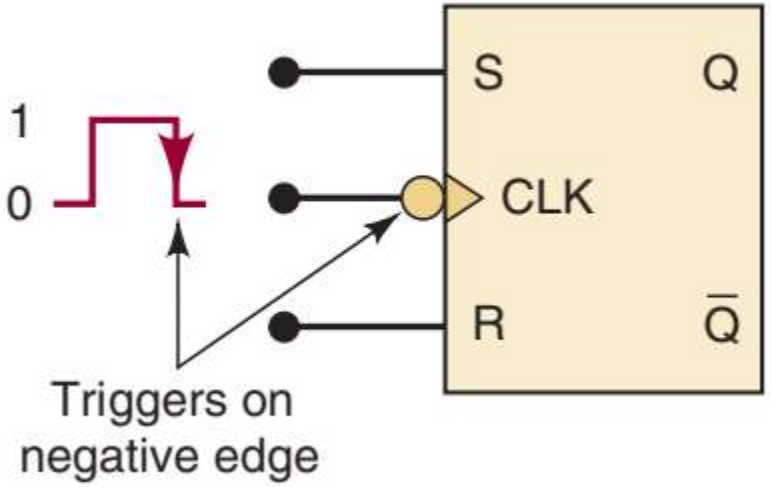
# Flip Flop

- Flip Flop S-R




| Inputs | | | Output |
|---|---|---|---|
| S | R | CLK | Q |
| 0 | 0 | ↑ | $Q_0$ (no change) |
| 1 | 0 | ↑ | 1 |
| 0 | 1 | ↑ | 0 |
| 1 | 1 | ↑ | Ambiguous |

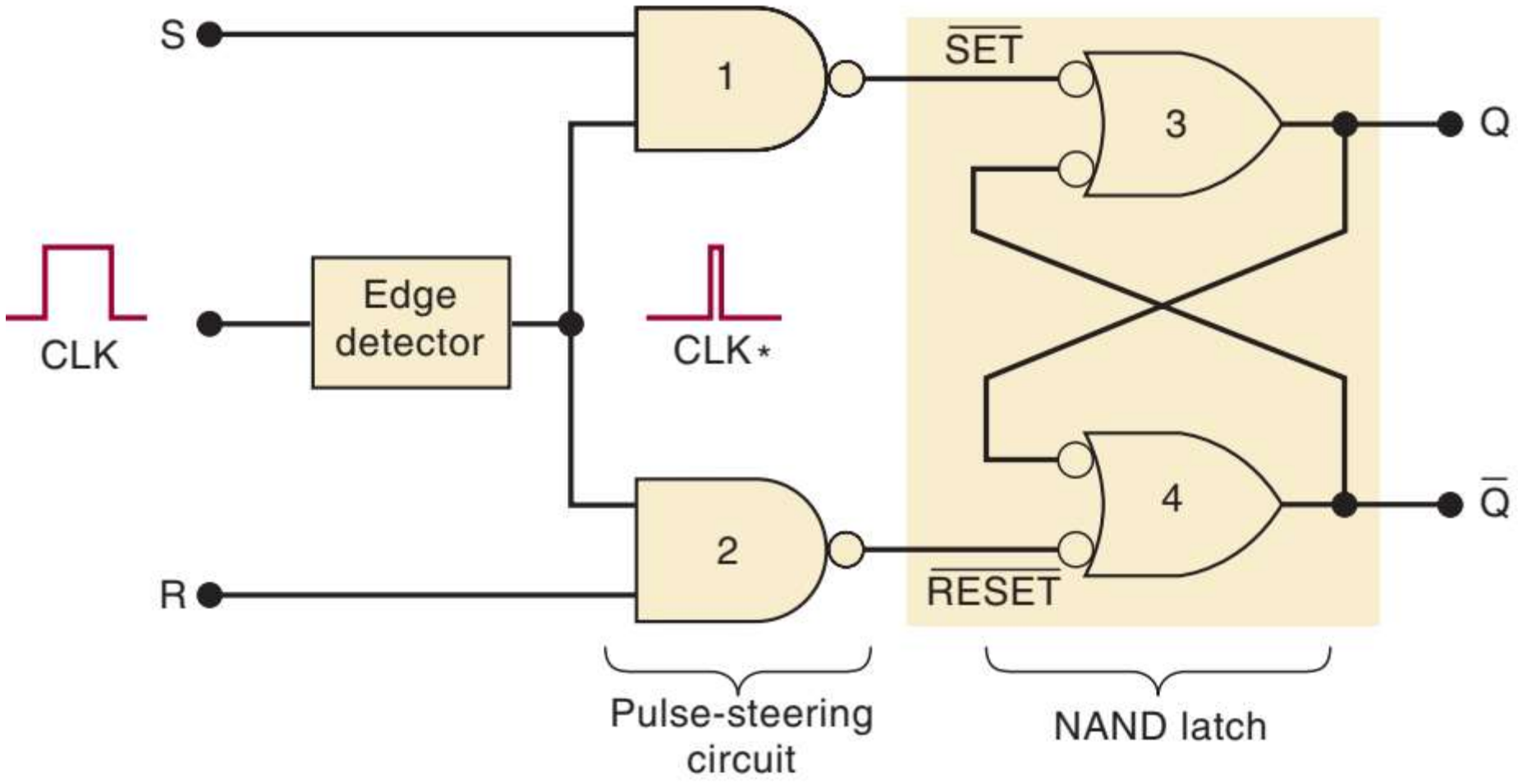
$Q_0$ is output level prior to ↑ of CLK.
↓ of CLK produces no change in Q.




| Inputs | | | Output |
|---|---|---|---|
| S | R | CLK | Q |
| 0 | 0 | ↓ | $Q_0$ (no change) |
| 1 | 0 | ↓ | 1 |
| 0 | 1 | ↓ | 0 |
| 1 | 1 | ↓ | Ambiguous |

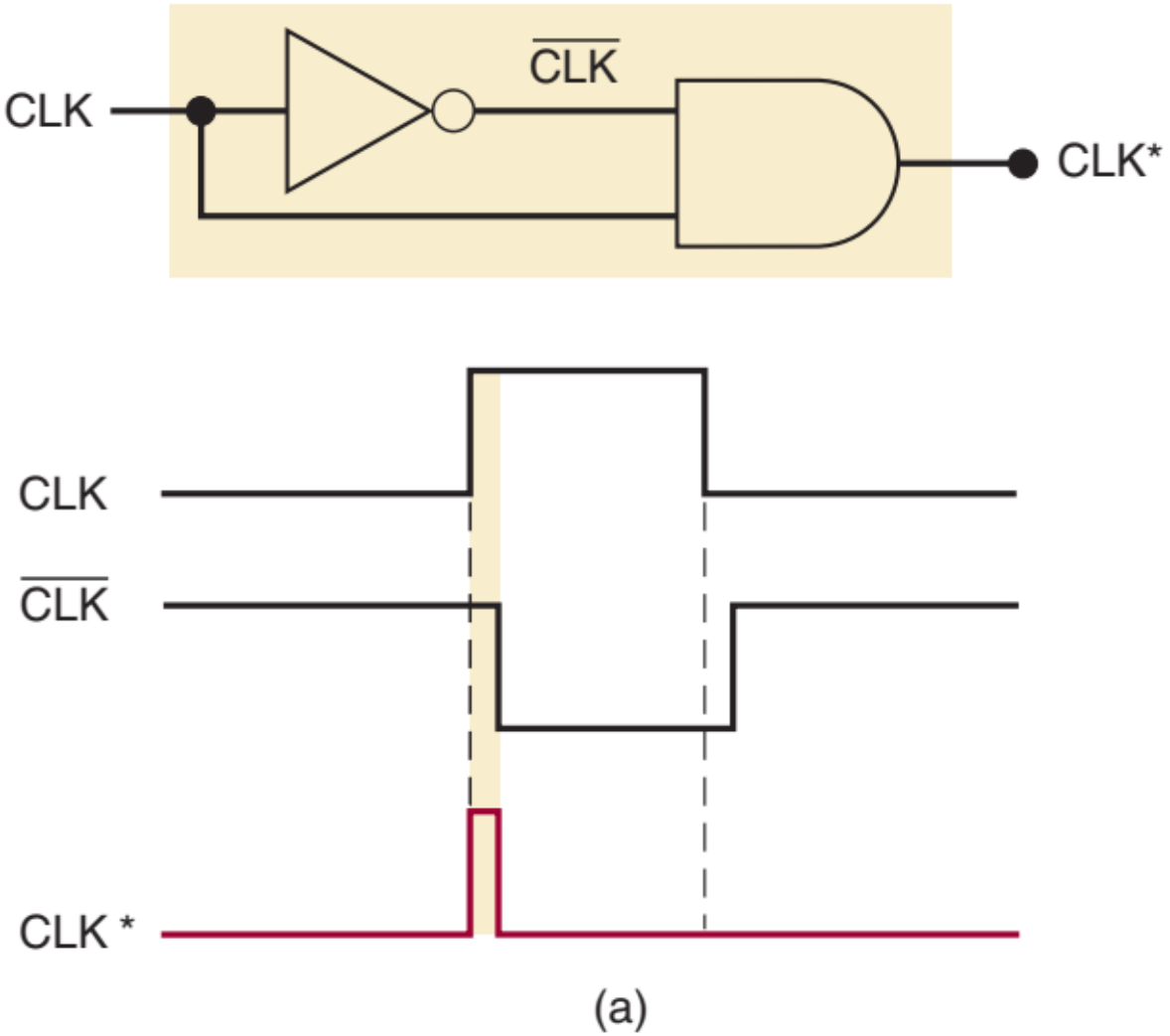
# Flip Flop

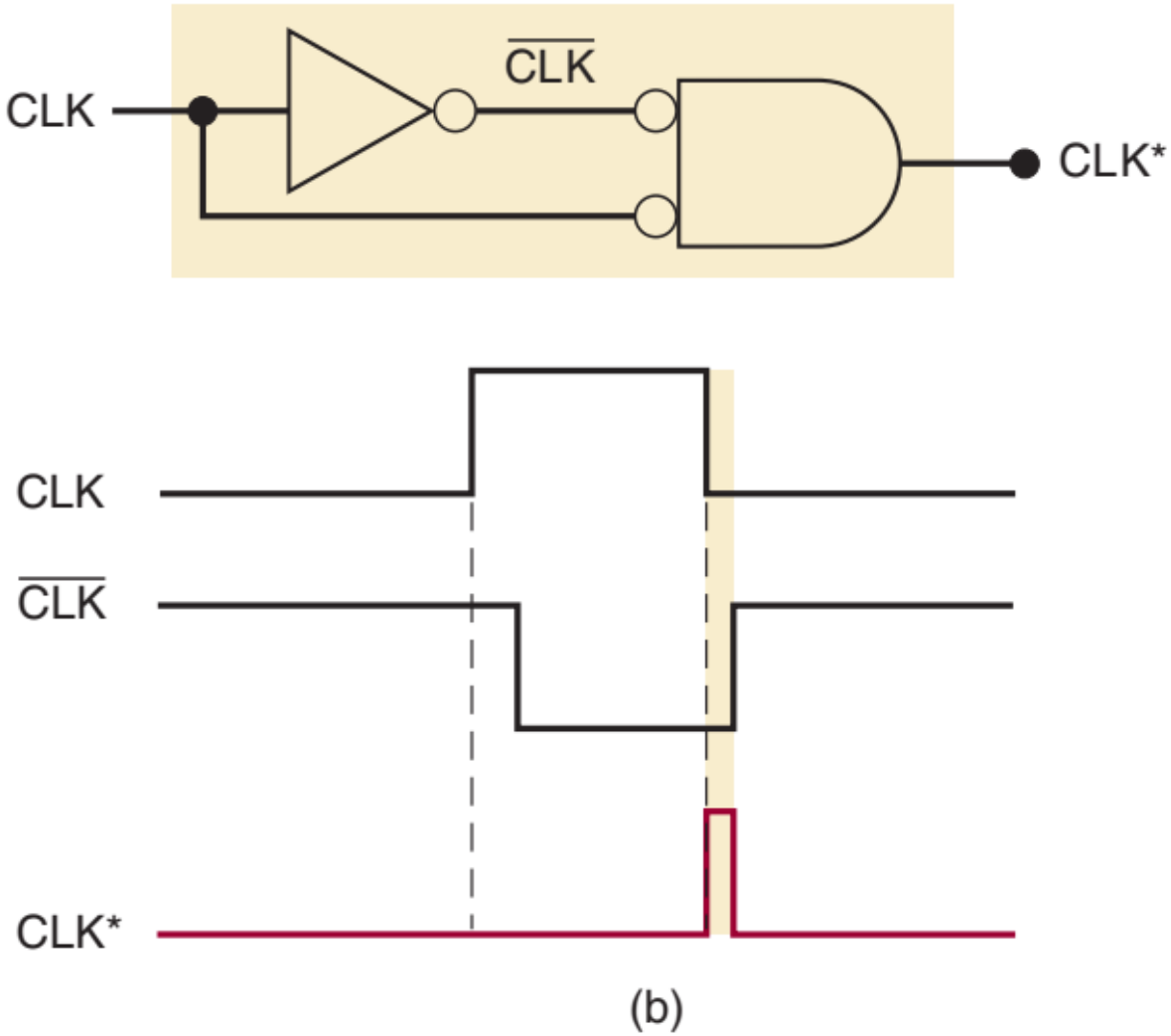
- Flip Flop S-R



# Flip Flop

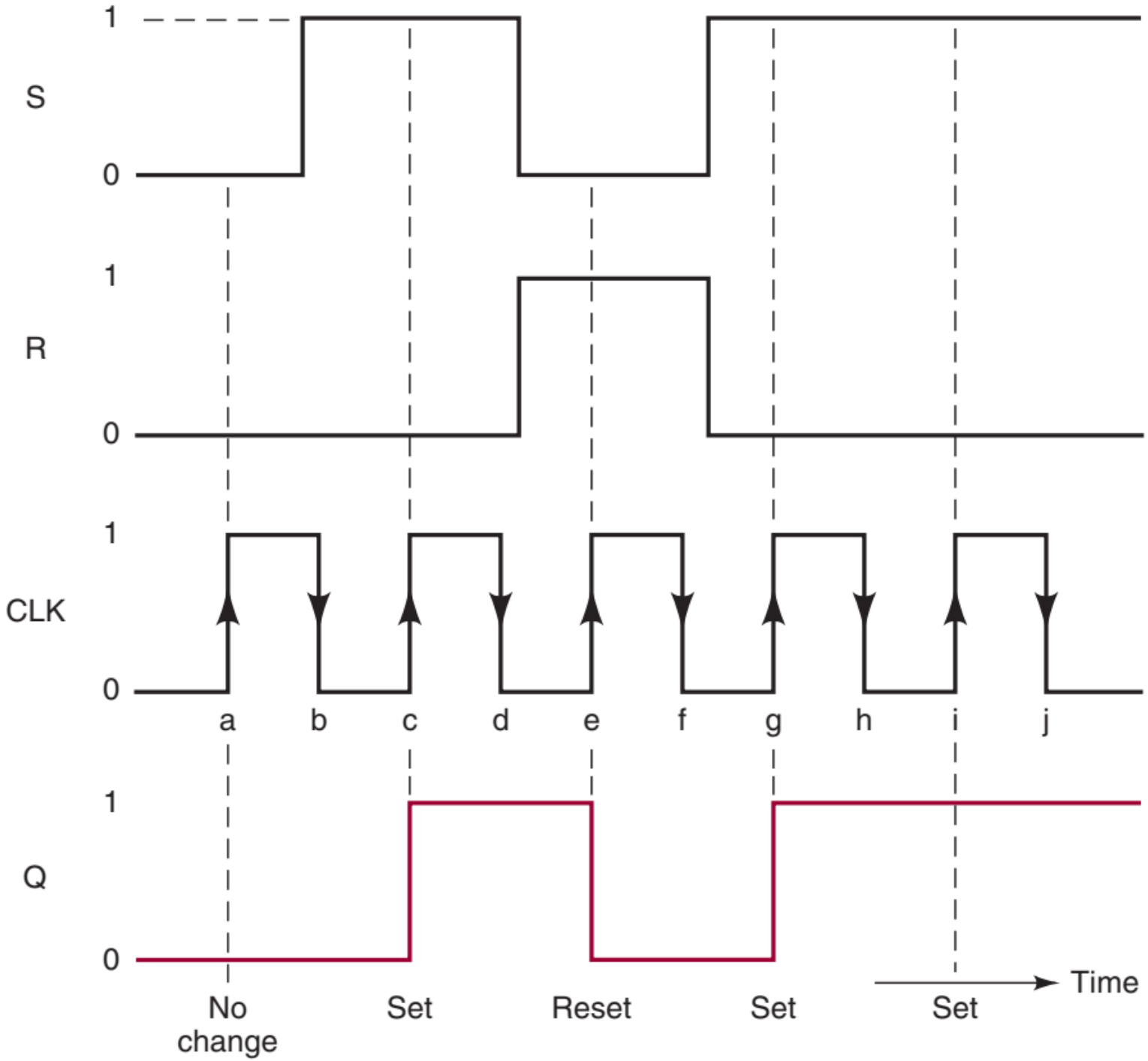
- Flip Flop S-R



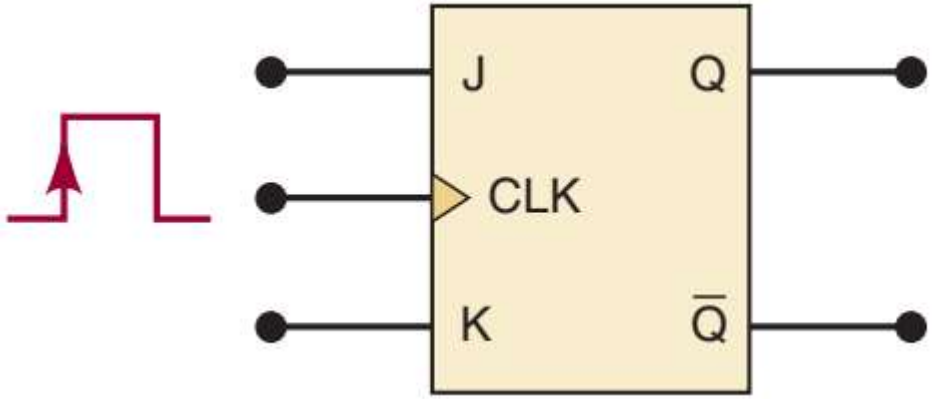
# Fli

- Flip I

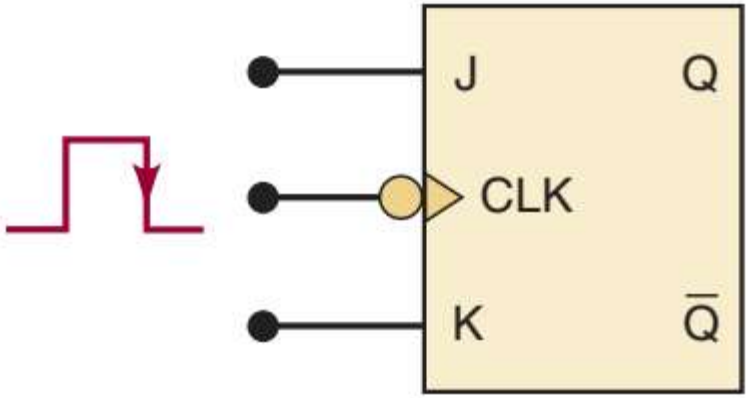


# Flip Flop

- Flip Flop J-K



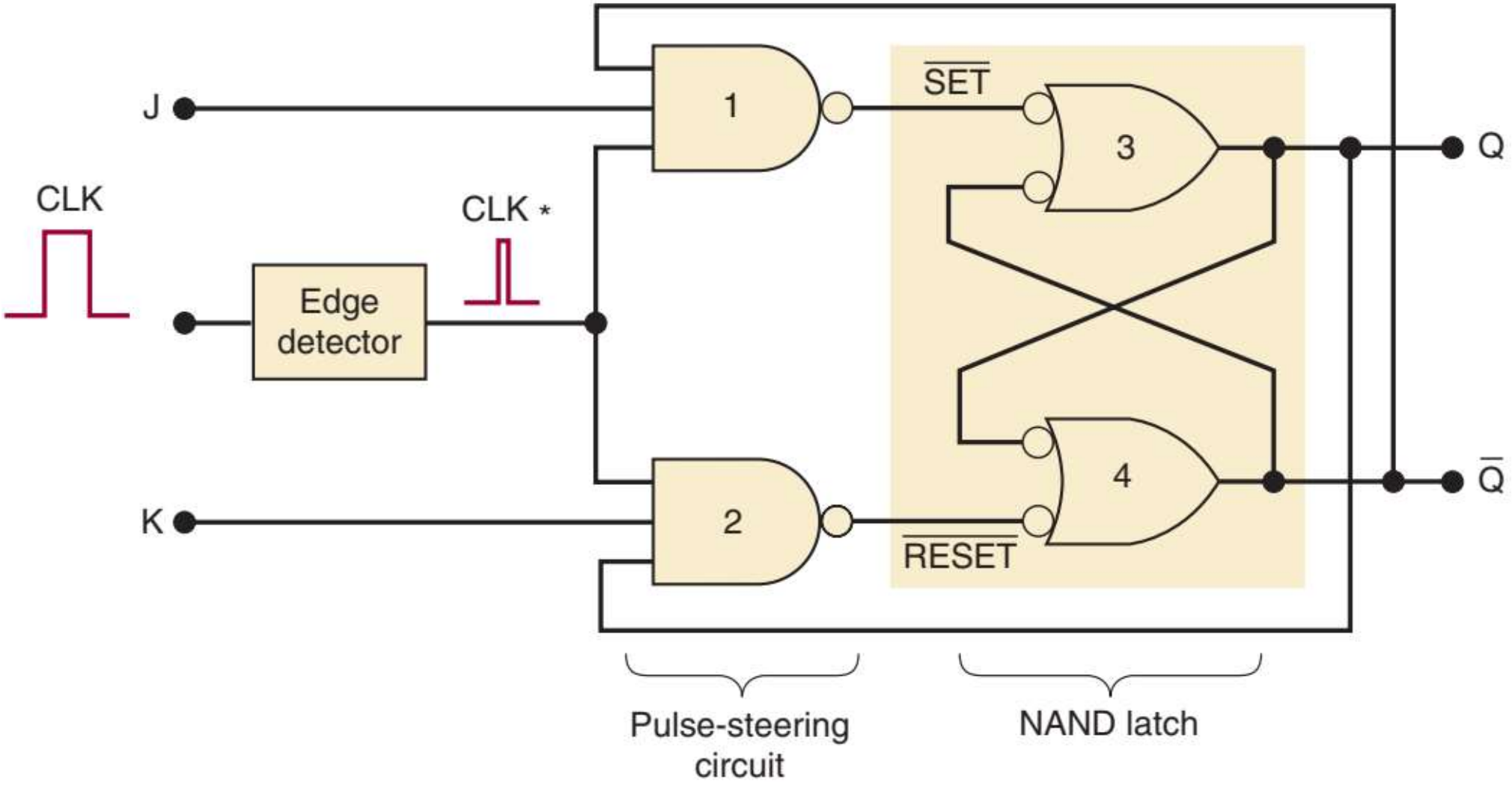

| J | K | CLK | Q |
|---|---|---|---|
| 0 | 0 | ↑ | $Q_0$ (no change) |
| 1 | 0 | ↑ | 1 |
| 0 | 1 | ↑ | 0 |
| 1 | 1 | ↑ | $\overline{Q}_0$ (toggles) |

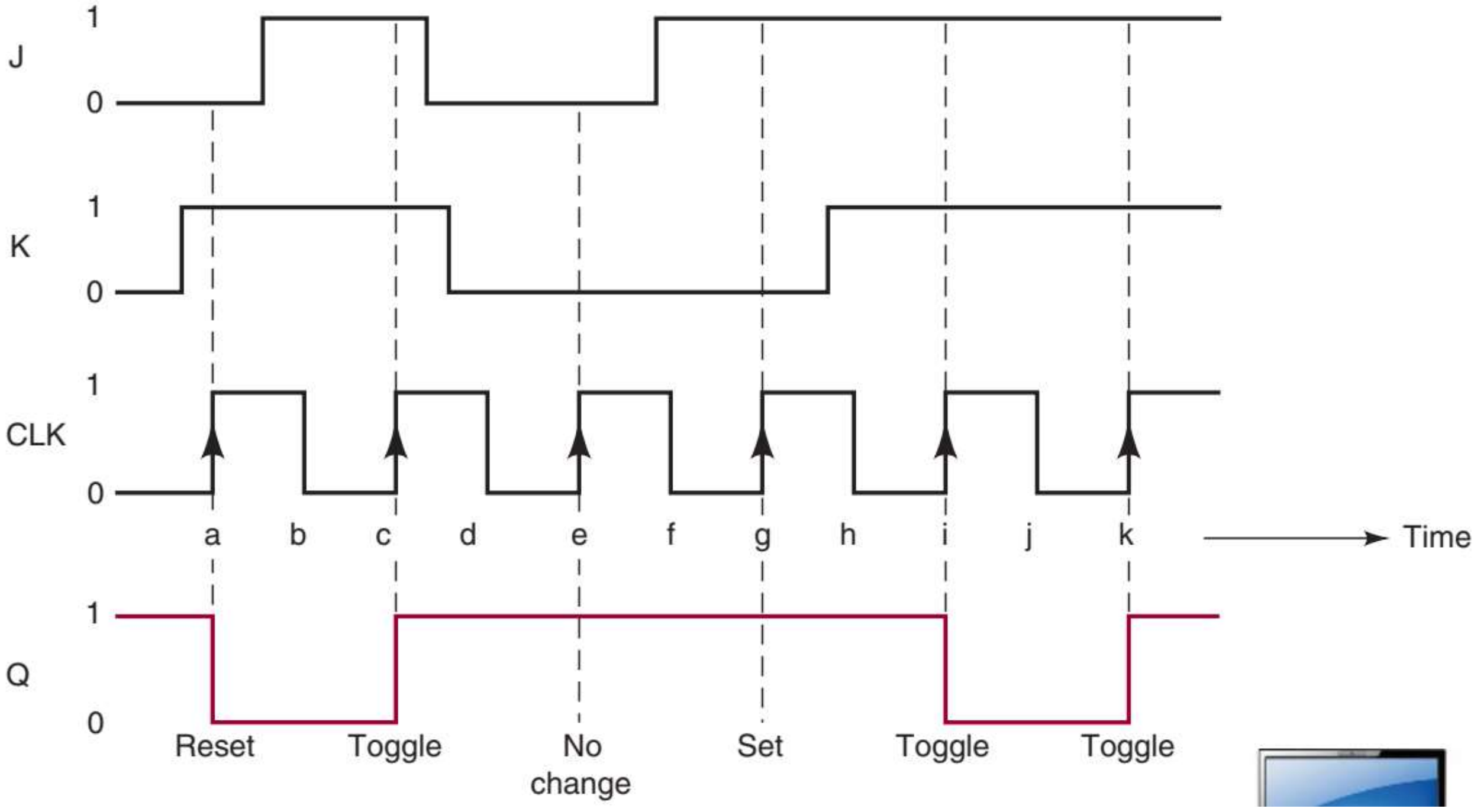



| J | K | CLK | Q |
|---|---|---|---|
| 0 | 0 | ↓ | $Q_0$ (no change) |
| 1 | 0 | ↓ | 1 |
| 0 | 1 | ↓ | 0 |
| 1 | 1 | ↓ | $\overline{Q}_0$ (toggles) |

# Flip Flop

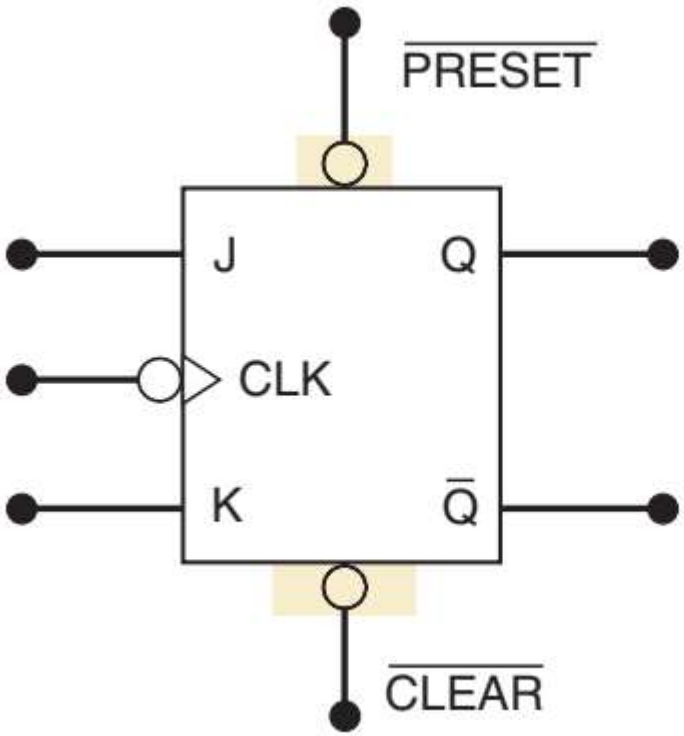
- Flip Flop J-K



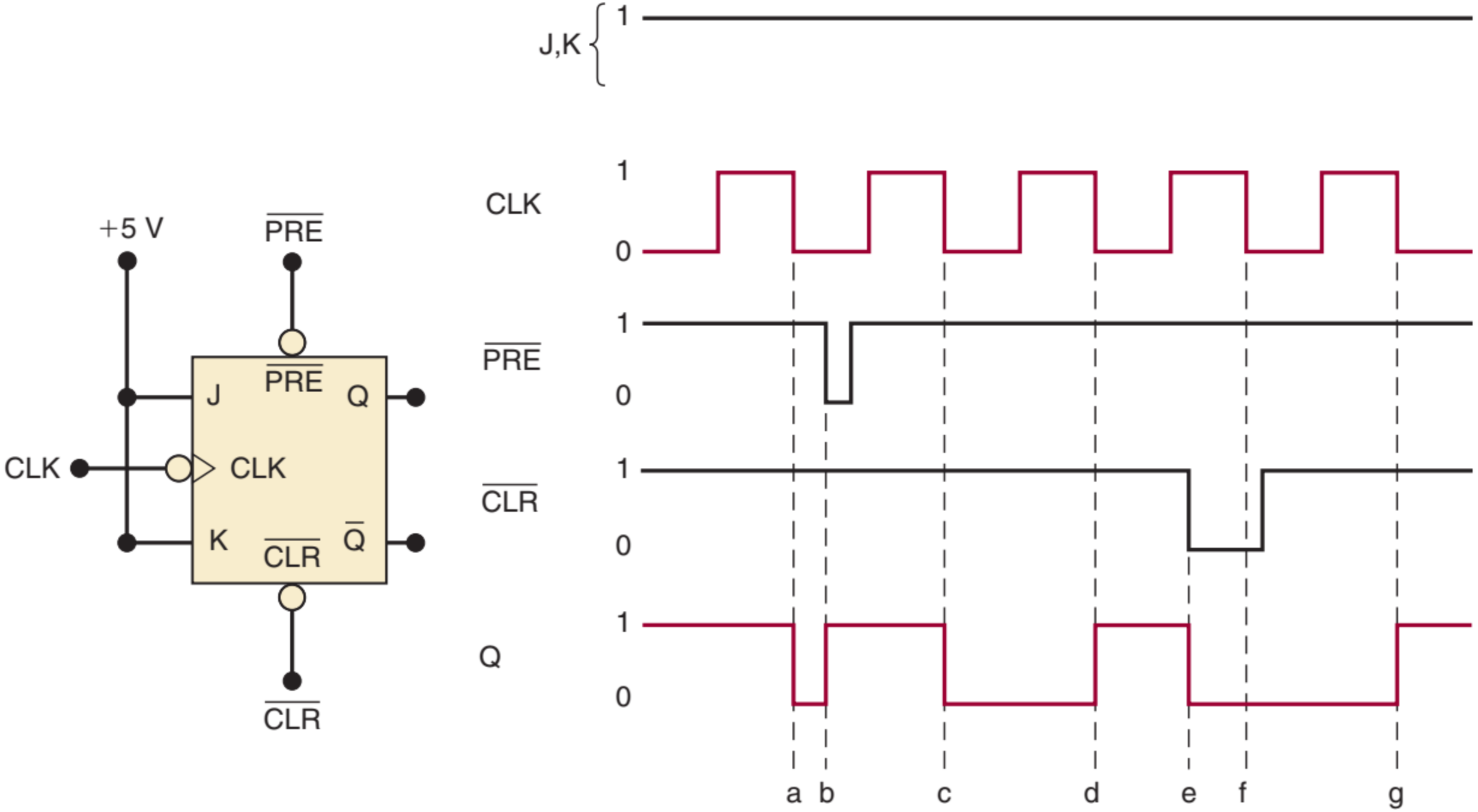

J
K
CLK
Q
1
0
a
b
c
d
e
f
g
h
i
j
k
Time
Reset
Toggle
No change
Set
Toggle
Toggle

# Flip Flop

- Flip Flop J-K




| J | K | Clk | $\overline{PRE}$ | $\overline{CLR}$ | Q |
|---|---|---|---|---|---|
| 0 | 0 | ↓ | 1 | 1 | Q (no change) |
| 0 | 1 | ↓ | 1 | 1 | 0 (Synch reset) |
| 1 | 0 | ↓ | 1 | 1 | 1 (Synch set) |
| 1 | 1 | ↓ | 1 | 1 | $\overline{Q}$ (Synch toggle) |
| x | x | x | 1 | 1 | Q (no change) |
| x | x | x | 1 | 0 | 0 (asynch clear) |
| x | x | x | 0 | 1 | 1 (asynch preset) |
| x | x | x | 0 | 0 | (Invalid) |


+5 V
PRE
J
PRE
Q
CLK
CLK
K
CLR
Q
CLR
J,K
1
CLK
1
0
PRE
1
0
CLR
1
0
Q
1
0
a
b
c
d
e
f
g

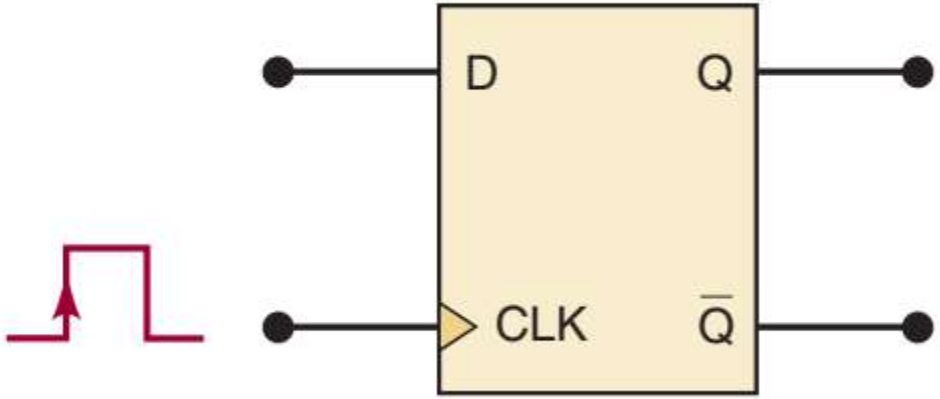
# Flip Flop

- Flip Flop J-K

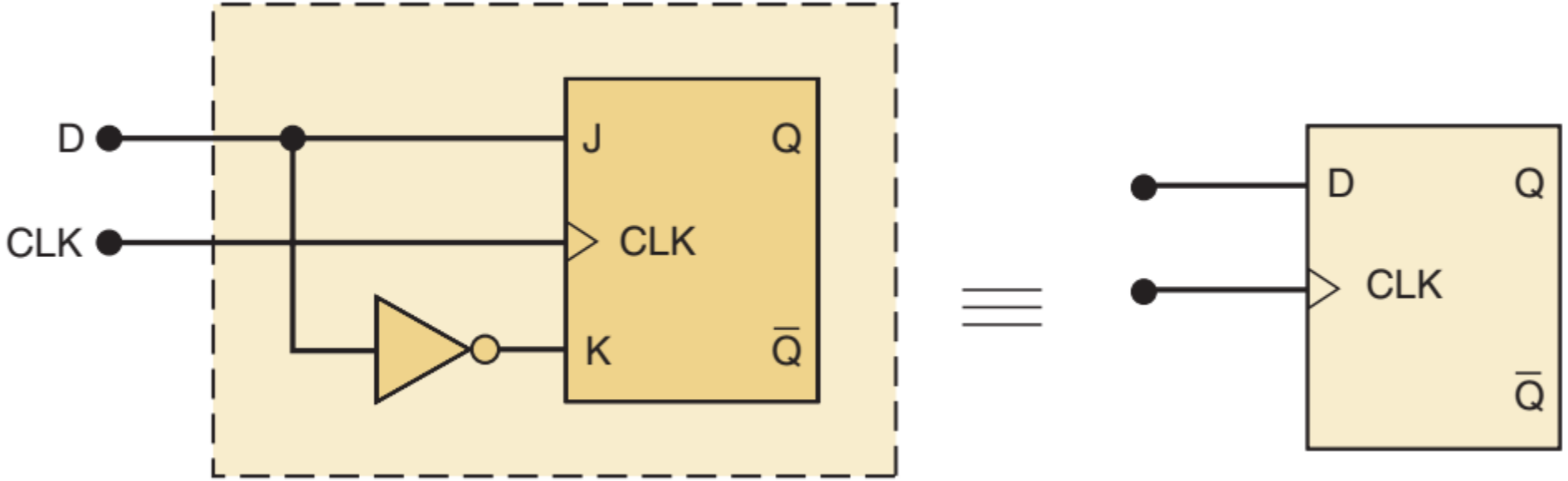
| Point | Operation |
|---|---|
| a | Synchronous toggle on NGT of CLK |
| b | Asynchronous set on $\overline{PRE} = 0$ |
| c | Synchronous toggle |
| d | Synchronous toggle |
| e | Asynchronous clear on $\overline{CLR} = 0$ |
| f | $\overline{CLR}$ overrides the NGT of CLK |
| g | Synchronous toggle |

# Flip Flop

- Flip Flop D

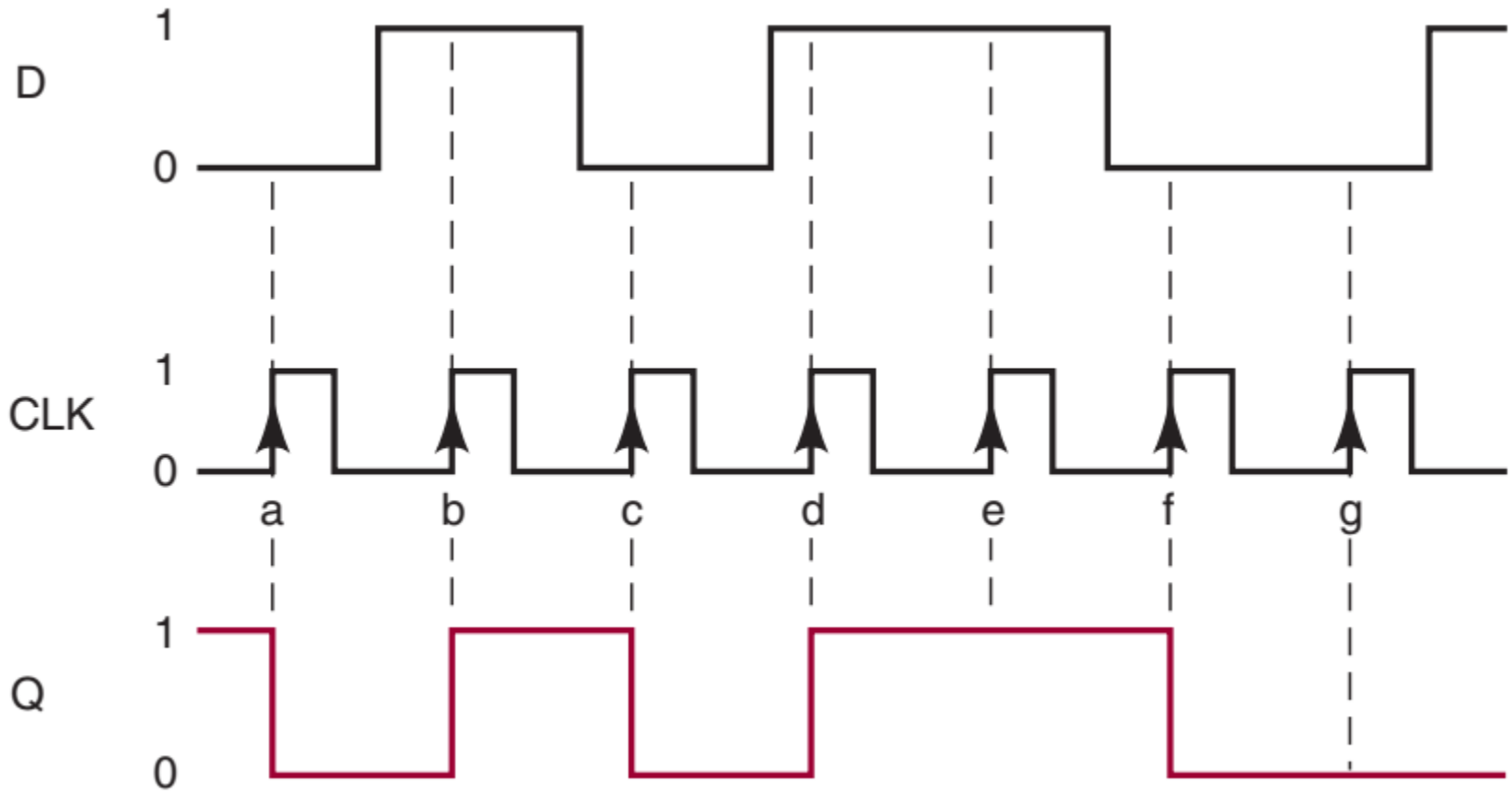



| D | CLK | Q |
|---|---|---|
| 0 | ↑ | 0 |
| 1 | ↑ | 1 |



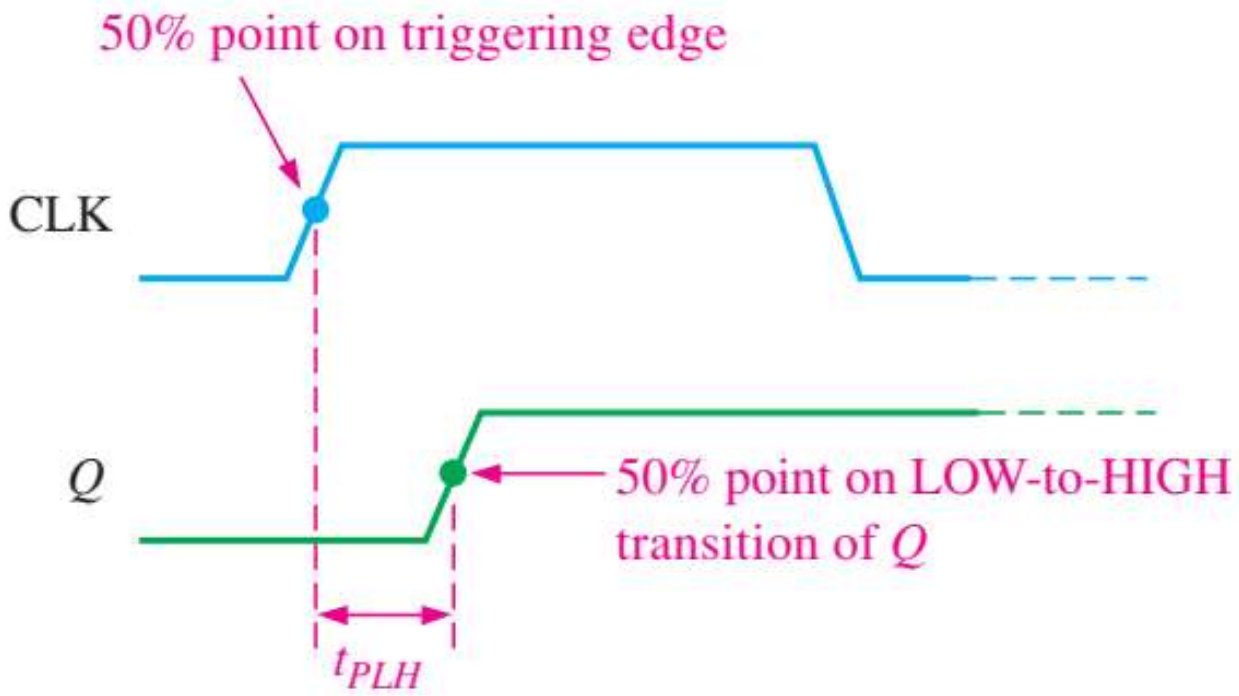
# Flip Flop

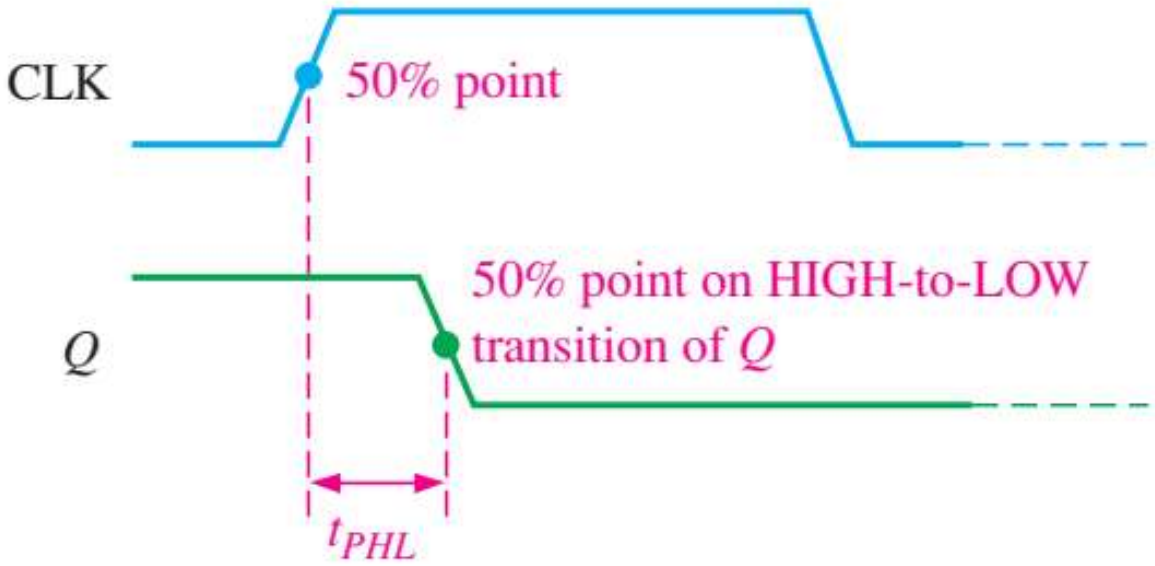
- Flip Flop D



# Flip Flop

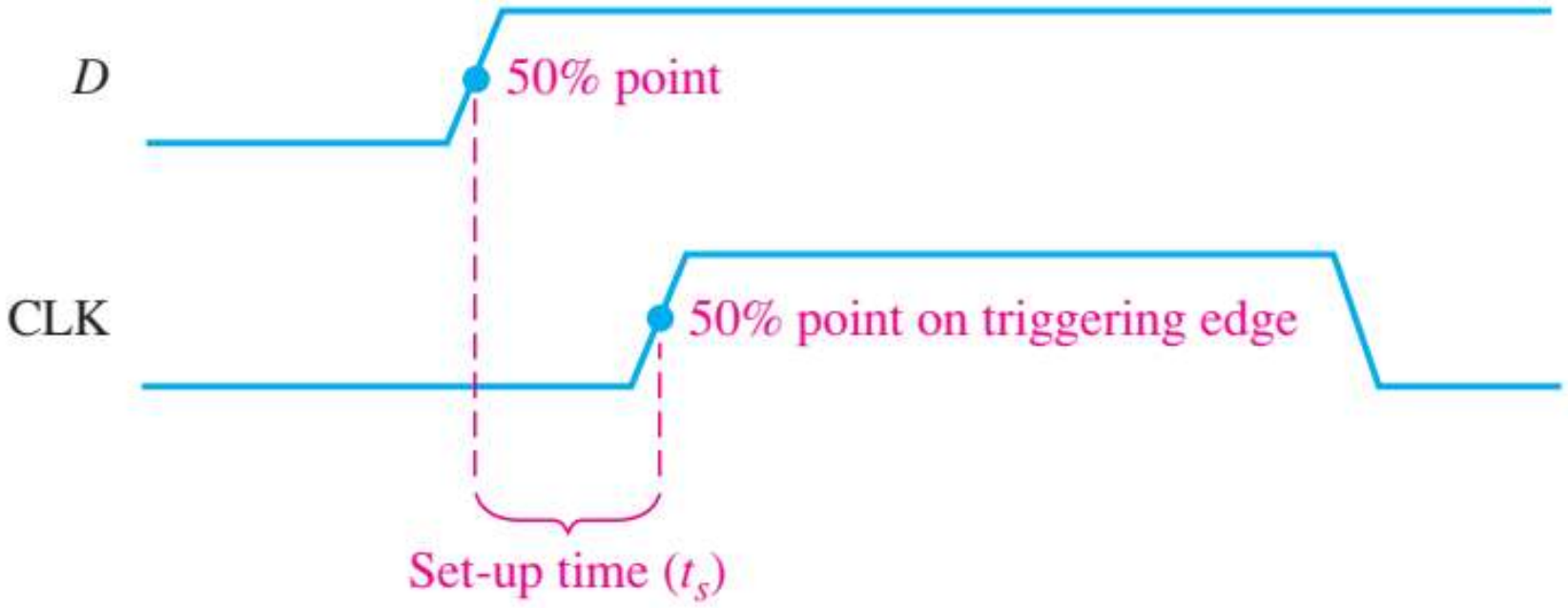
- Thời gian Delay



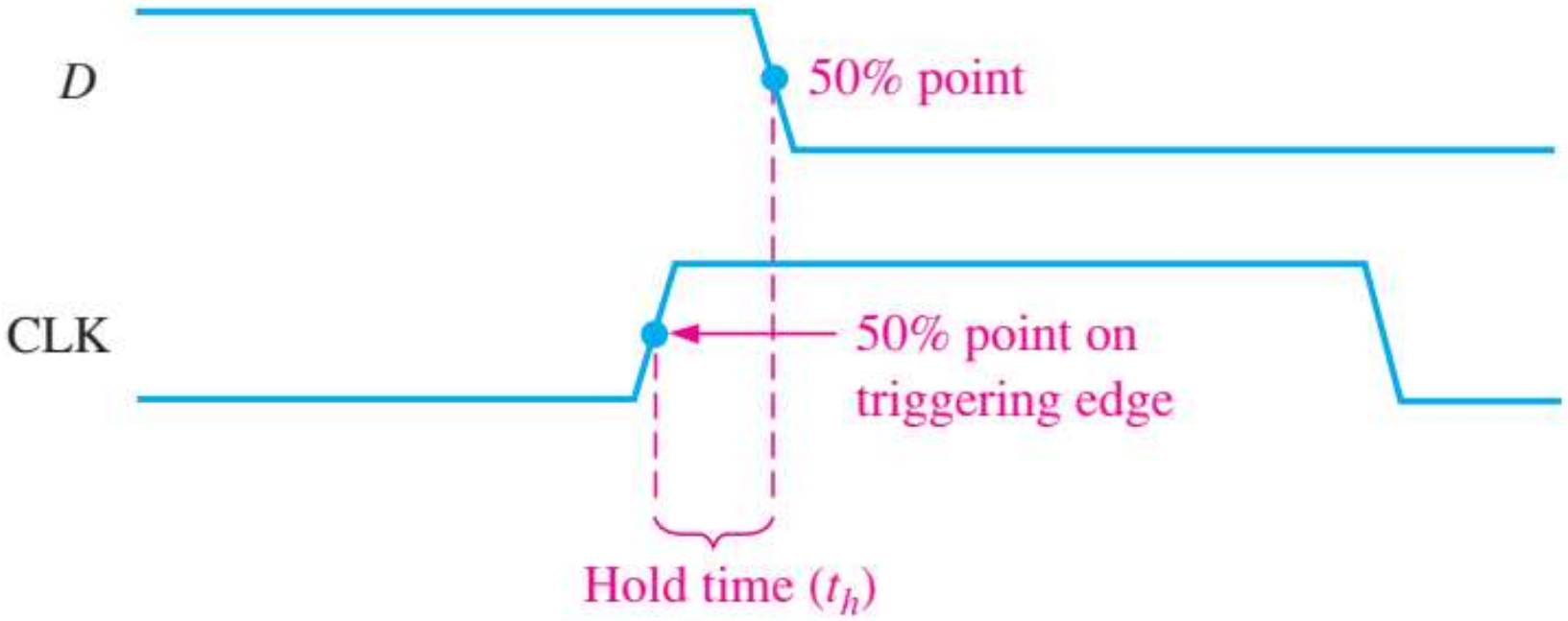
# Flip Flop

- Thời gian Set-up



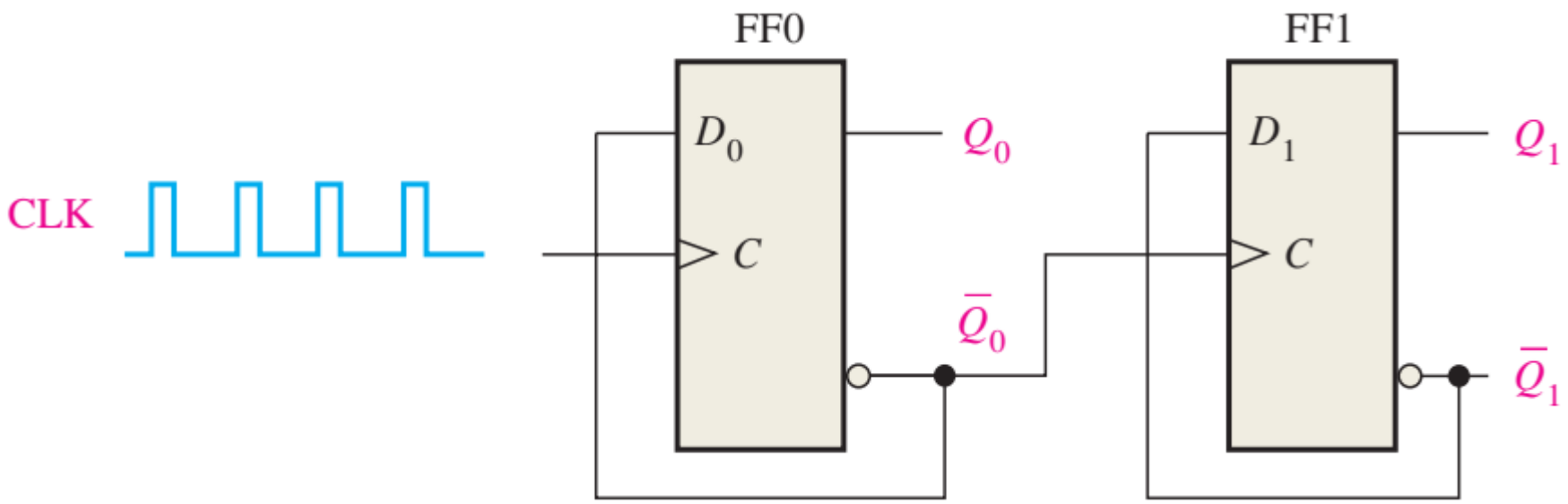
# Flip Flop

- Thời gian Hold



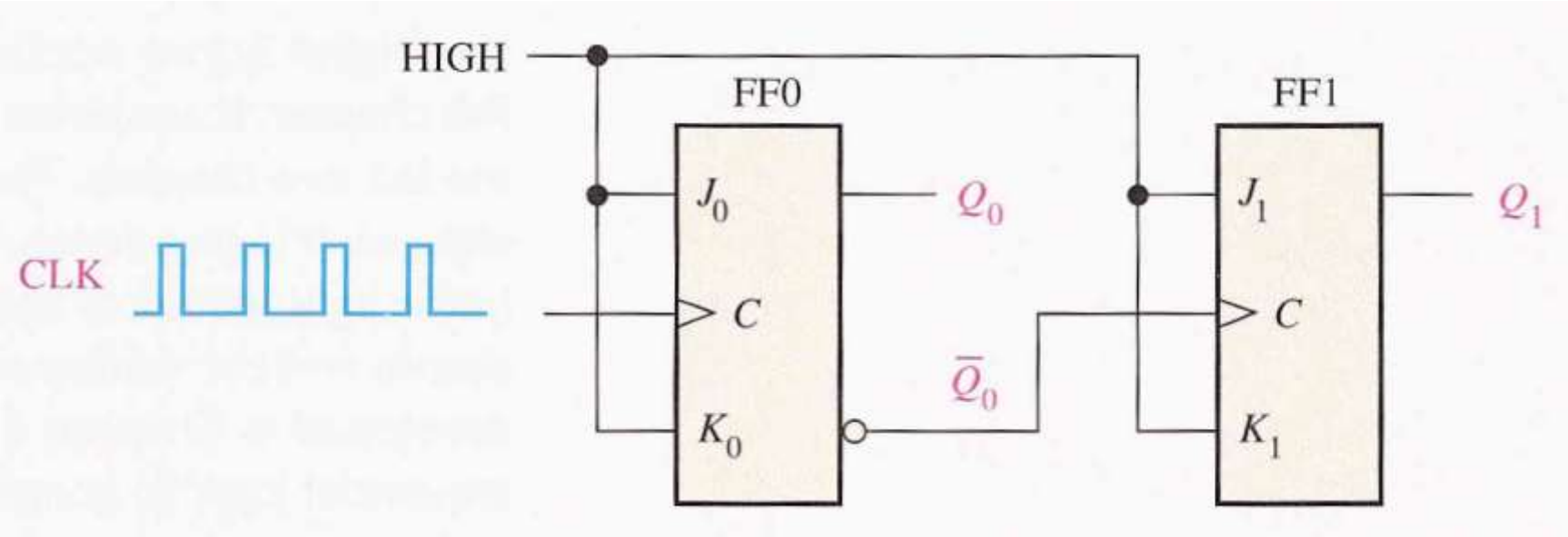
# Mạch đếm

- Mạch đếm không đồng bộ 2 bit



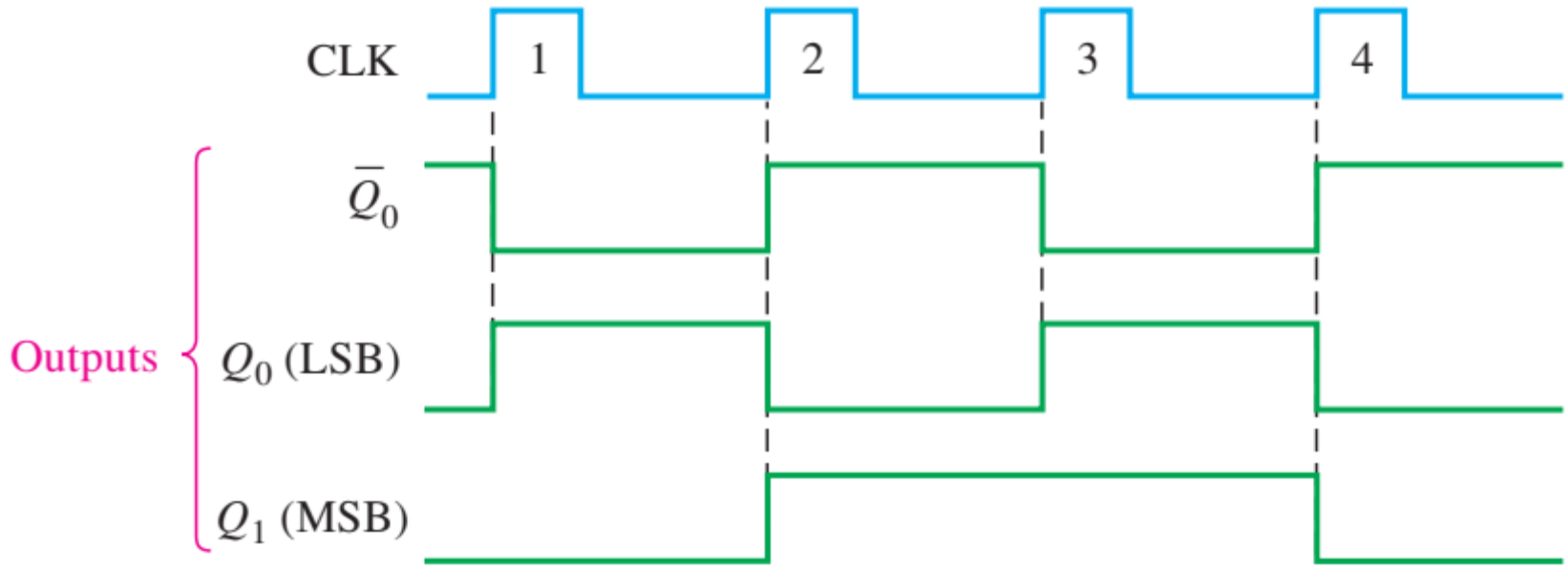
# Mạch đếm

- Mạch đếm không đồng bộ 2 bit



# Mạch đếm

- Mạch đếm không đồng bộ 2 bit



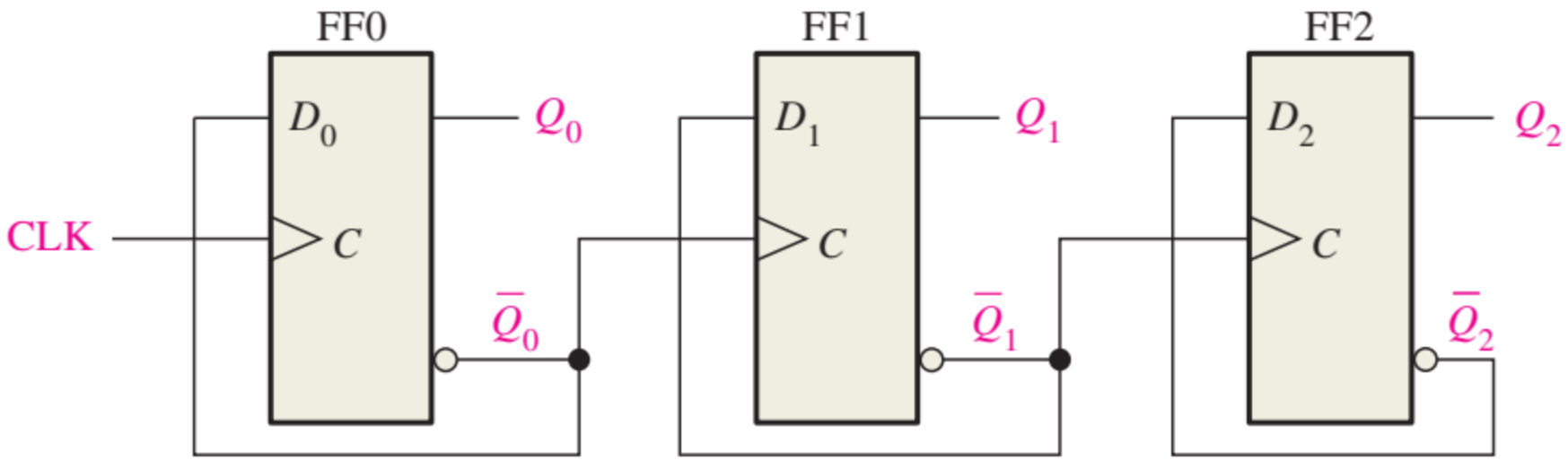
# Mạch đếm

- Mạch đếm không đồng bộ 2 bit

| Clock Pulse | $Q_1$ | $Q_0$ |
|---|---|---|
| Initially | 0 | 0 |
| 1 | 0 | 1 |
| 2 | 1 | 0 |
| 3 | 1 | 1 |
| 4 (recycles) | 0 | 0 |

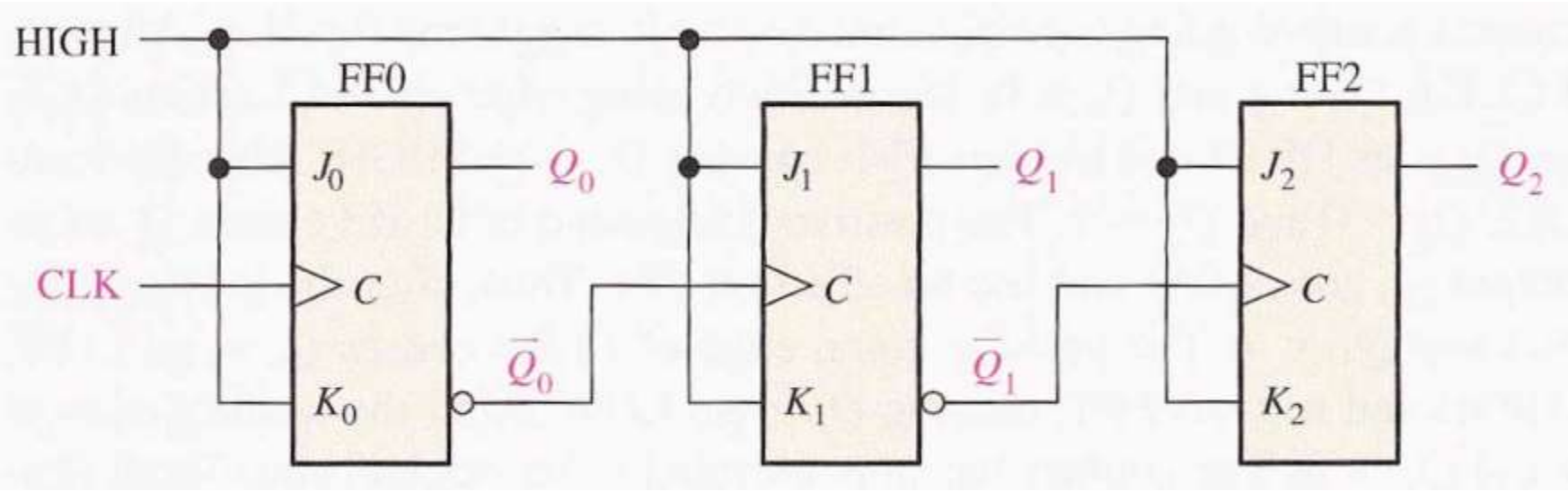
# Mạch đếm

- Mạch đếm không đồng bộ 3 bit



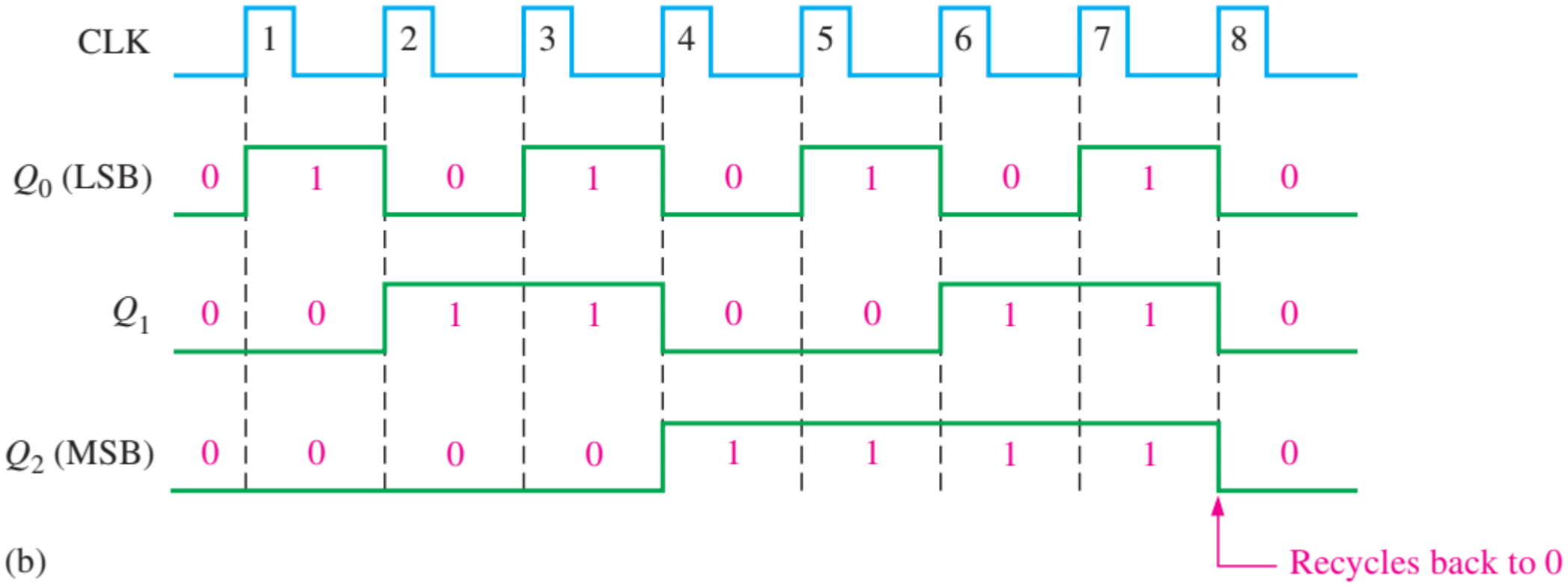
# Mạch đếm

- Mạch đếm không đồng bộ 3 bit



# Mạch đếm

- Mạch đếm không đồng bộ 3 bit




(b)

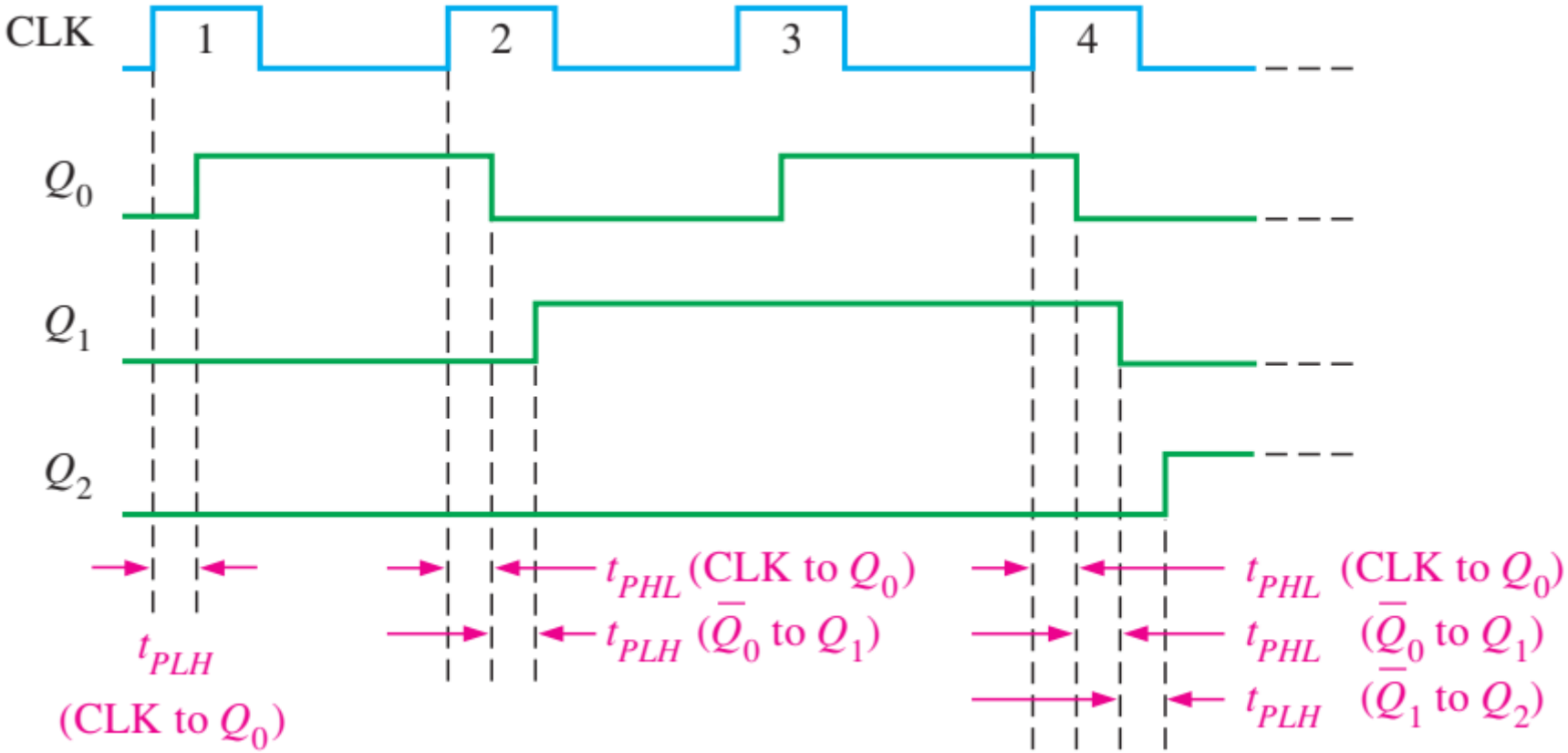
# Mạch đếm

- Mạch đếm không đồng bộ 3 bit

| Clock Pulse | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|
| Initially | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 2 | 0 | 1 | 0 |
| 3 | 0 | 1 | 1 |
| 4 | 1 | 0 | 0 |
| 5 | 1 | 0 | 1 |
| 6 | 1 | 1 | 0 |
| 7 | 1 | 1 | 1 |
| 8 (recycles) | 0 | 0 | 0 |

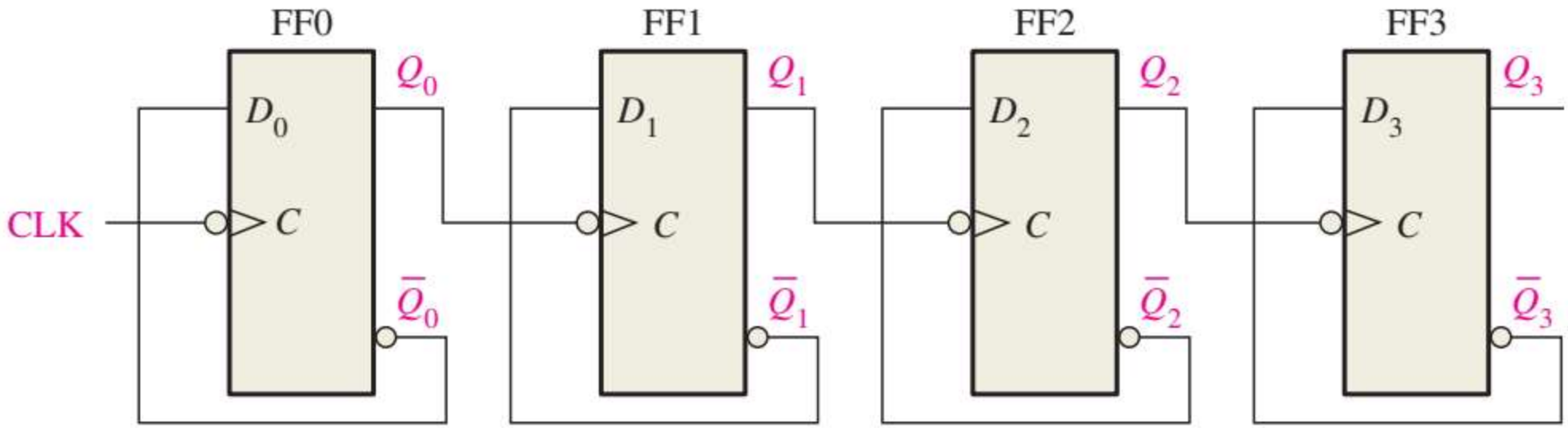
# Mạch đếm

- Mạch đếm không đồng bộ 3 bit



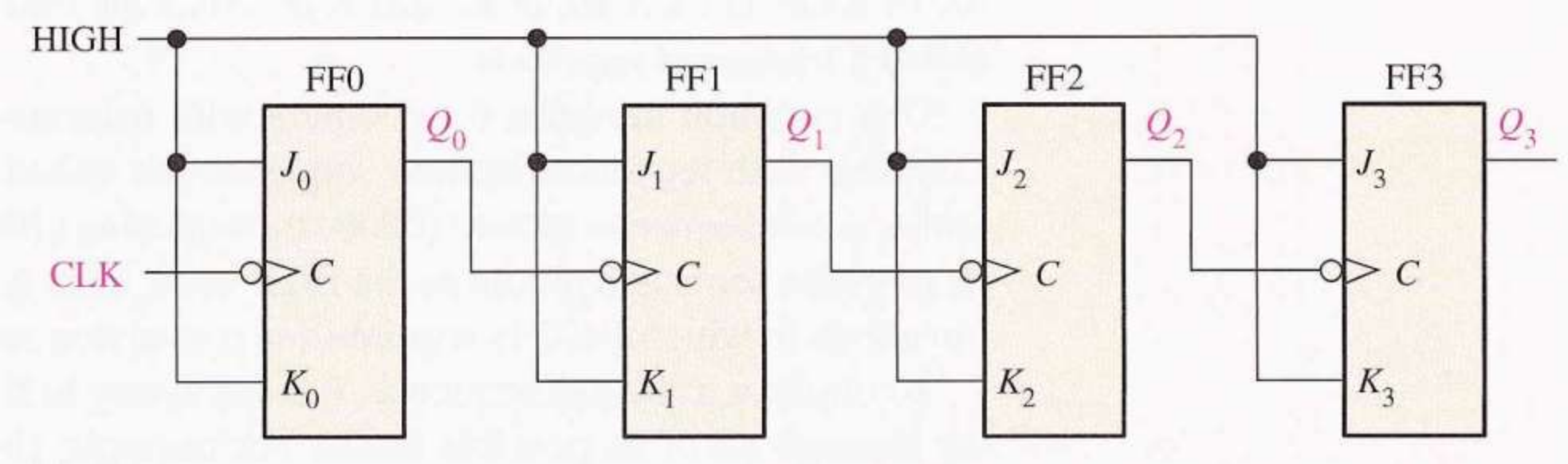
# Mạch đếm

- Mạch đếm không đồng bộ 4 bit



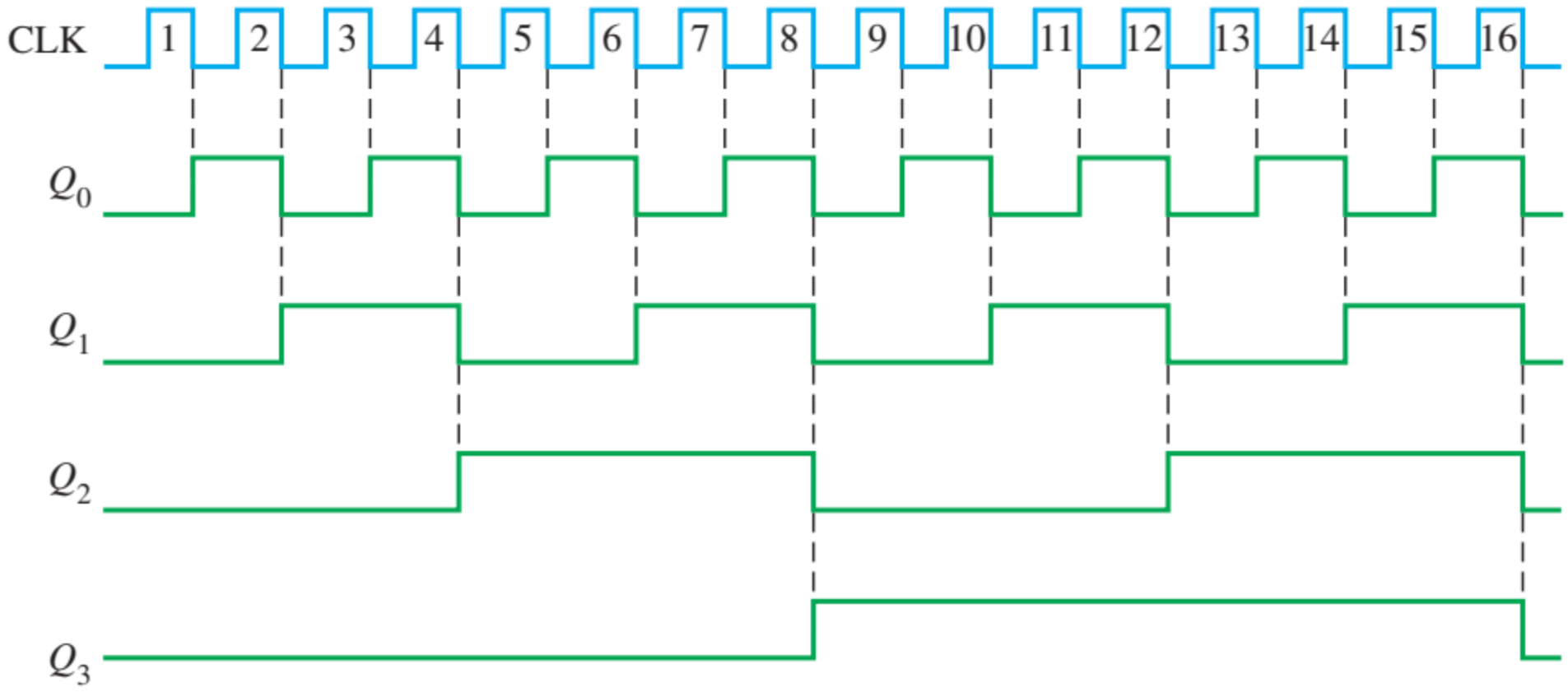
# Mạch đếm

- Mạch đếm không đồng bộ 4 bit



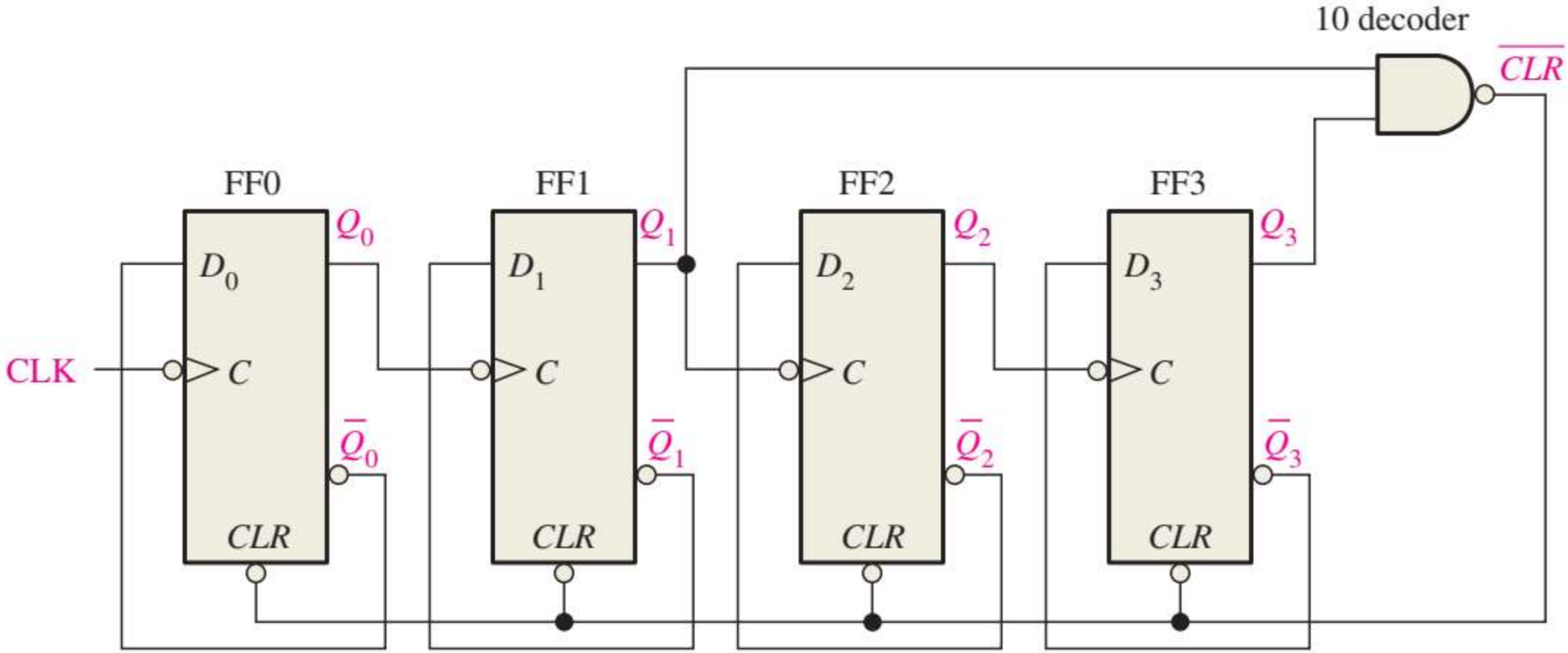
# Mạch đếm

- Mạch đếm không đồng bộ 4 bit



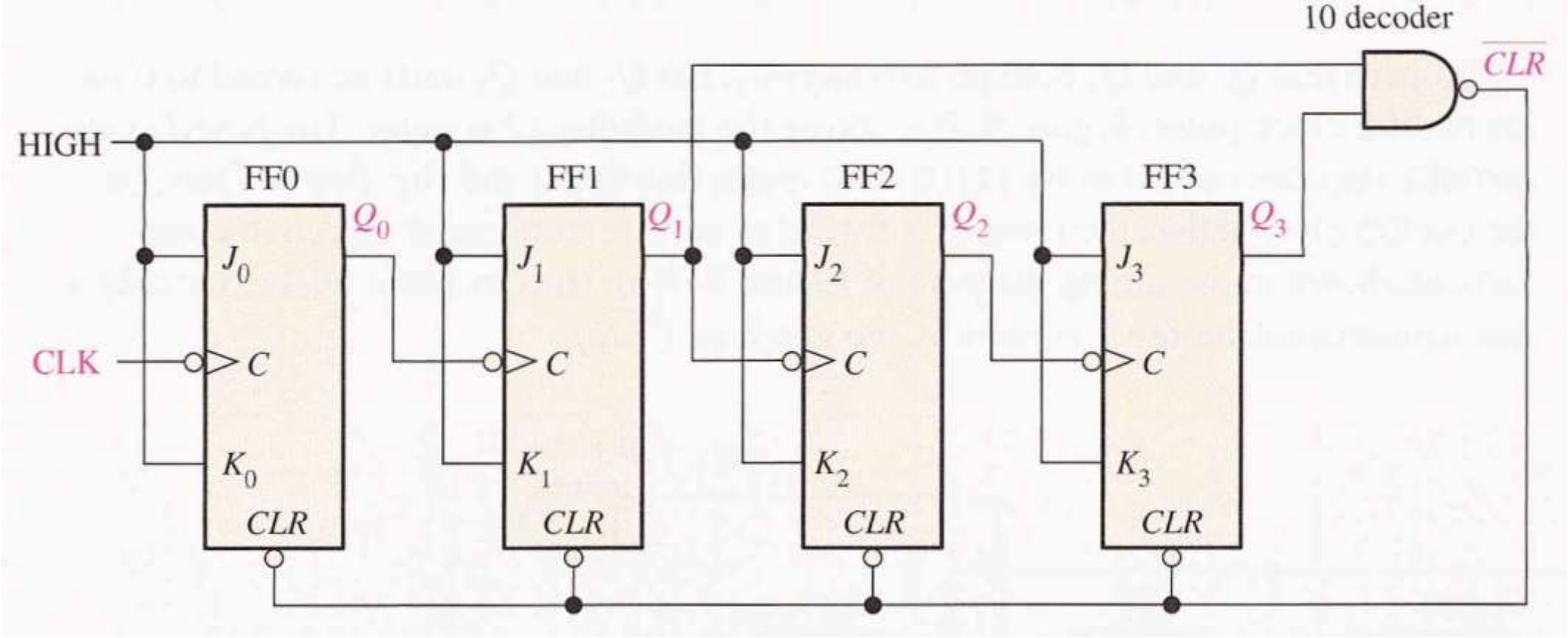
# Mạch đếm

- Mạch đếm không đồng bộ MOD 10



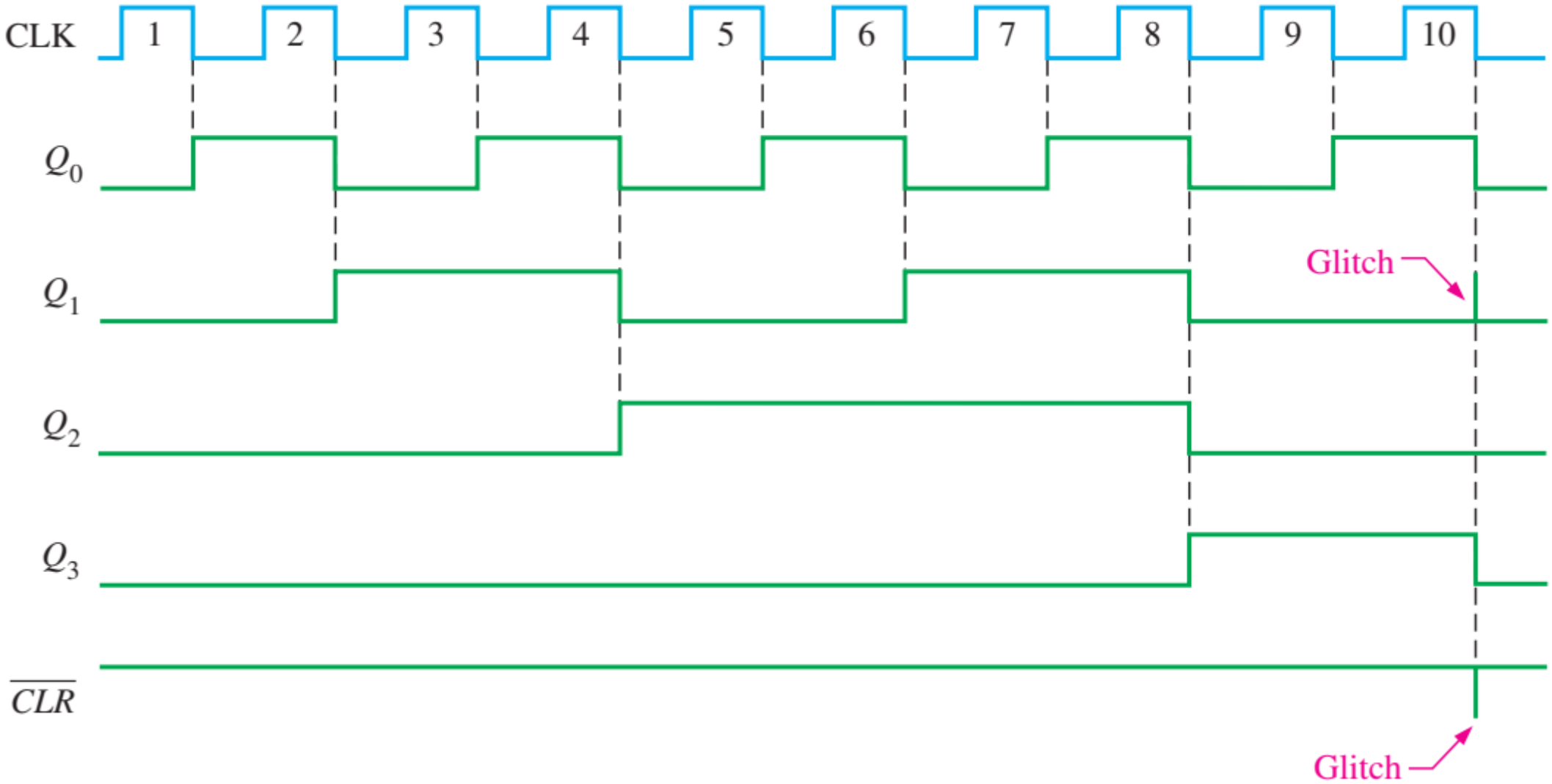
# Mạch đếm

- Mạch đếm không đồng bộ MOD 10



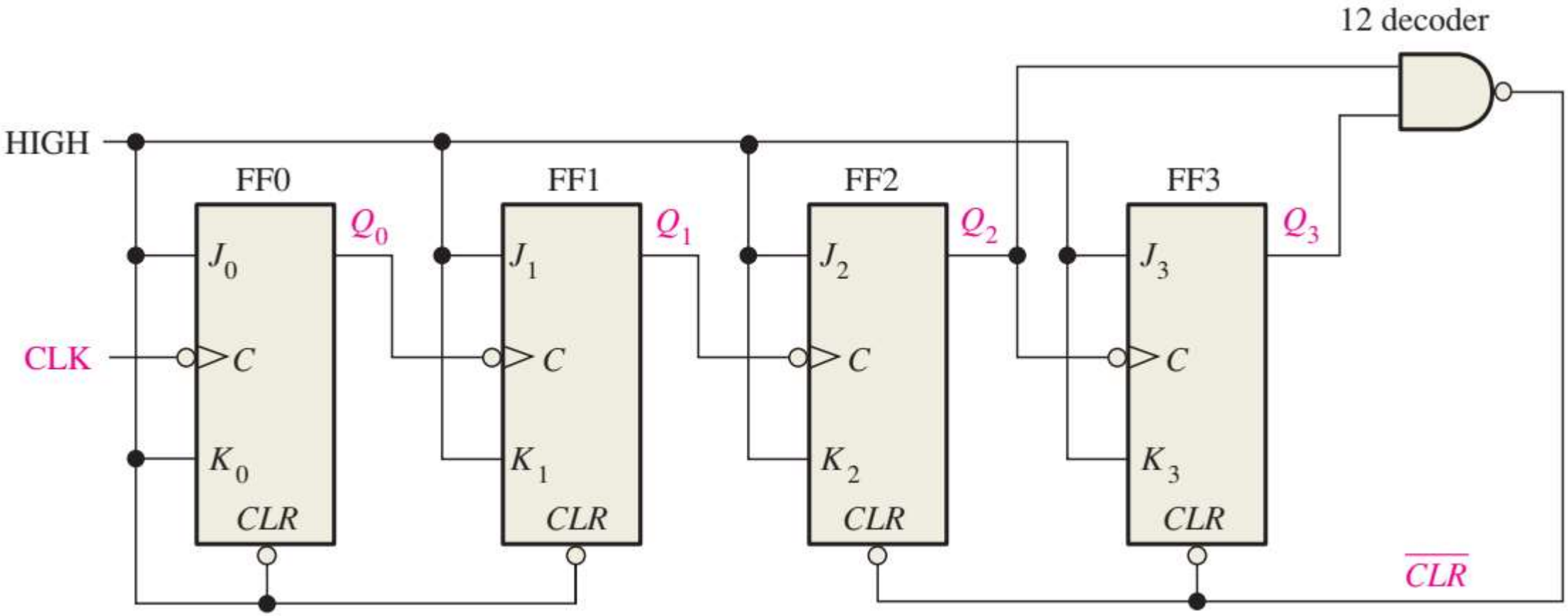
# Mạch đếm

- Mạch đếm không đồng bộ MOD 10



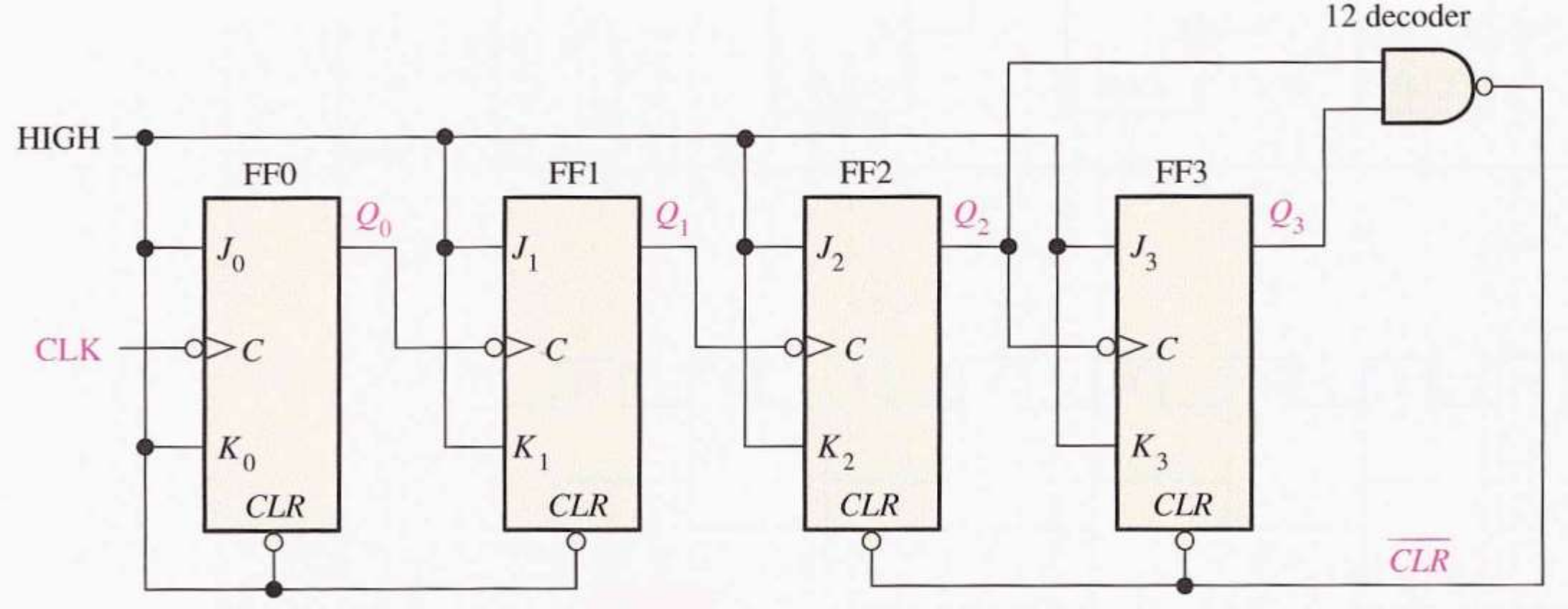
# Mạch đếm

- Mạch đếm không đồng bộ MOD 12



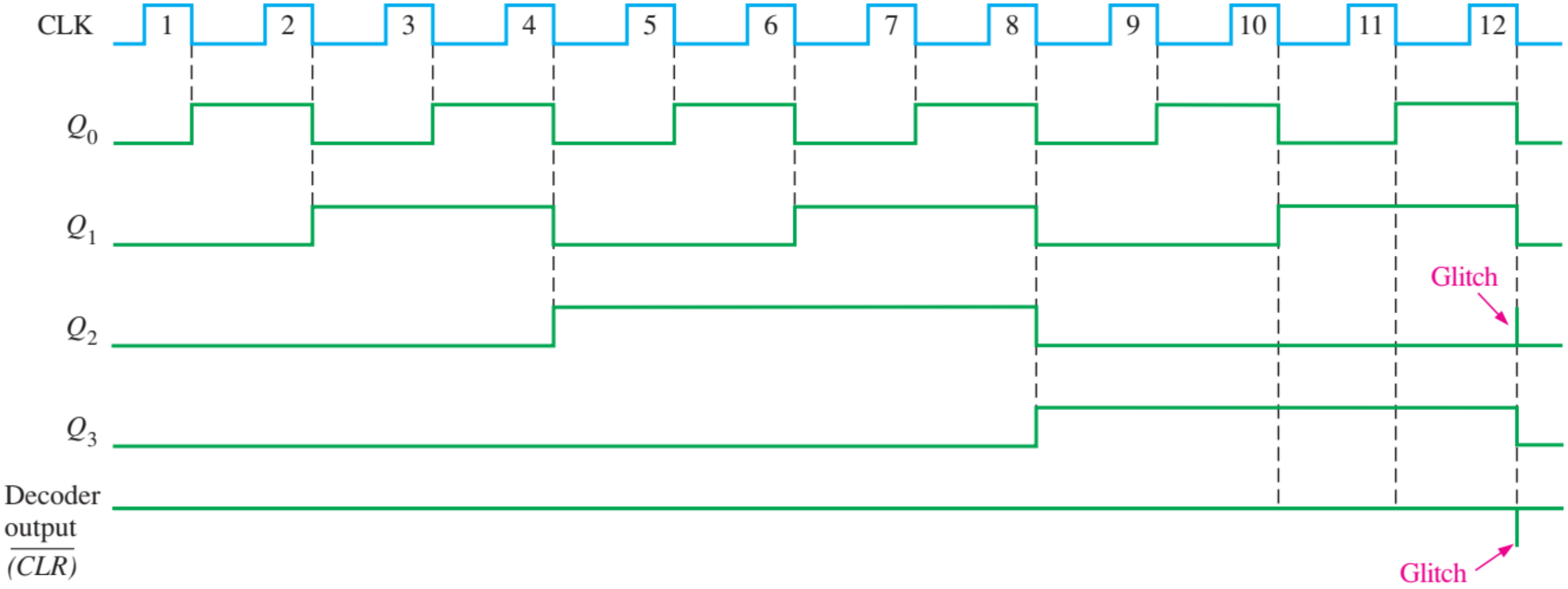
# Mạch đếm

- Mạch đếm không đồng bộ MOD 12



# Mạch đếm

- Mạch đếm không đồng bộ MOD 12



# Mạch đếm

- Mạch đếm không đồng bộ

– Với mạch đếm lên sử dụng FF với CK tích cực cạnh xuống thì $Q_I$ nối với ngõ vào $CK_{I+1}$.

– Với mạch đếm xuống sử dụng FF với CK tích cực cạnh xuống thì $\overline{Q_I}$ nối với ngõ vào $CK_{I+1}$.

– Với mạch đếm lên sử dụng FF với CK tích cực cạnh lên thì $\overline{Q_I}$ nối với ngõ vào $CK_{I+1}$.

– Với mạch đếm xuống sử dụng FF với CK tích cực cạnh lên thì $Q_I$ nối với ngõ vào $CK_{I+1}$.

# Mạch đếm

- Thiết kế mạch đếm không đồng bộ

  Lập biểu đồ trạng thái

  Lập bảng trạng thái

  Xác định loại FF

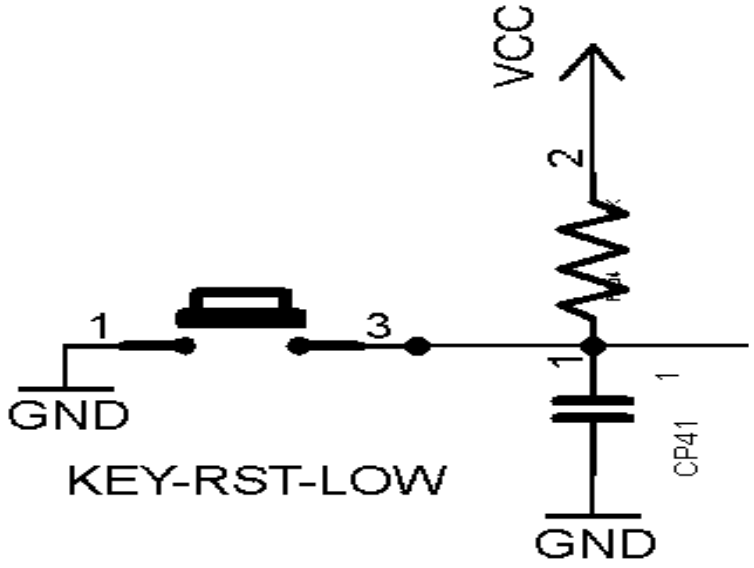
  Xác định mạch Reset

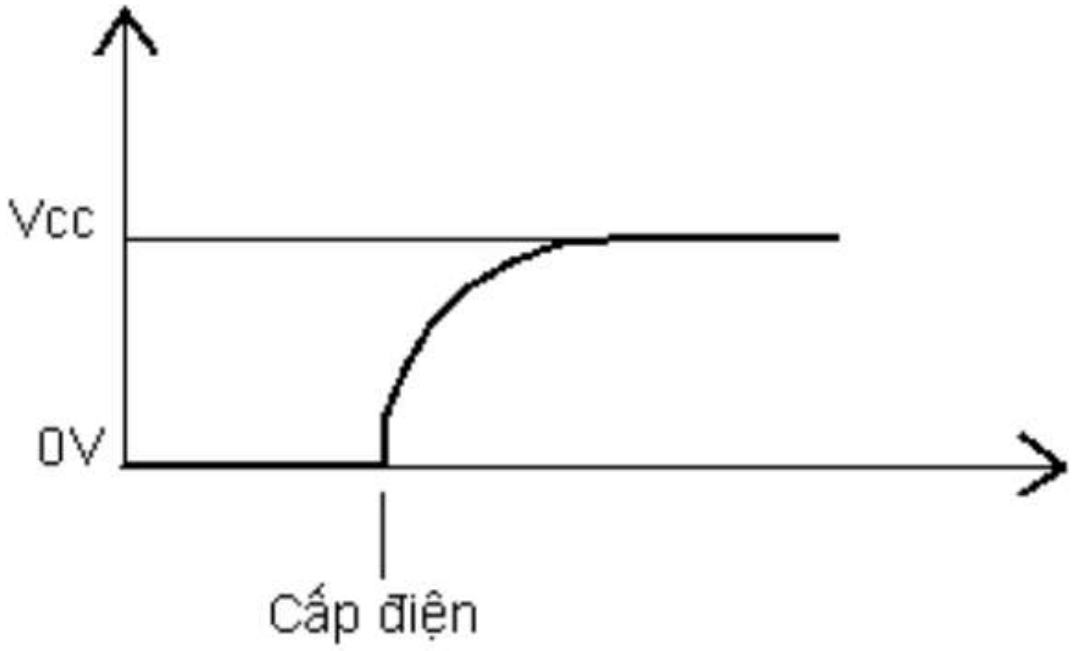
  Vẽ mạch hoàn chỉnh

  Vẽ dạng sóng

# Mạch đếm

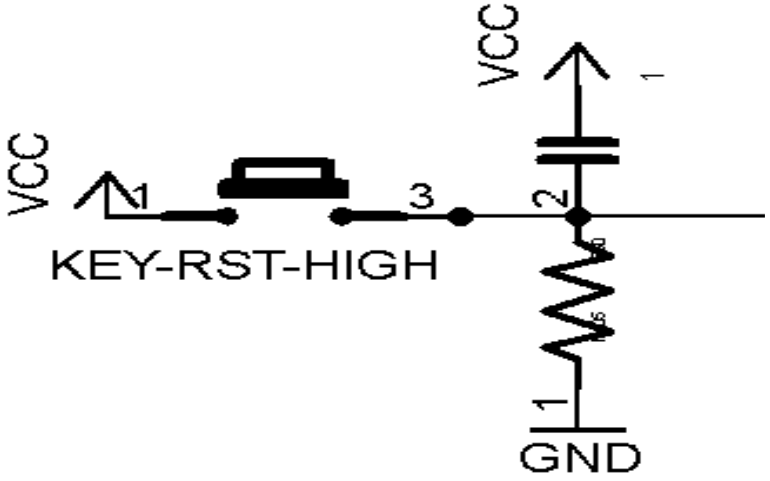
- Nút nhấn RST-L, RST-H

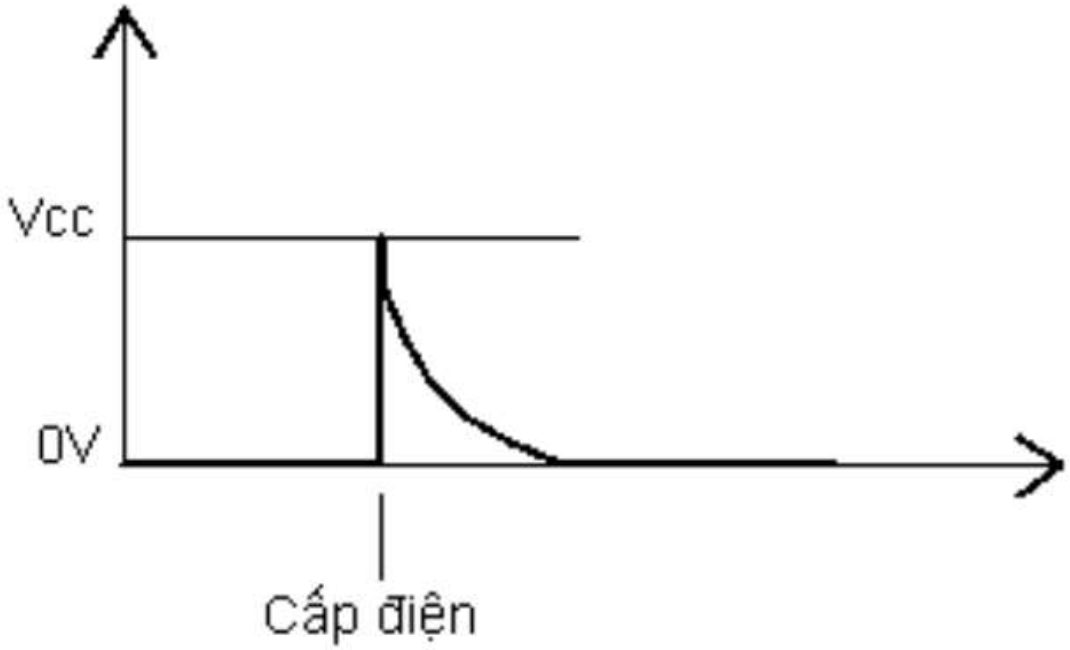


# Mạch đếm

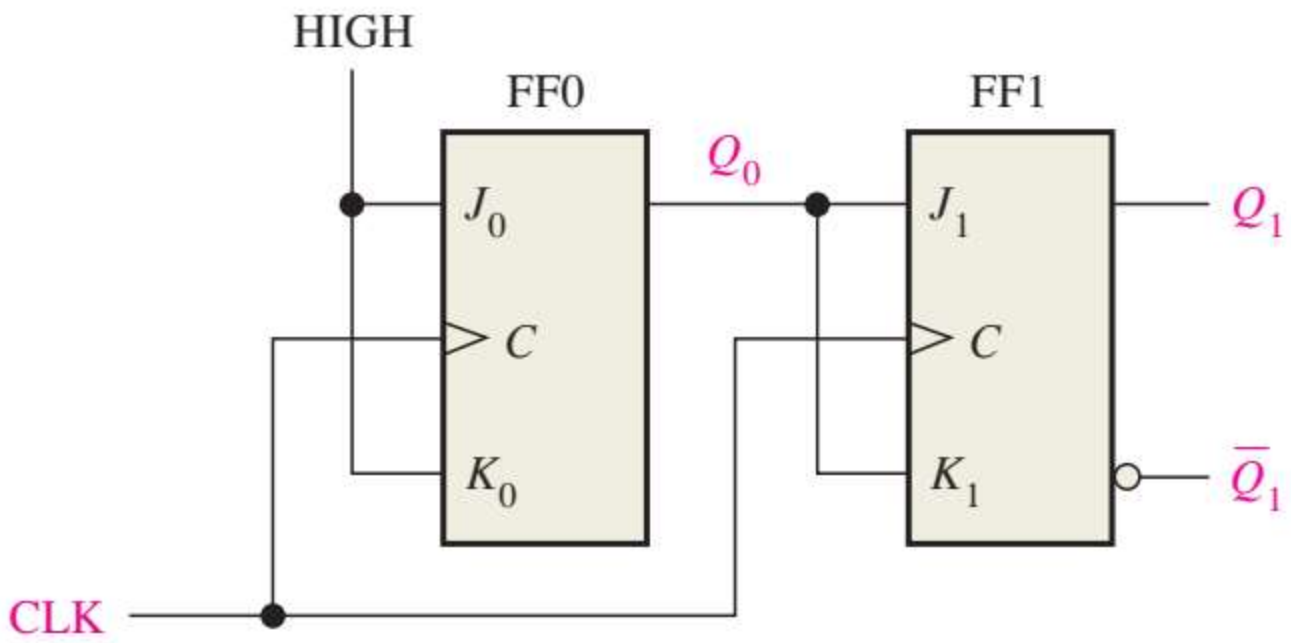
- Nút nhấn RST-L, RST-H



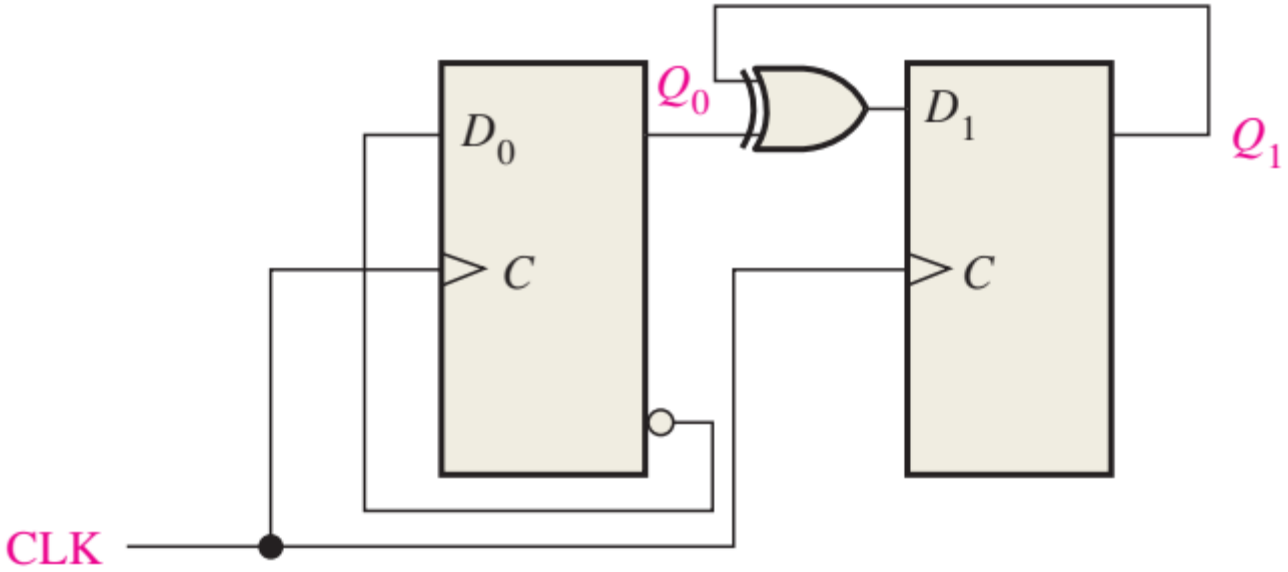
# Mạch đếm

- Mạch đếm đồng bộ 2 bit



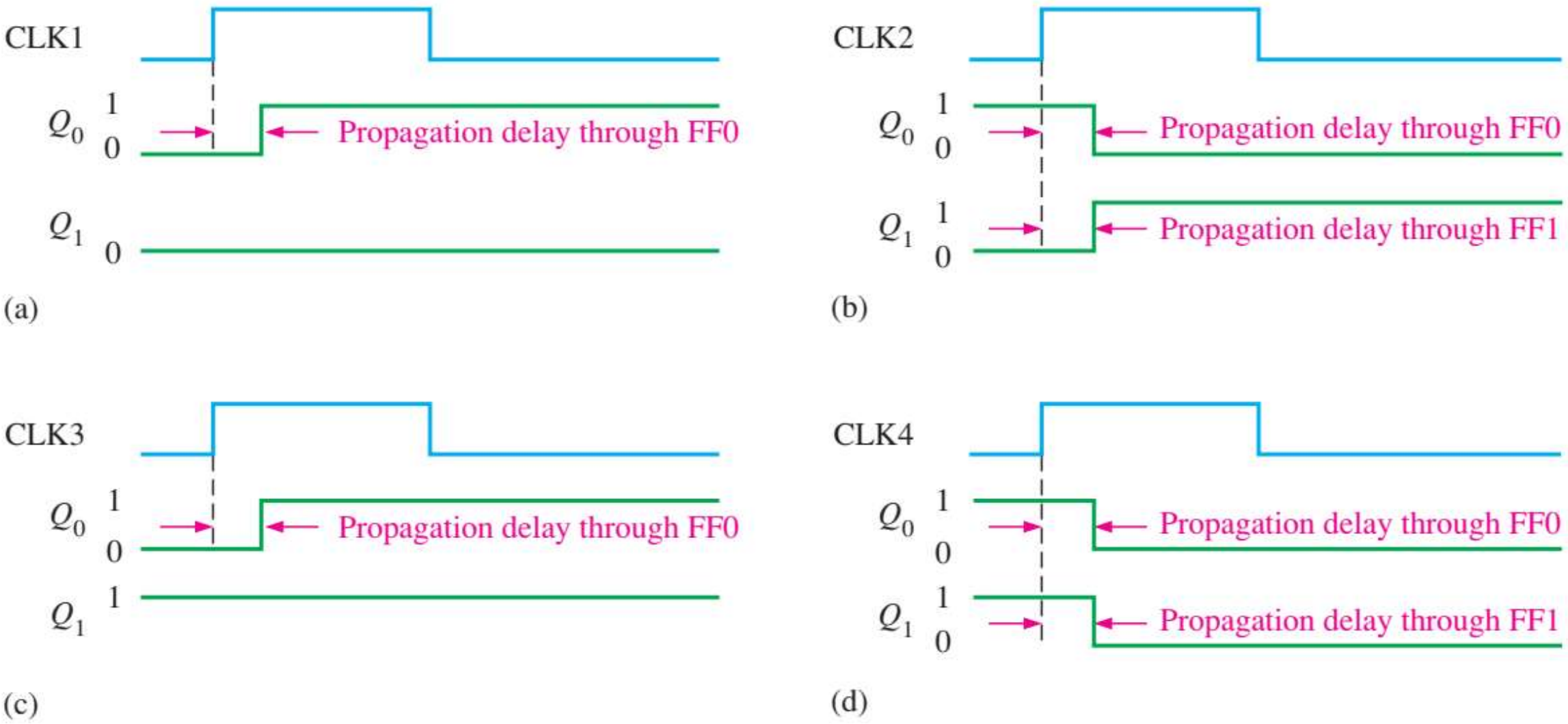
# Mạch đếm

- Mạch đếm đồng bộ 2 bit



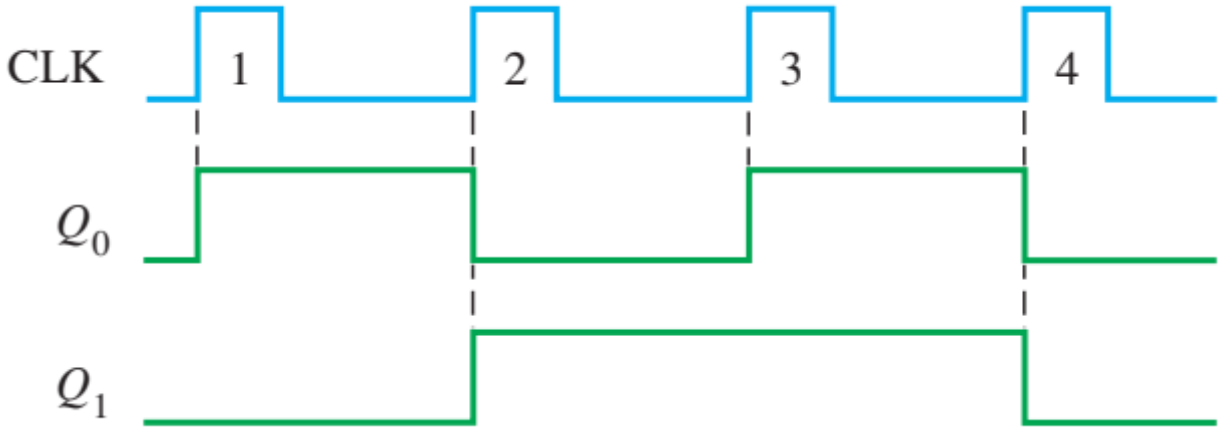
# Mạch đếm

- Mạch đếm đồng bộ 2 bit



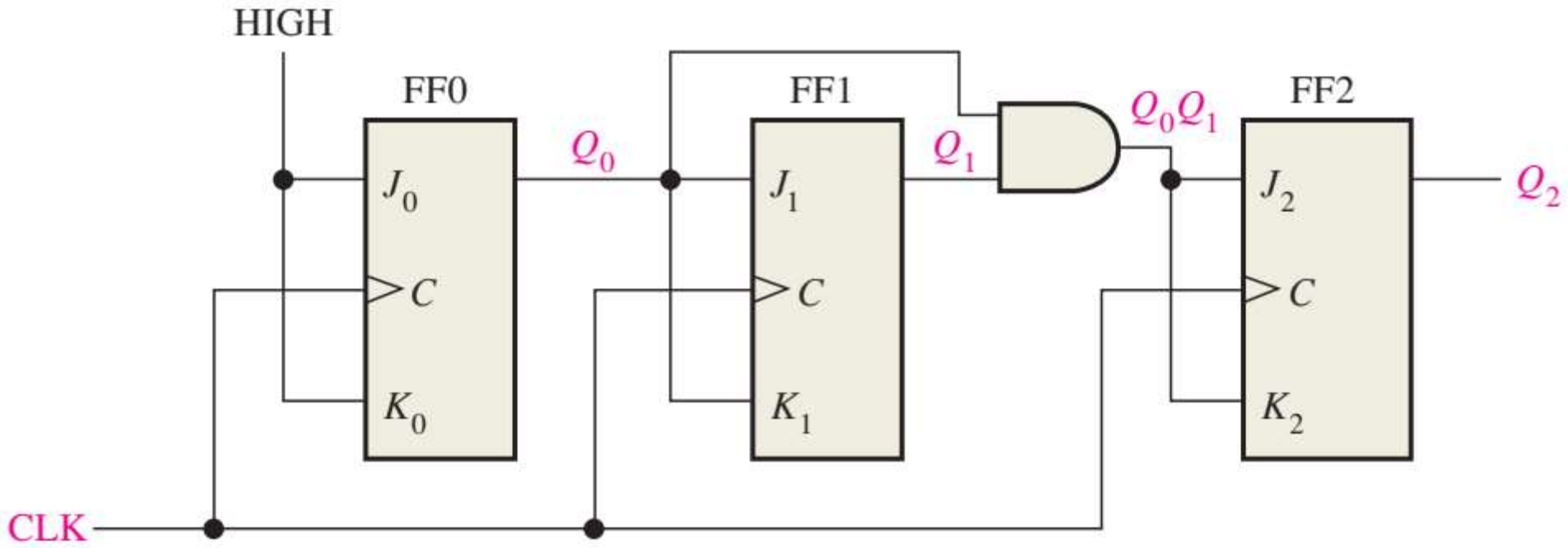
# Mạch đếm

- Mạch đếm đồng bộ 2 bit



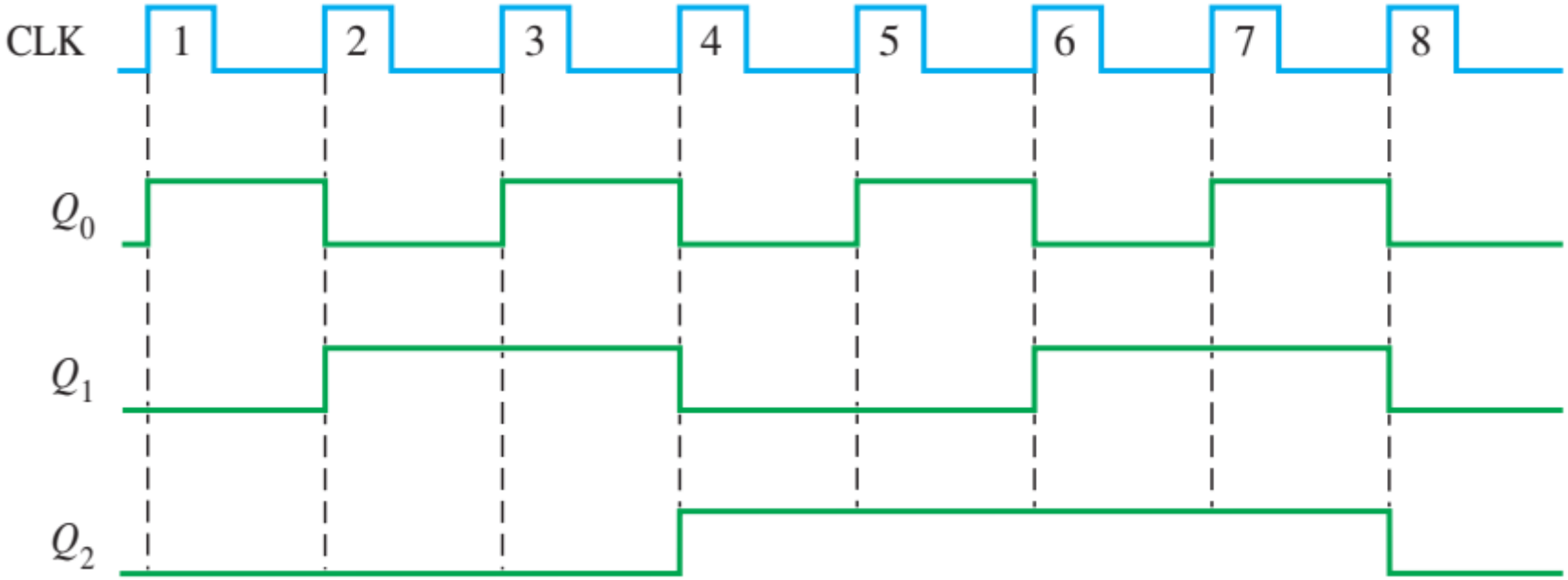
# Mạch đếm

- Mạch đếm đồng bộ 3 bit



# Mạch đếm

- Mạch đếm đồng bộ 3 bit



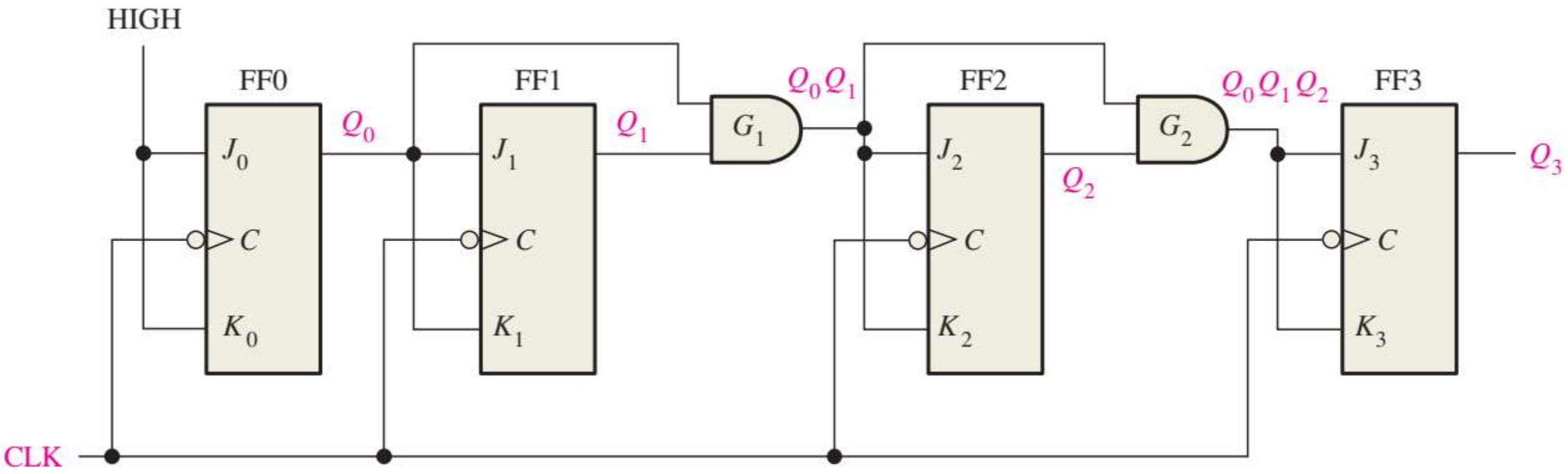
# Mạch đếm

- Mạch đếm đồng bộ 3 bit

| Clock Pulse | Outputs | | | J-K Inputs | | | | | | At the Next Clock Pulse | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $Q_2$ | $Q_1$ | $Q_0$ | $J_2$ | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ | FF2 | FF1 | FF0 |
| Initially | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | NC* | NC | Toggle |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | NC | Toggle | Toggle |
| 2 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | NC | NC | Toggle |
| 3 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Toggle | Toggle | Toggle |
| 4 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | NC | NC | Toggle |
| 5 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | NC | Toggle | Toggle |
| 6 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | NC | NC | Toggle |
| 7 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Toggle | Toggle | Toggle |
| | | | | | | | | | | Counter recycles back to 000. | | |

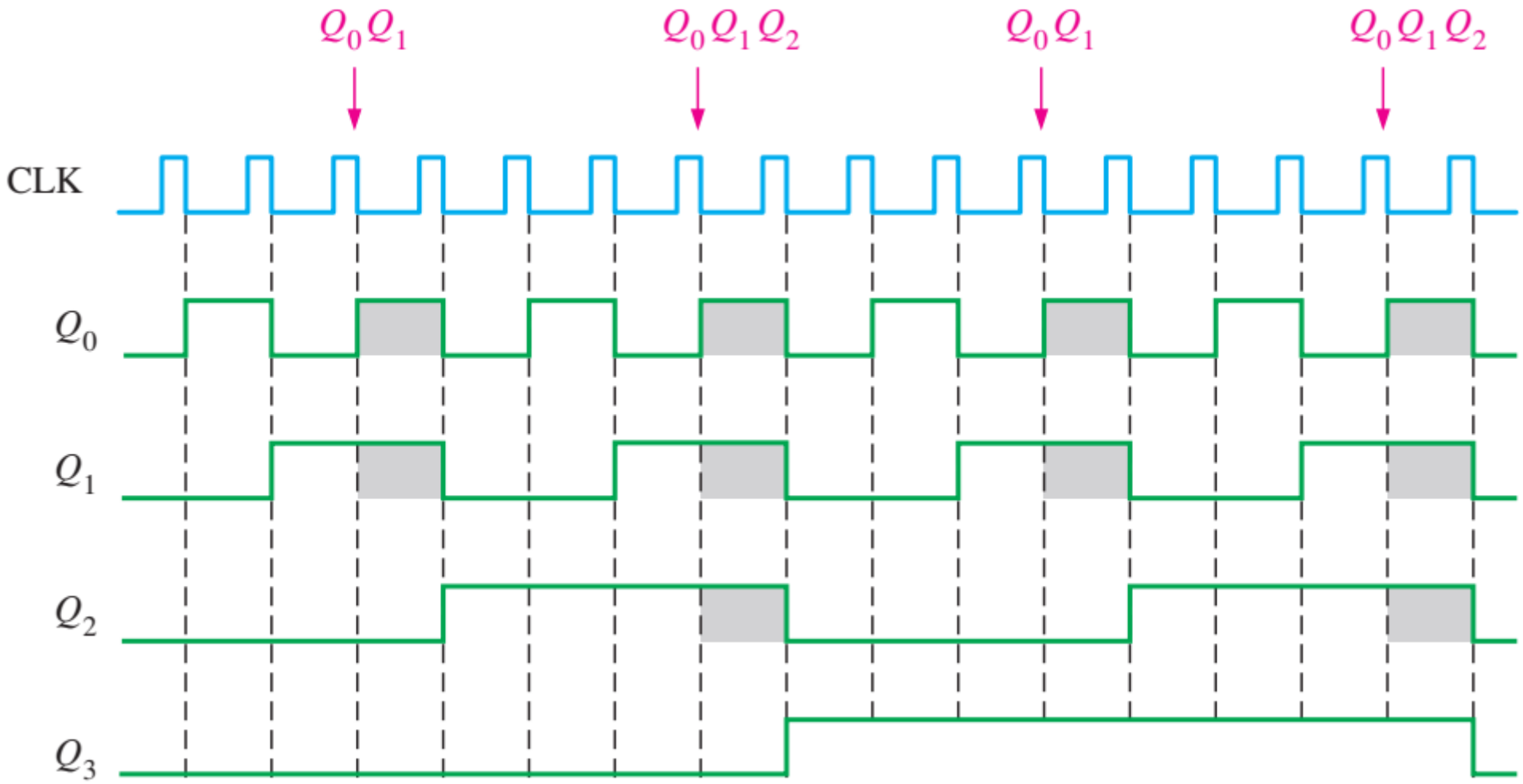
# Mạch đếm

- Mạch đếm đồng bộ 4 bit



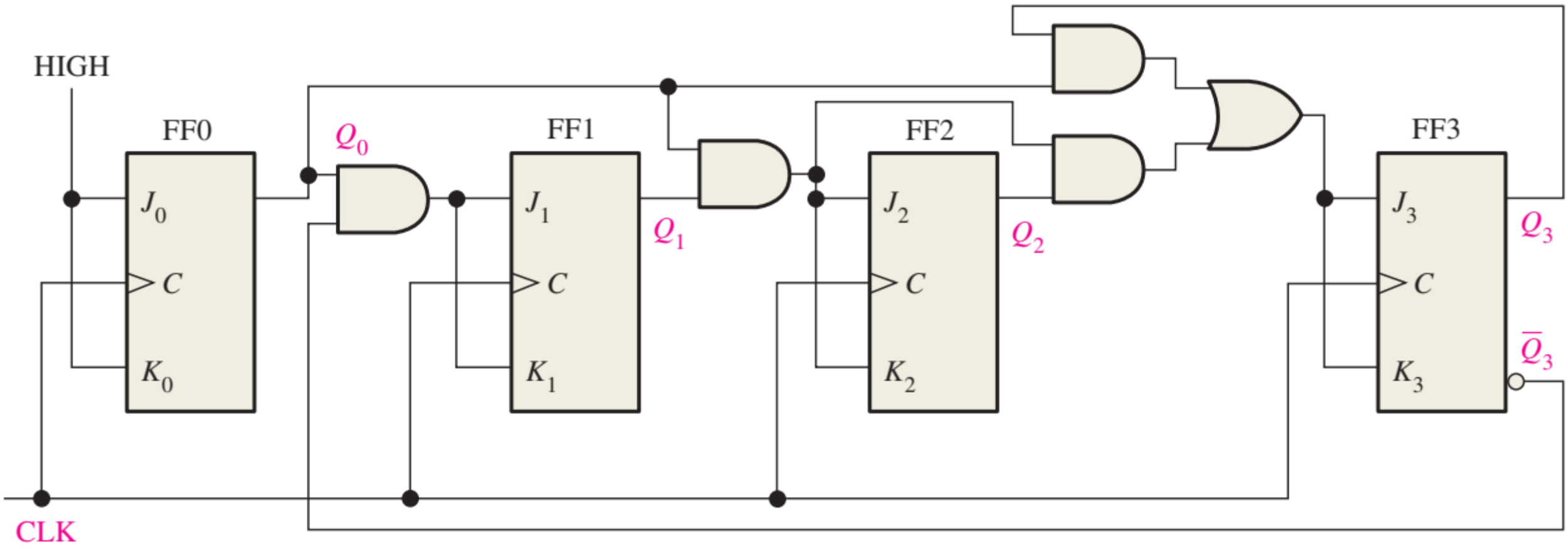
# Mạch đếm

- Mạch đếm đồng bộ 4 bit



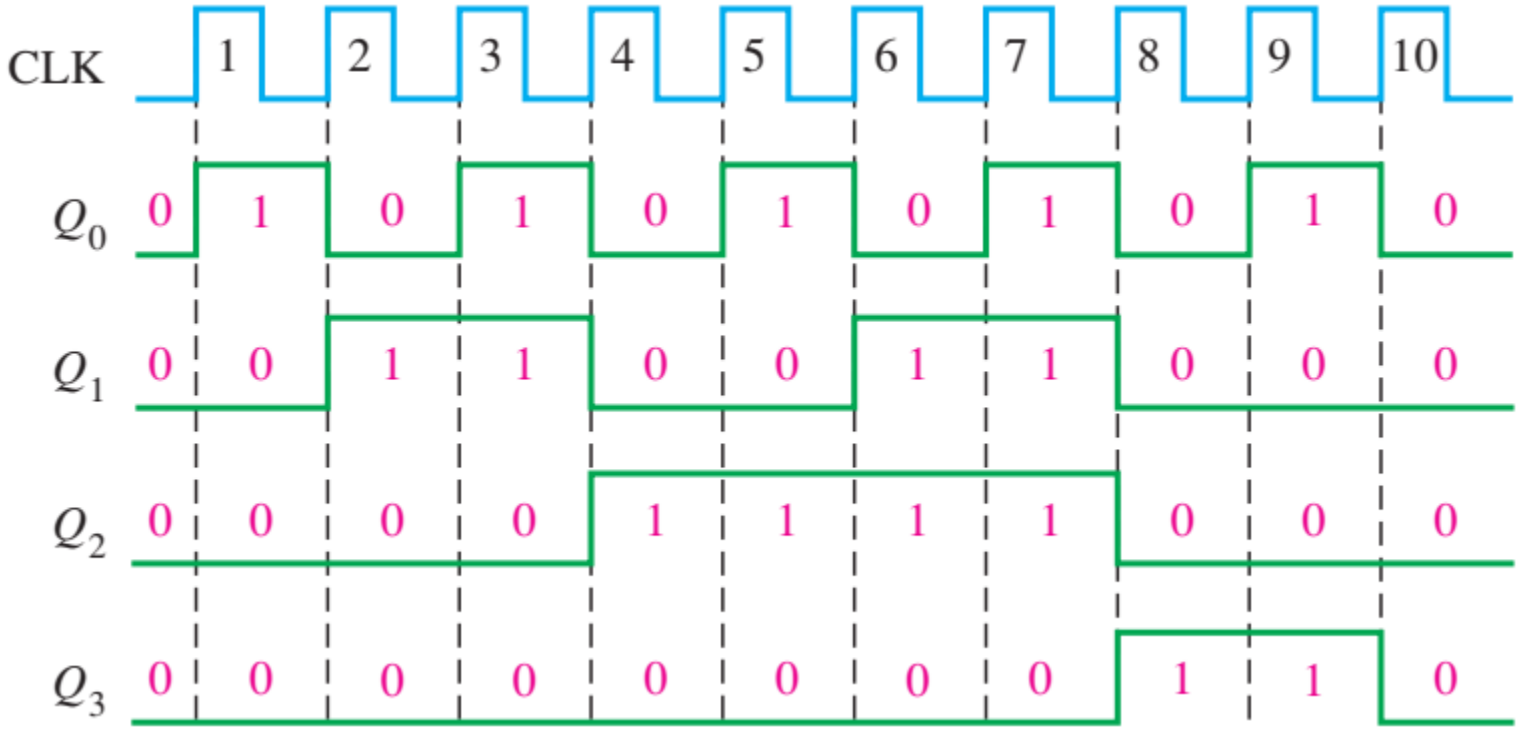
# Mạch đếm

- Mạch đếm đồng bộ MOD 10



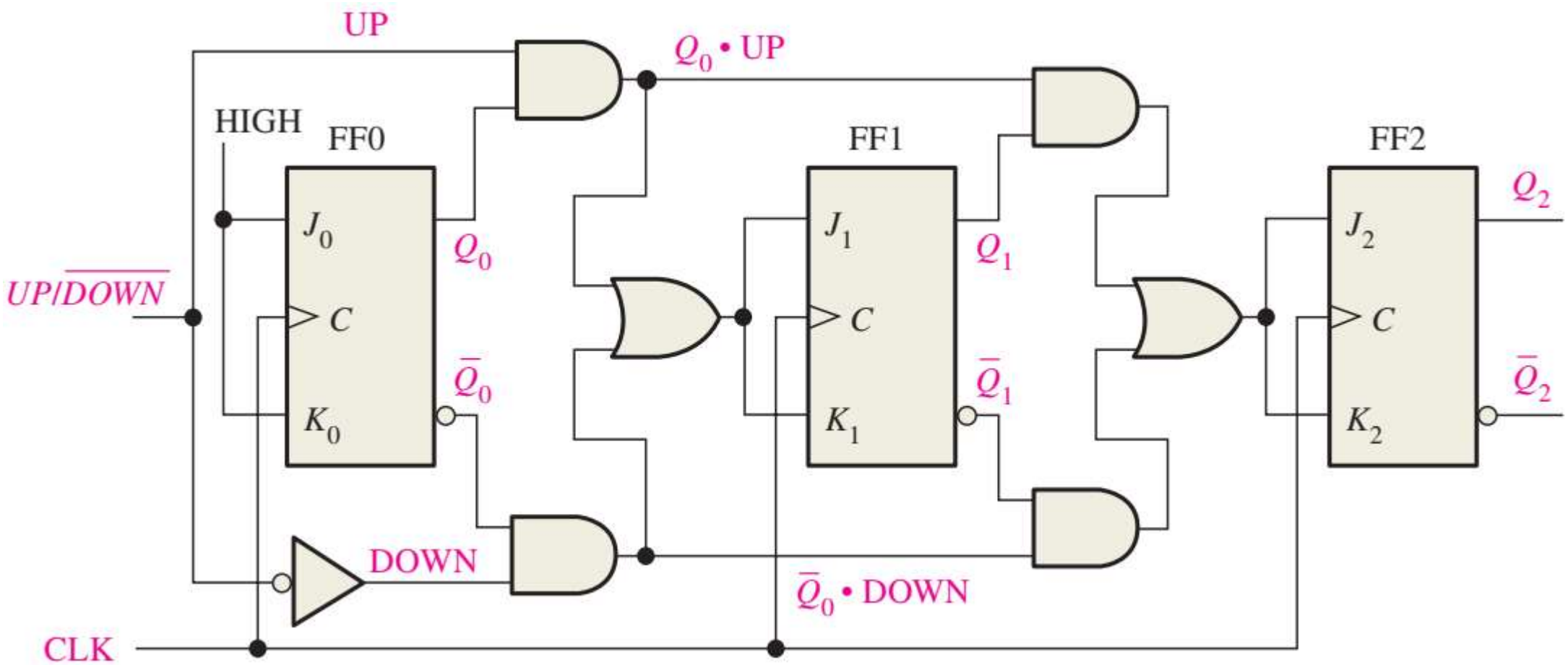
# Mạch đếm

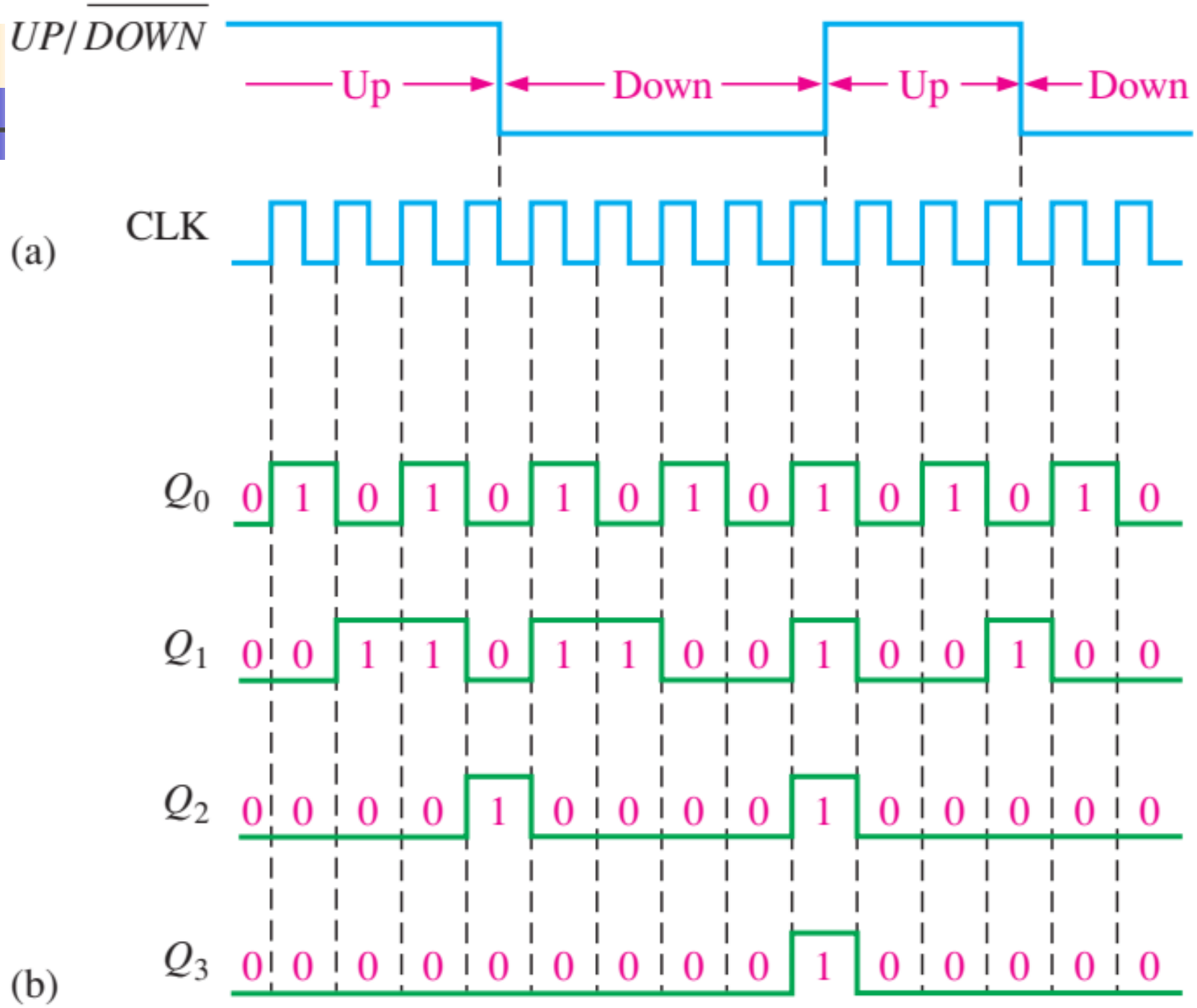
- Mạch đếm đồng bộ MOD 10



# Mạch đếm

- Mạch đếm đồng bộ UP/DOWN




UP/DOWN
Up
Down
Up
Down
CLK
(a)
Q0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
Q1 0 0 1 1 0 1 1 0 0 1 0 0 1 0 0
Q2 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0
Q3 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
(b)

# Mạch đếm

- Thiết kế mạch đếm đồng bộ

  Lập biểu đồ trạng thái

  Lập bảng trạng thái

  Xác định loại FF

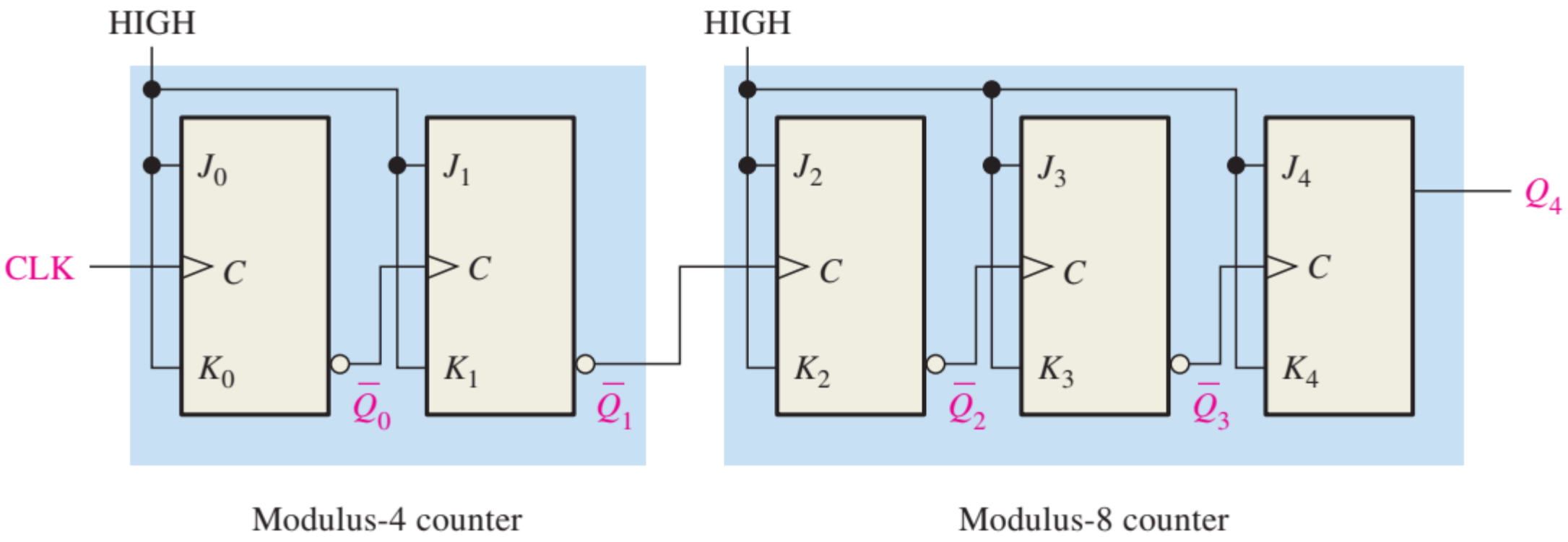
  Xác định mạch ngõ vào FF

  Vẽ mạch hoàn chỉnh

  Vẽ dạng sóng

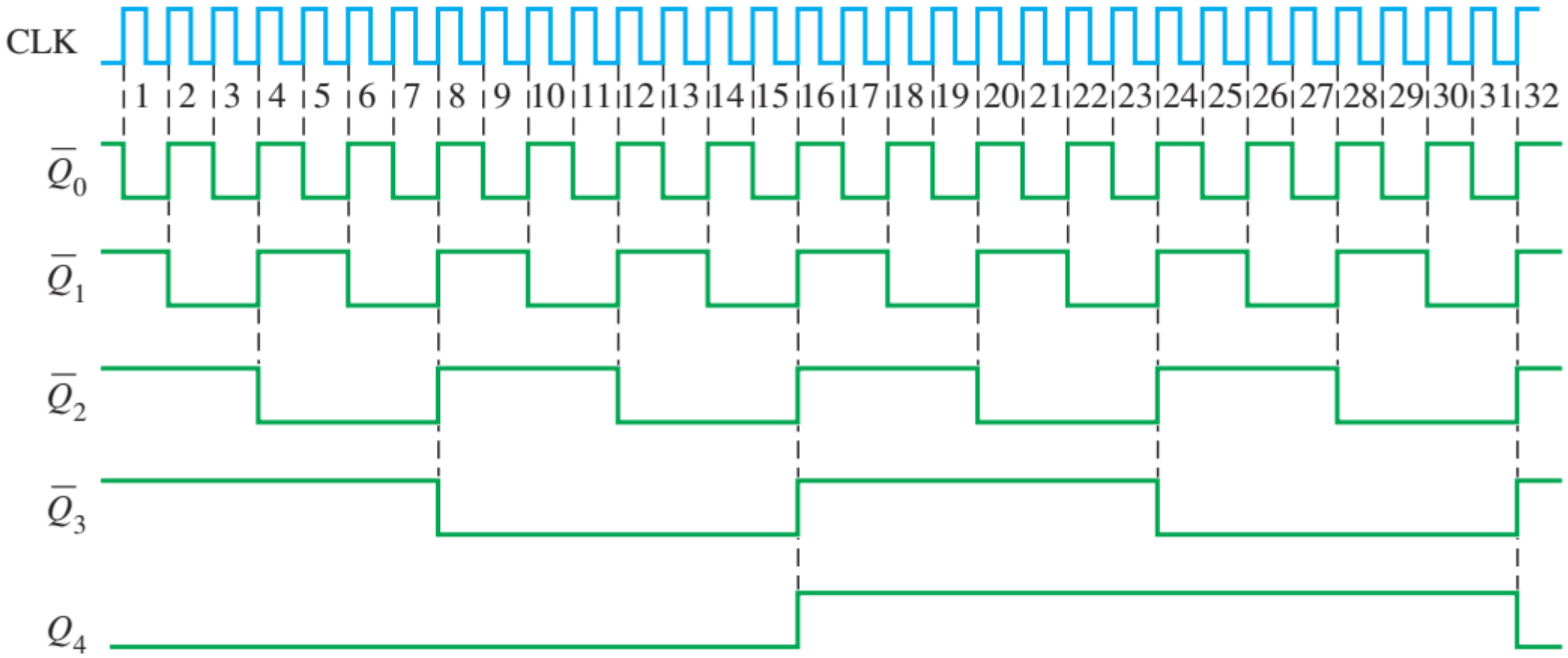
# Mạch đếm

- Ghép mạch đếm



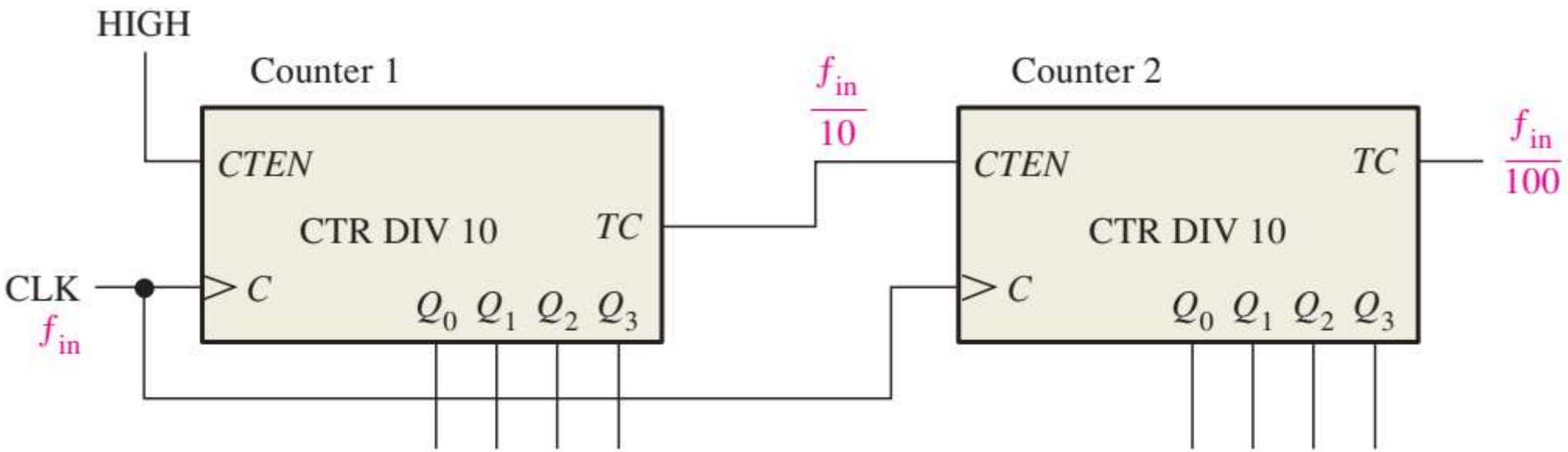
# Mạch đếm

- Ghép mạch đếm



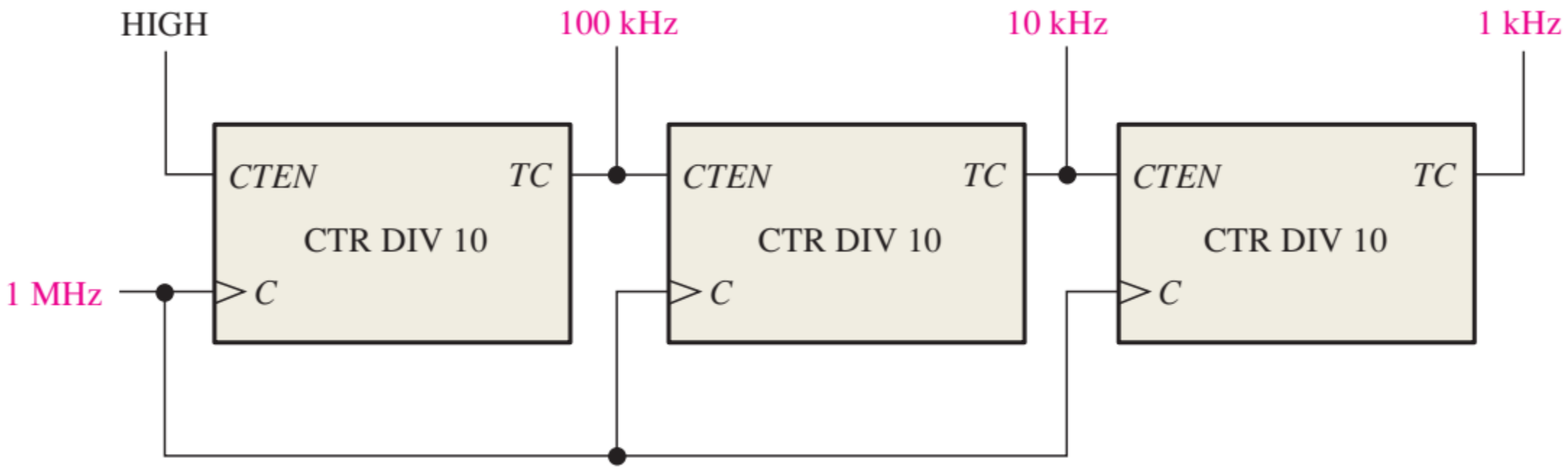
# Mạch đếm

- Ghép mạch đếm



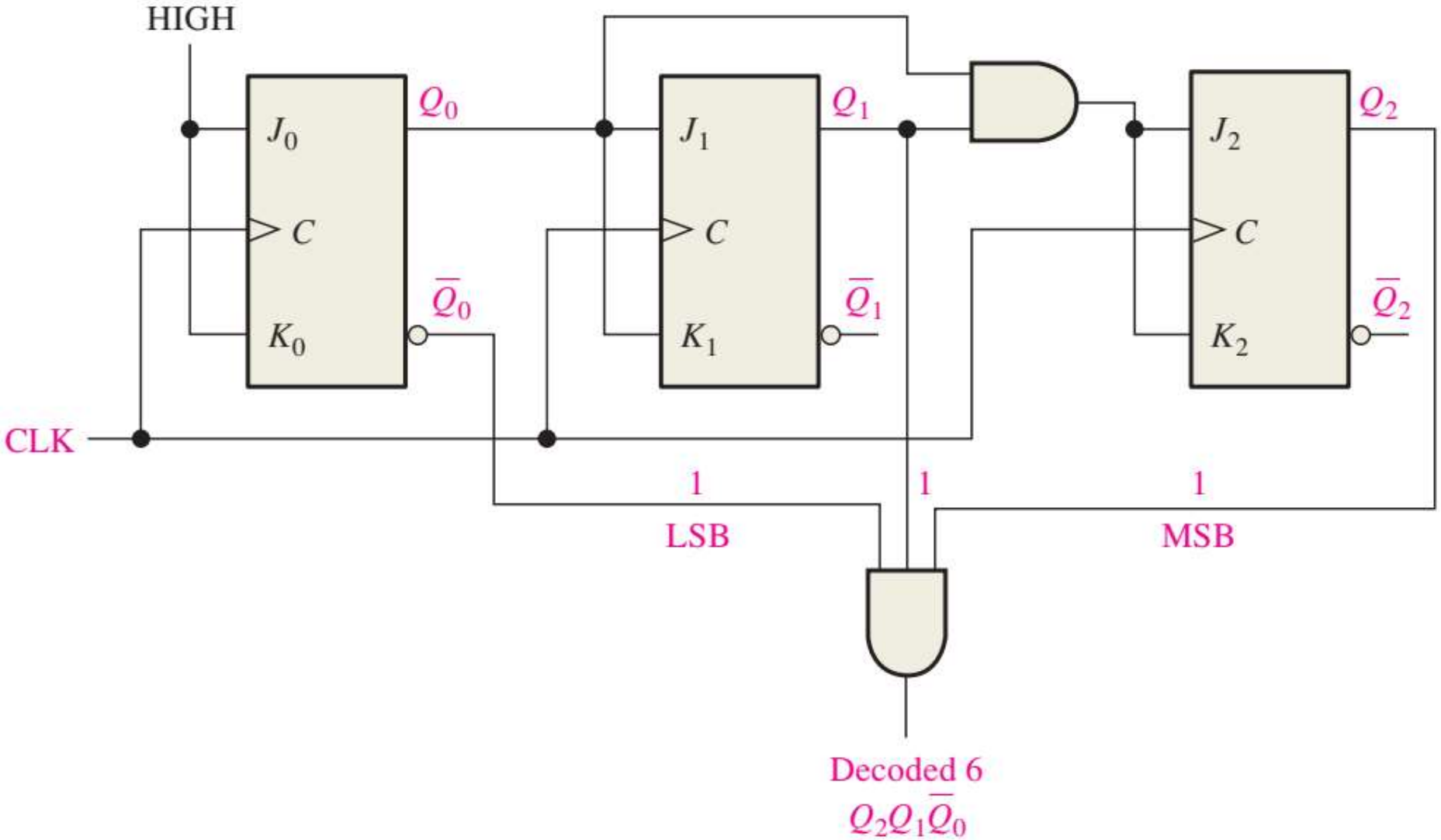
# Mạch đếm

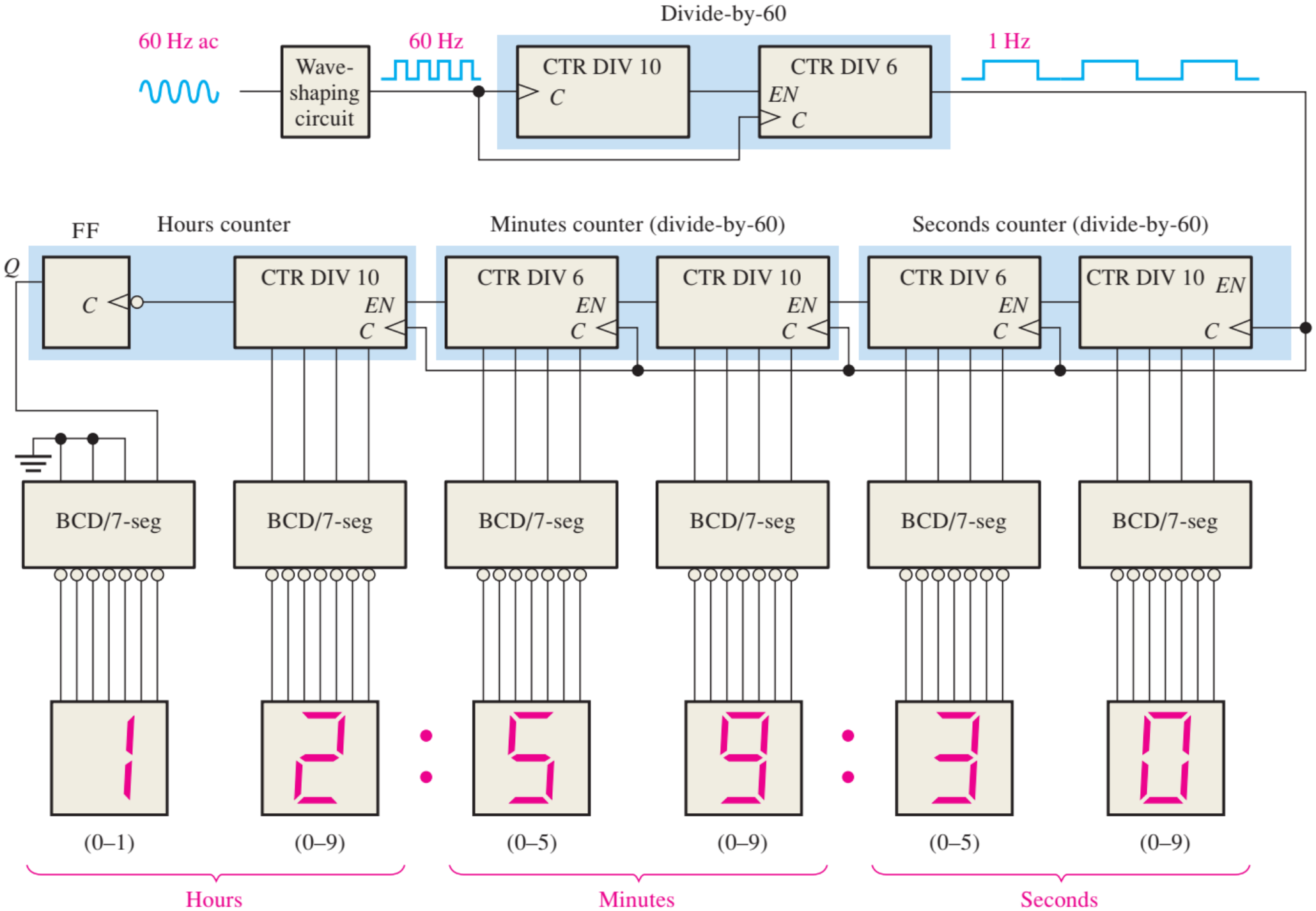
- Ghép mạch đếm



# Mạch đếm

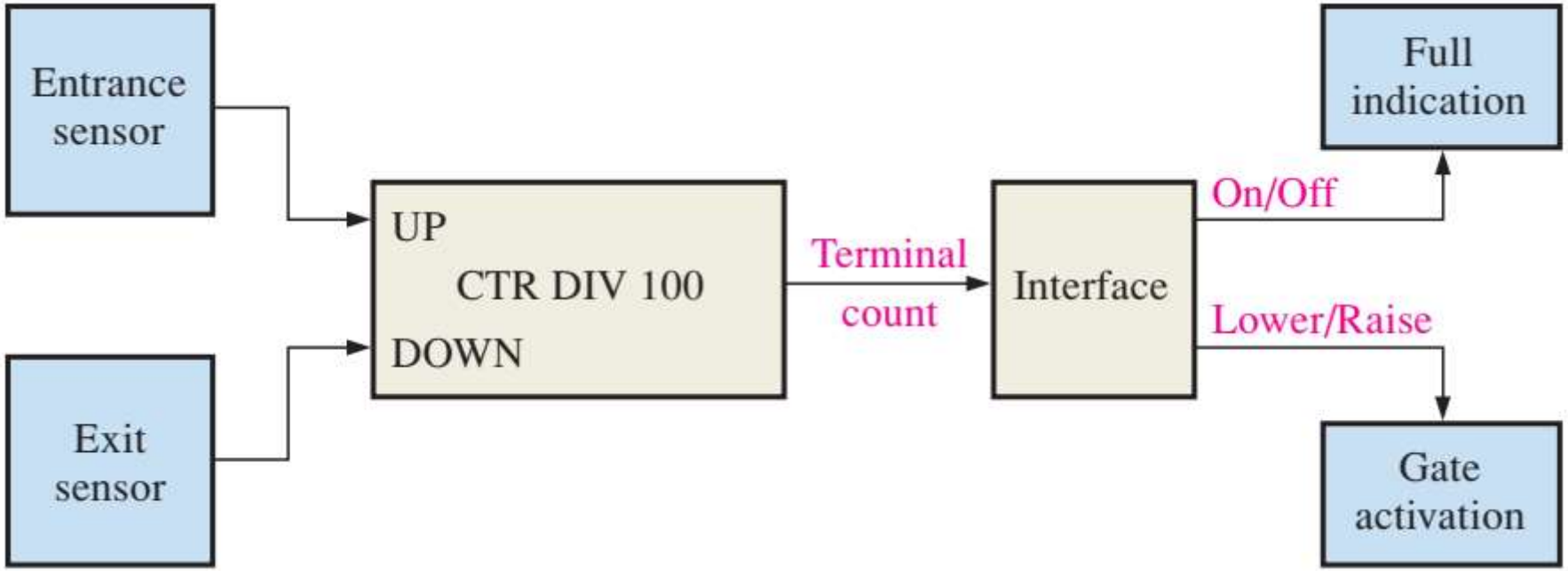
- Giải mã mạch đếm




Divide-by-60
60 Hz ac
Wave-shaping circuit
60 Hz
CTR DIV 10
C
CTR DIV 6
EN
C
1 Hz
FF
Hours counter
Minutes counter (divide-by-60)
Seconds counter (divide-by-60)
Q
C
CTR DIV 10
EN
C
CTR DIV 6
EN
C
CTR DIV 10
EN
C
CTR DIV 6
EN
C
CTR DIV 10
EN
C
BCD/7-seg
BCD/7-seg
BCD/7-seg
BCD/7-seg
BCD/7-seg
BCD/7-seg
1
2
5
9
3
0
(0–1)
(0–9)
(0–5)
(0–9)
(0–5)
(0–9)
Hours
Minutes
Seconds

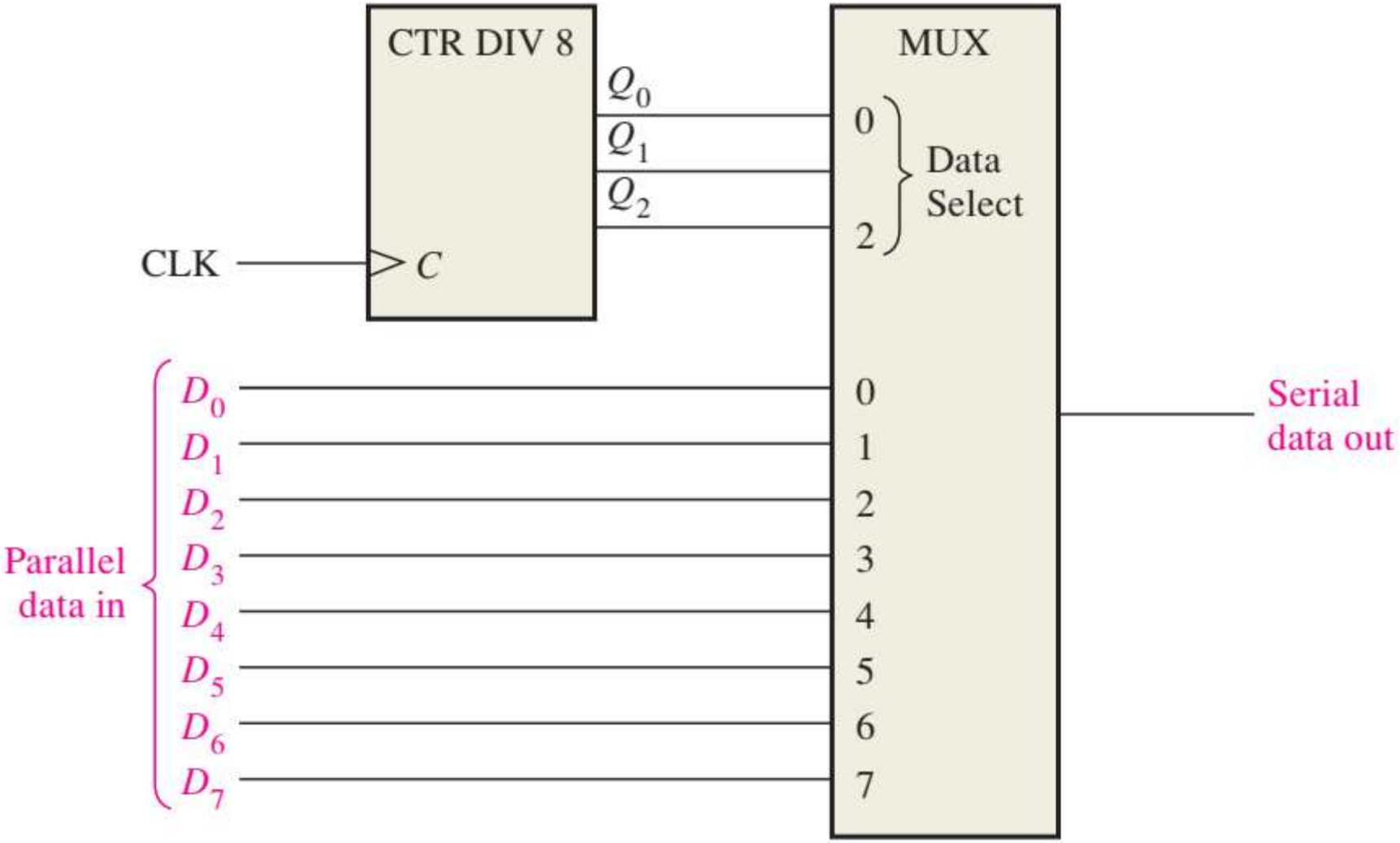
# Mạch đếm

- Ứng dụng mạch đếm



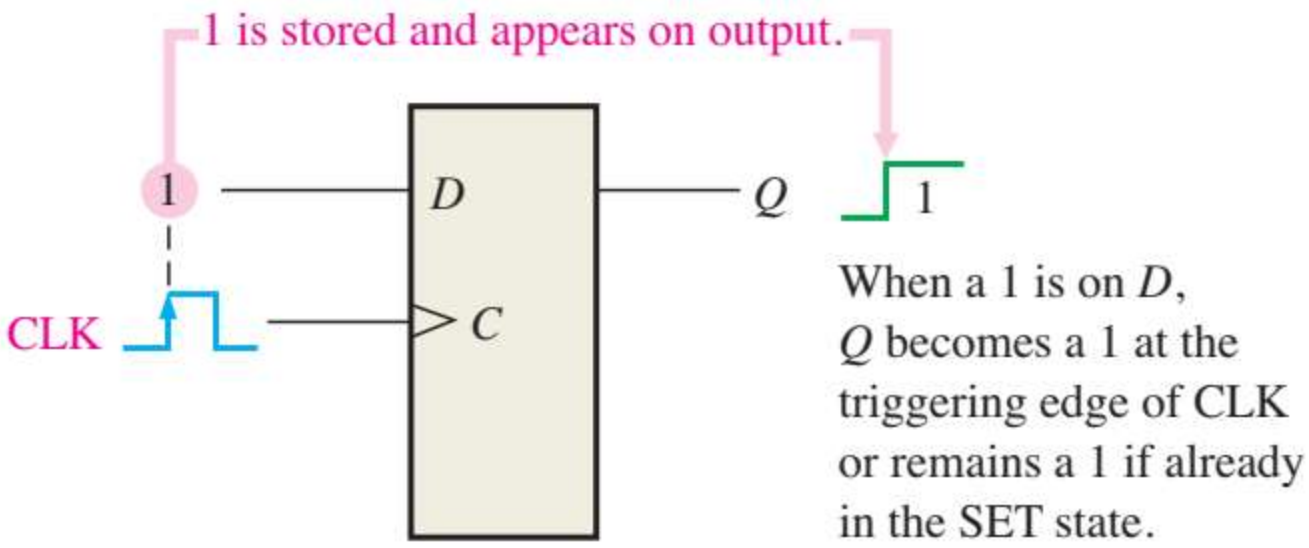
# Mạch đếm

- Ứng dụng mạch đếm



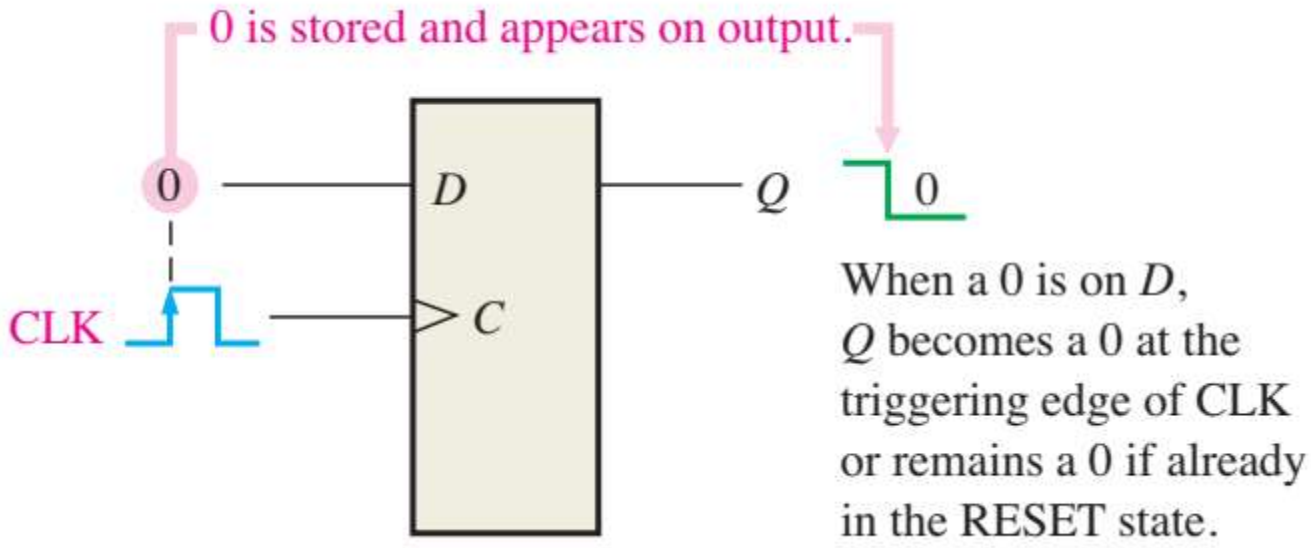
# Thanh ghi

- Flip Flop D



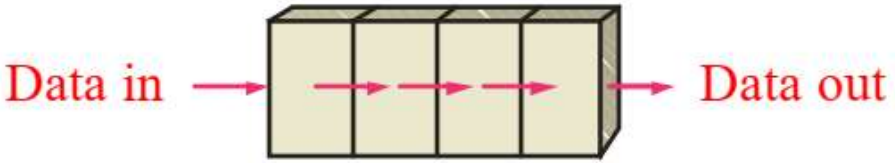
# Thanh ghi

- Flip Flop D

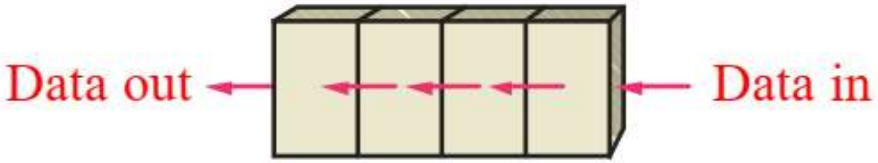


# Thanh ghi

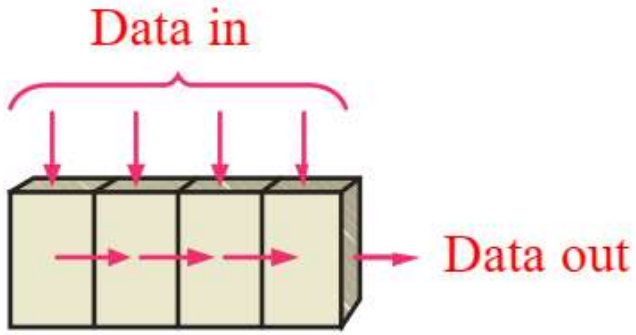
- Các dạng thanh ghi



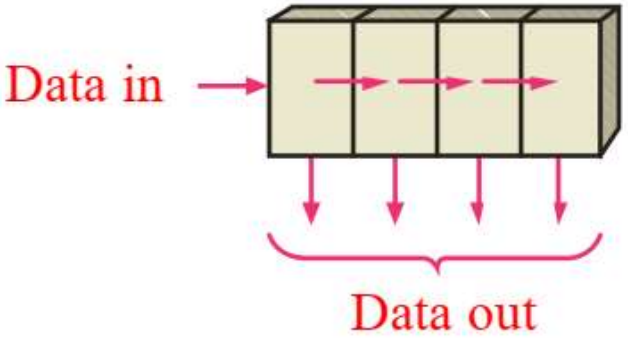

Serial in/shift right/serial out



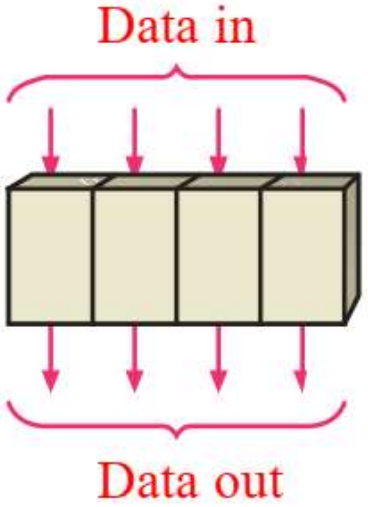

Serial in/shift left/serial out



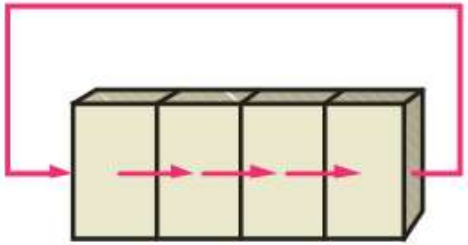

Parallel in/serial out



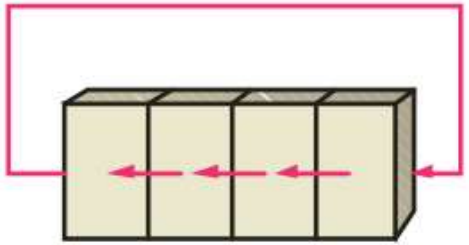

Serial in/parallel out




Parallel in/parallel out



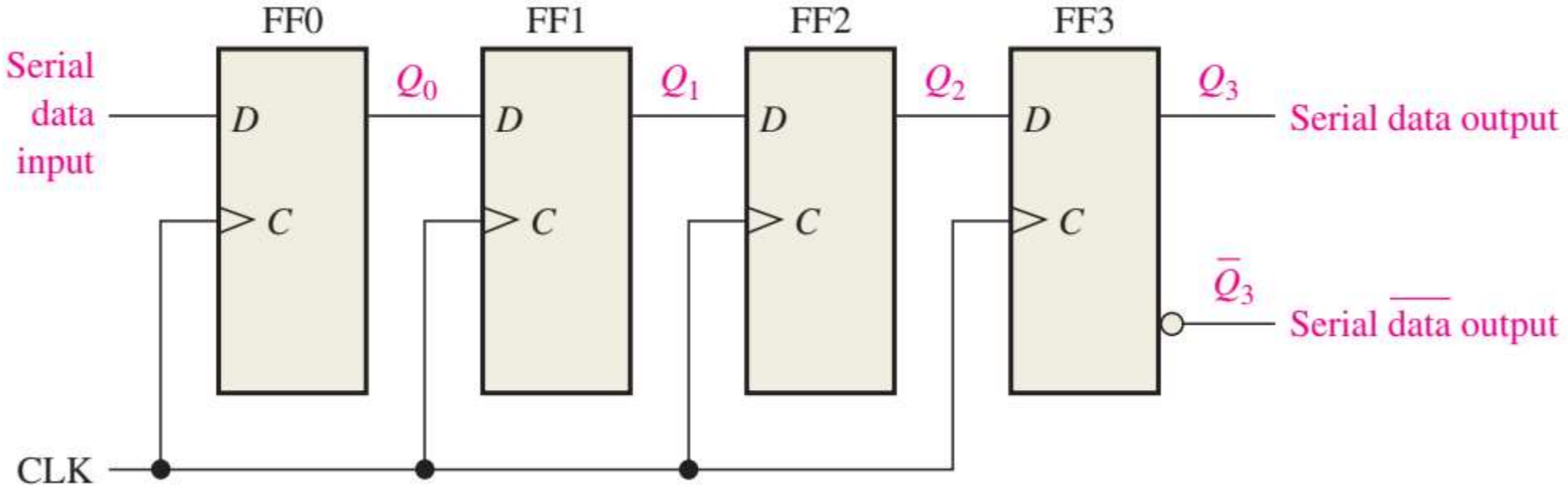
Rotate right

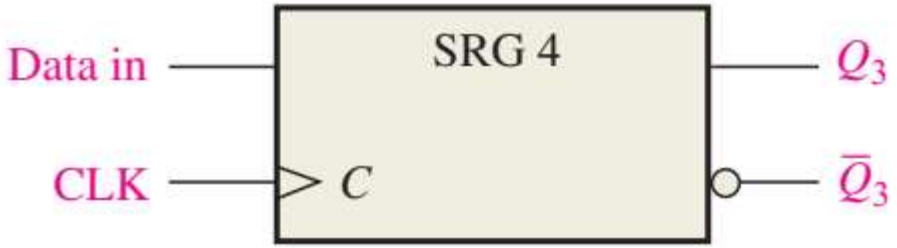


Rotate left

# Thanh ghi

- Thanh ghi vào nối tiếp ra nối tiếp



# Thanh ghi

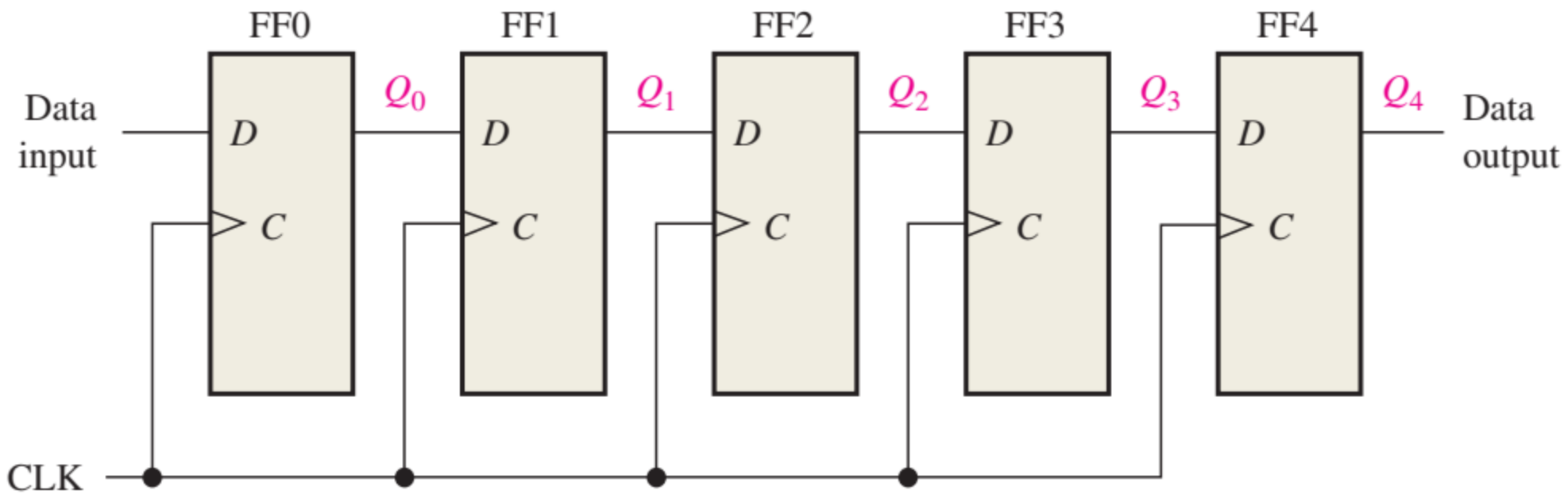
- Thanh ghi vào nối tiếp ra nối tiếp

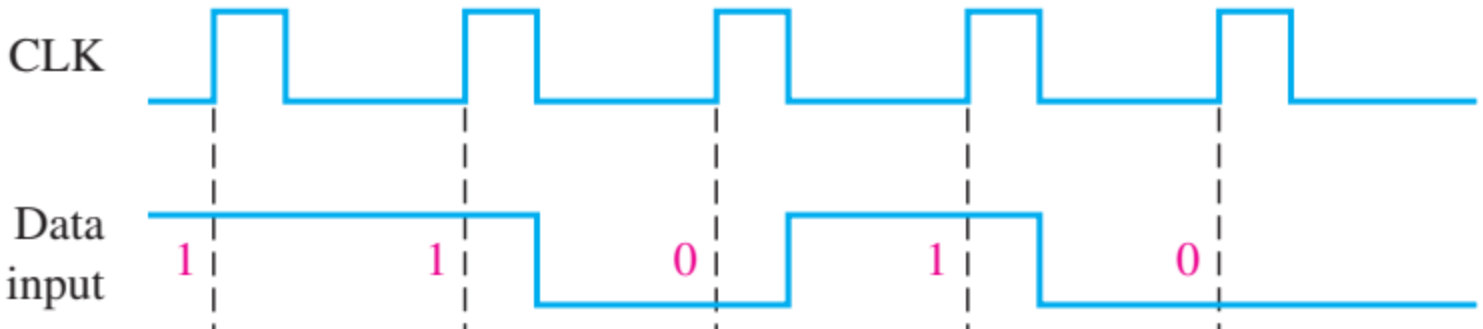
| CLK | FF0 ($Q_0$) | FF1 ($Q_1$) | FF2 ($Q_2$) | FF3 ($Q_3$) |
|---|---|---|---|---|
| Initial | 0 | 0 | 0 | 0 |
| 1 | **0** | 0 | 0 | 0 |
| 2 | **1** | **0** | 0 | 0 |
| 3 | **0** | **1** | **0** | 0 |
| 4 | **1** | **0** | **1** | **0** |

| CLK | FF0 ($Q_0$) | FF1 ($Q_1$) | FF2 ($Q_2$) | FF3 ($Q_3$) |
|---|---|---|---|---|
| Initial | **1** | **0** | **1** | **0** |
| 5 | 0 | **1** | **0** | **1** |
| 6 | 0 | 0 | **1** | **0** |
| 7 | 0 | 0 | 0 | **1** |
| 8 | 0 | 0 | 0 | 0 |

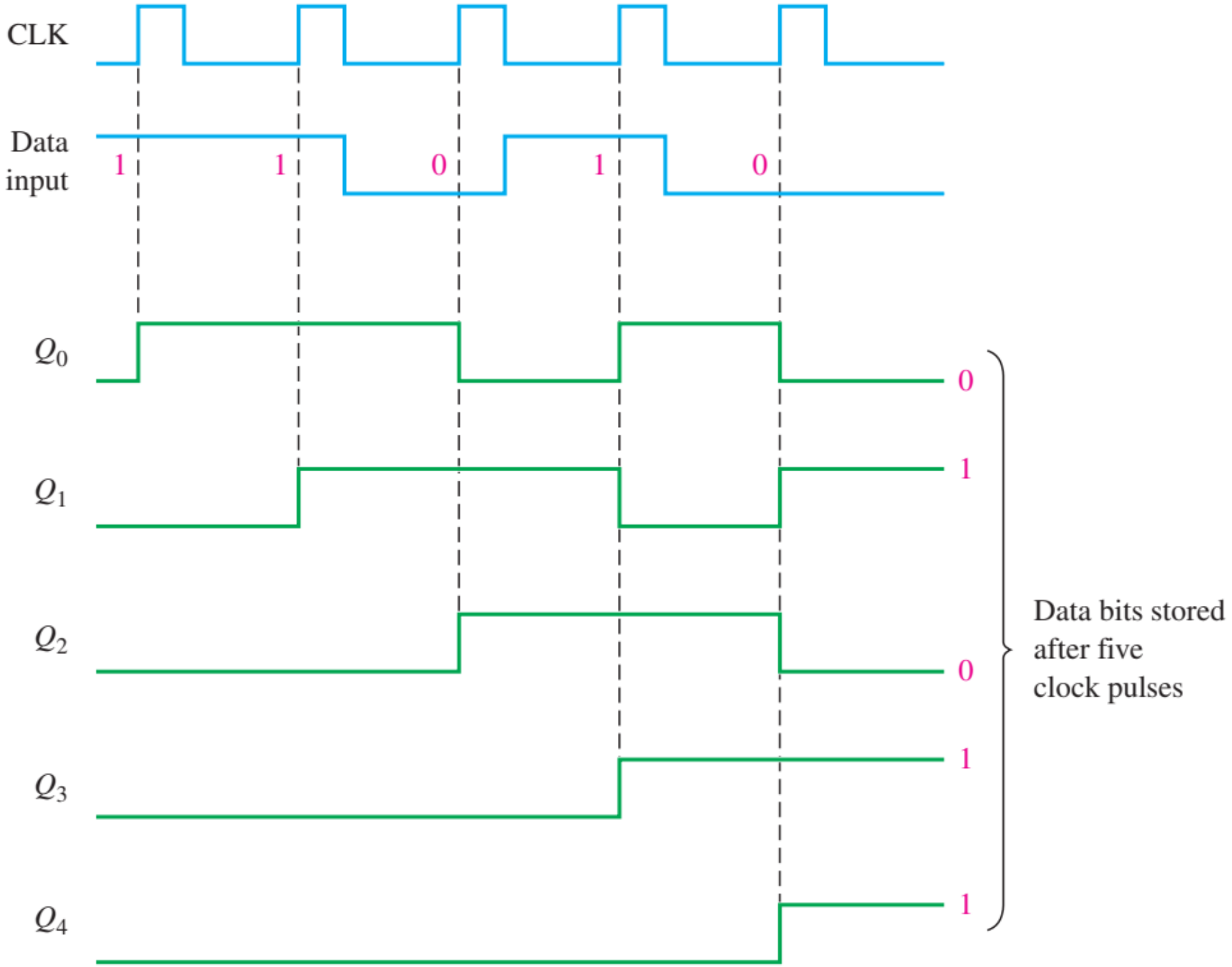
# Thanh ghi

- Thanh ghi vào nối tiếp ra nối tiếp



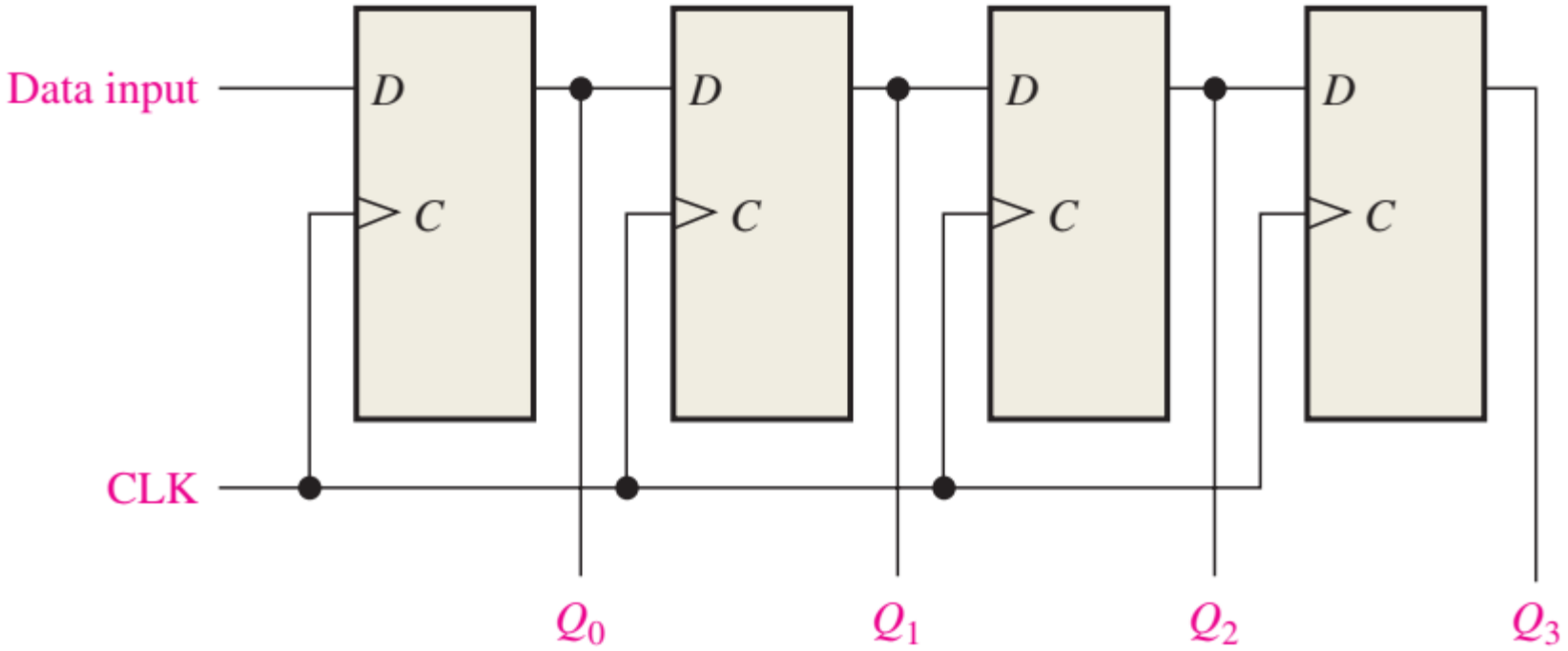



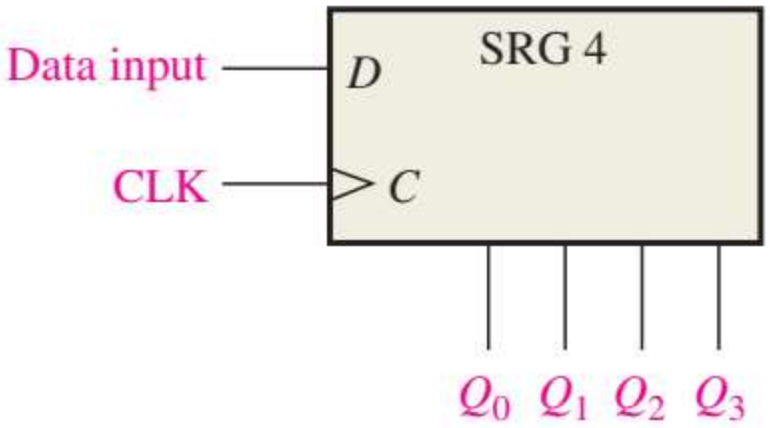
- Tl


CLK
Data input
1
1
0
1
0
$Q_0$
$Q_1$
$Q_2$
$Q_3$
$Q_4$
0
1
0
1
1
Data bits stored after five clock pulses

# Thanh ghi

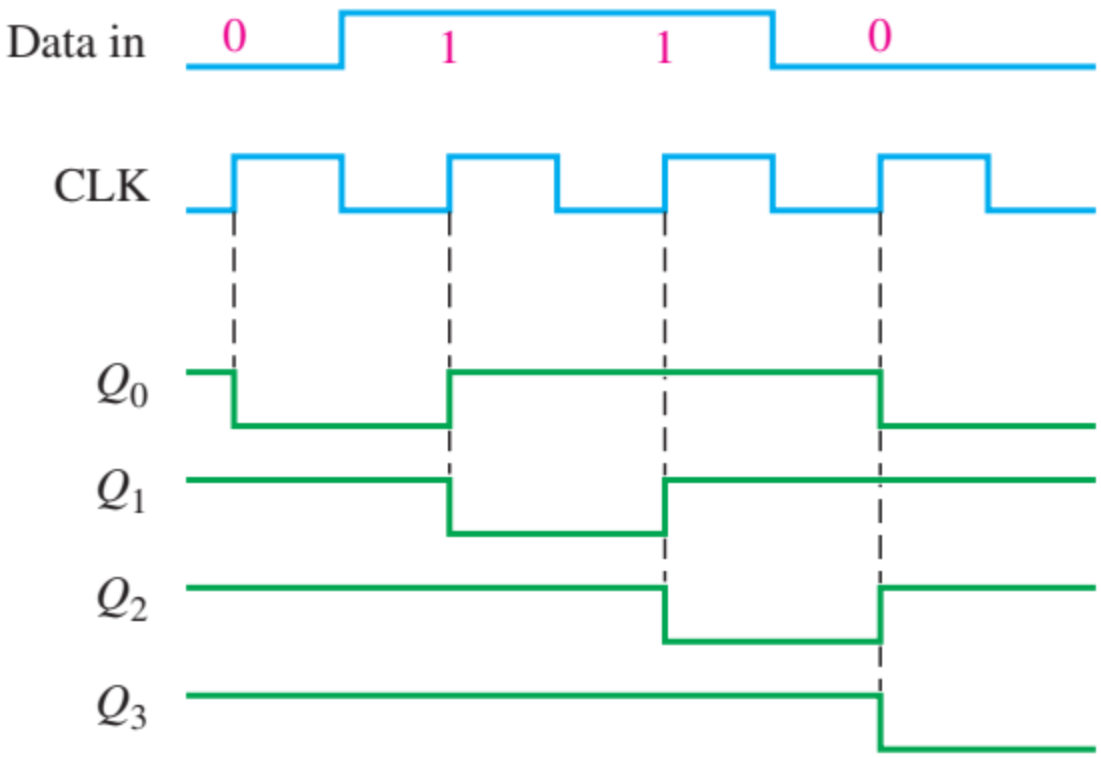
- Thanh ghi vào nối tiếp ra song song



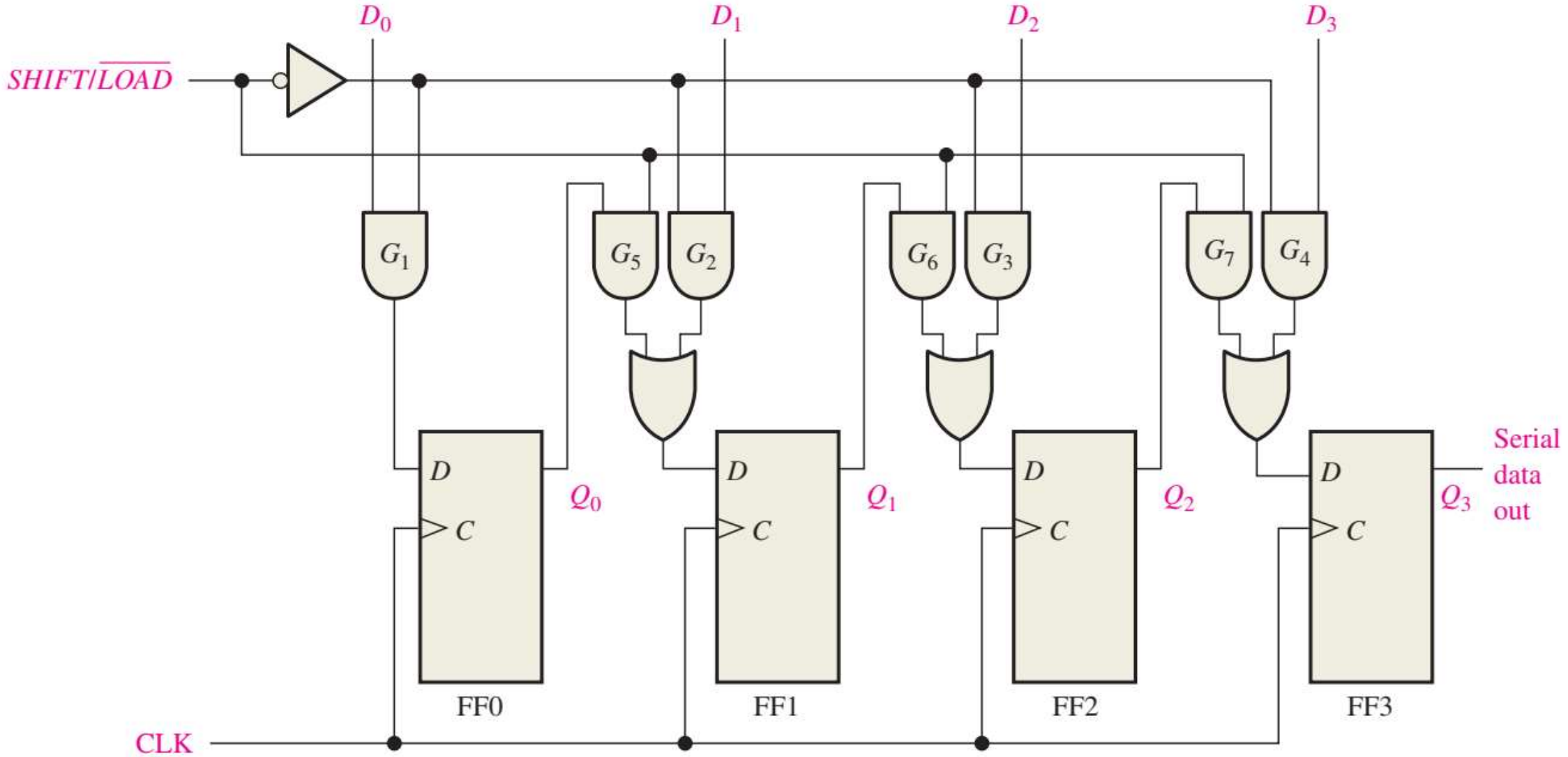
# Thanh ghi

- Thanh ghi vào nối tiếp ra song song



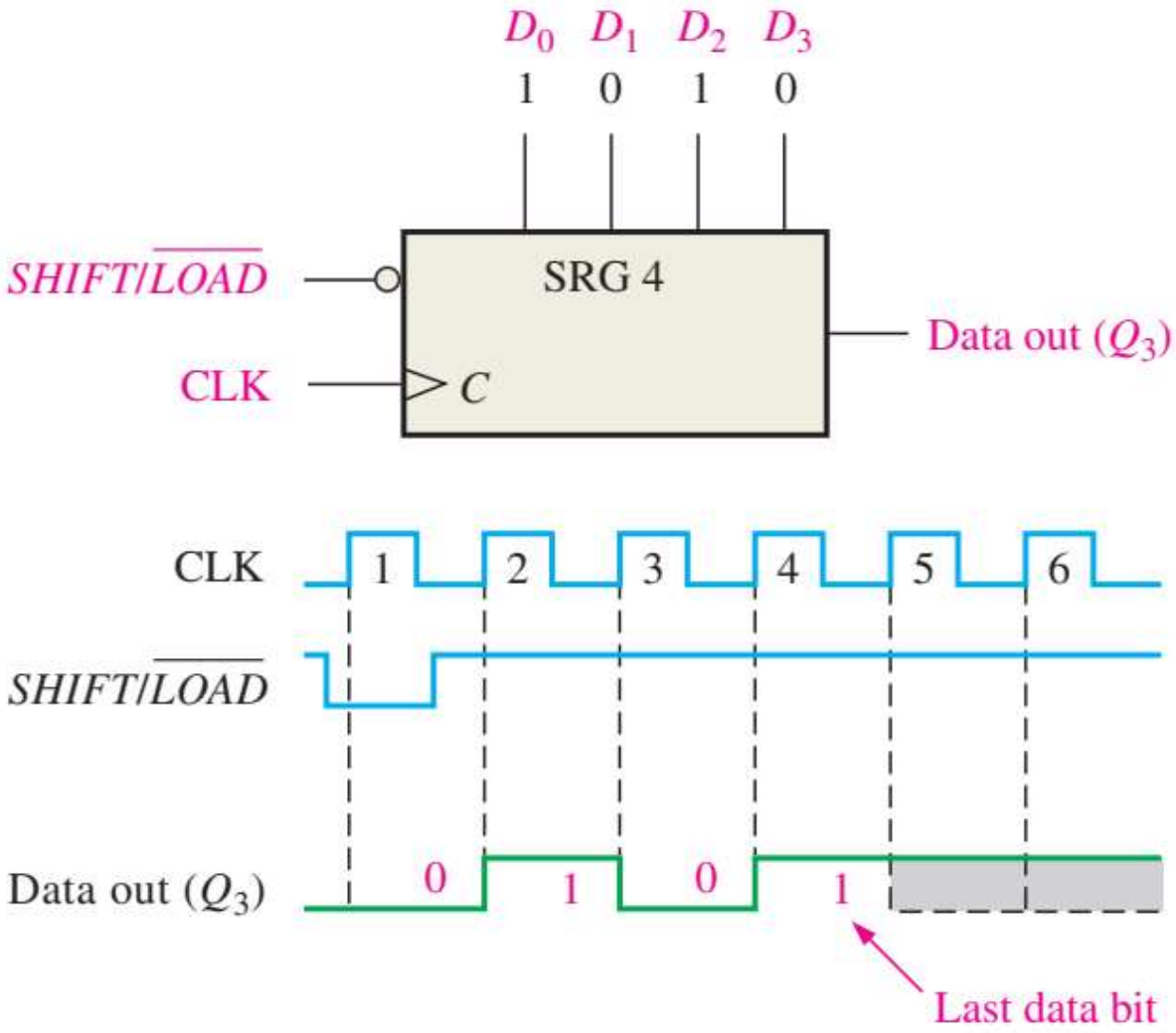
# Thanh ghi

- Thanh ghi vào song song ra nối tiếp



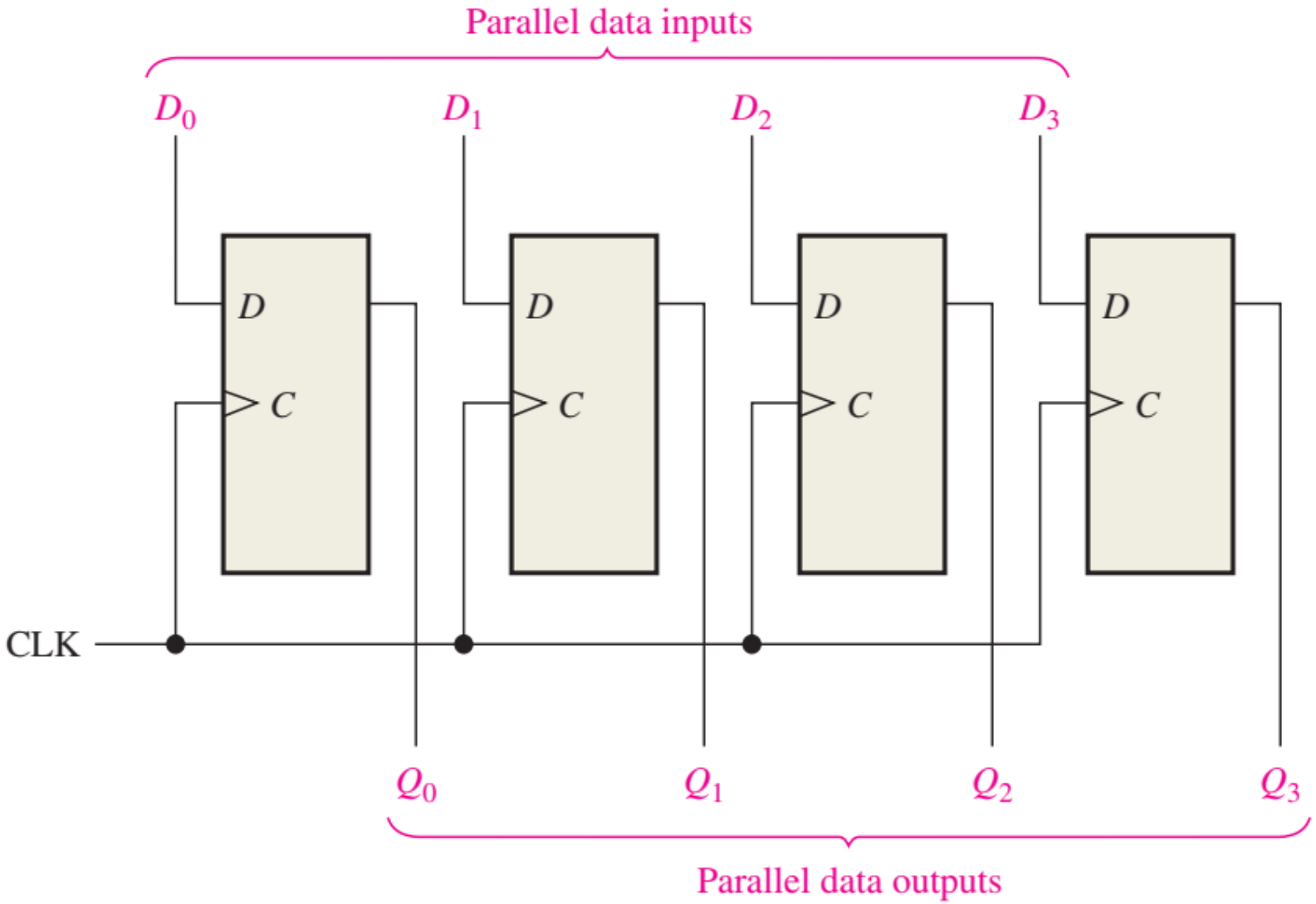
# Thanh ghi

- Thanh ghi vào song song ra nối tiếp

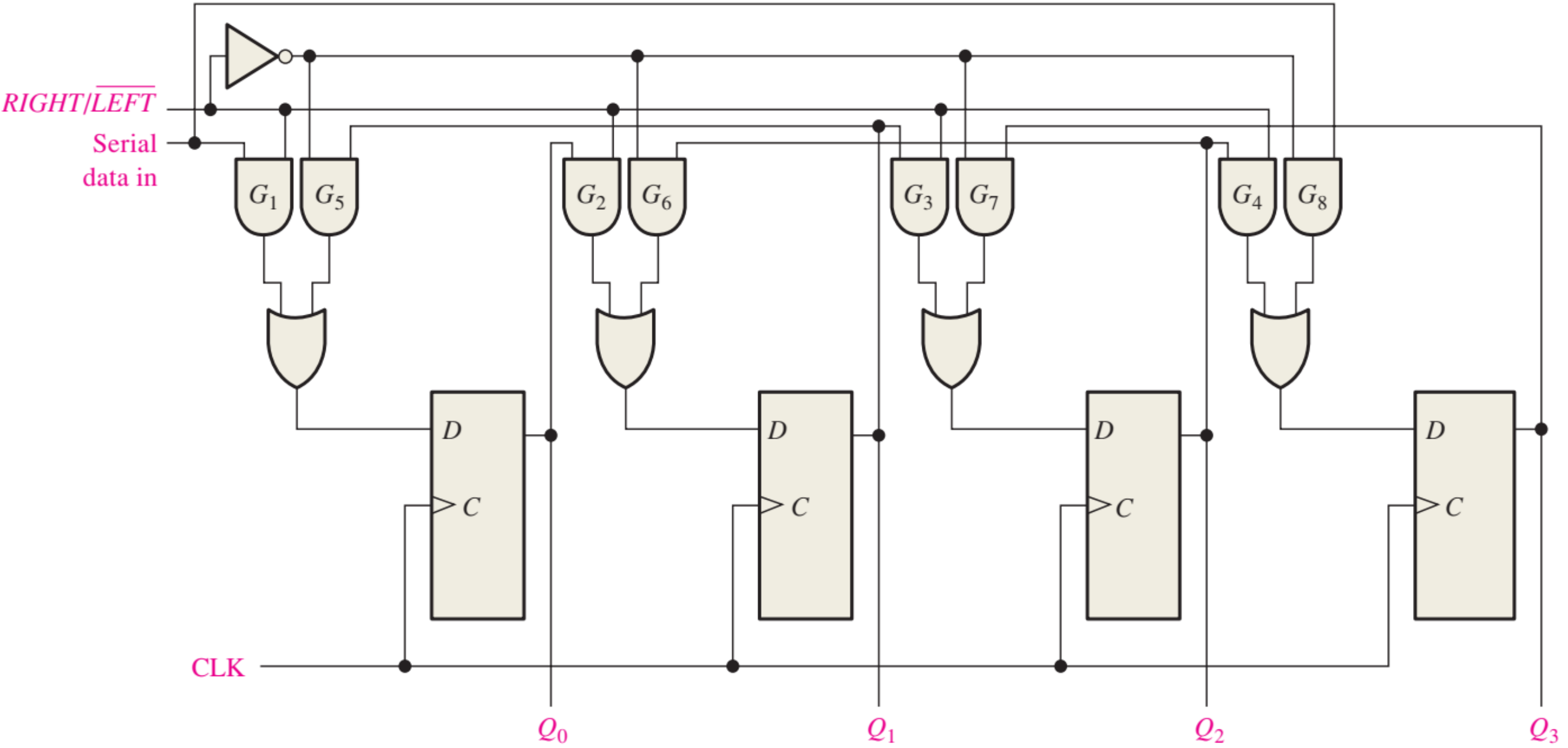


# Thanh ghi

- Thanh ghi vào song song ra song song



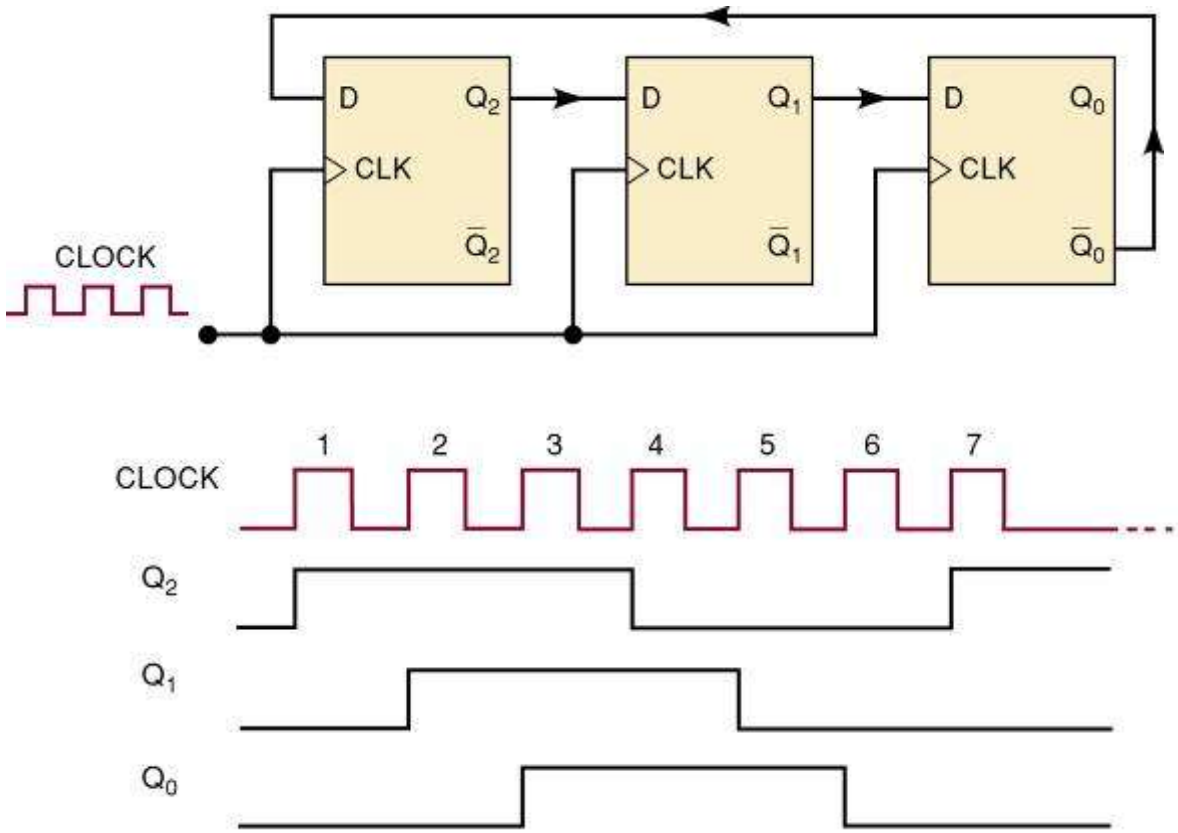
# Thanh ghi


RIGHT/$\overline{LEFT}$
Serial data in
$G_1$
$G_5$
$G_2$
$G_6$
$G_3$
$G_7$
$G_4$
$G_8$
D
C
D
C
D
C
D
C
CLK
$Q_0$
$Q_1$
$Q_2$
$Q_3$

# Thanh ghi

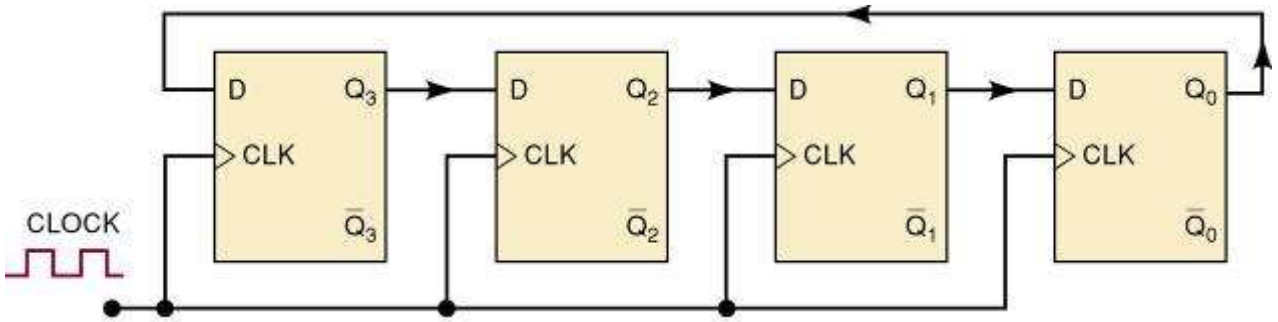
- Mạch đếm Johnson 4 bit

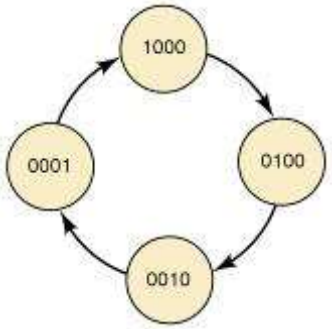
| $Q_2$ | $Q_1$ | $Q_0$ | CLOCK pulse |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 2 |
| 1 | 1 | 1 | 3 |
| 0 | 1 | 1 | 4 |
| 0 | 0 | 1 | 5 |
| 0 | 0 | 0 | 6 |
| 1 | 0 | 0 | 7 |
| 1 | 1 | 0 | 8 |
| . | . | . | . |
| . | . | . | . |
| . | . | . | . |



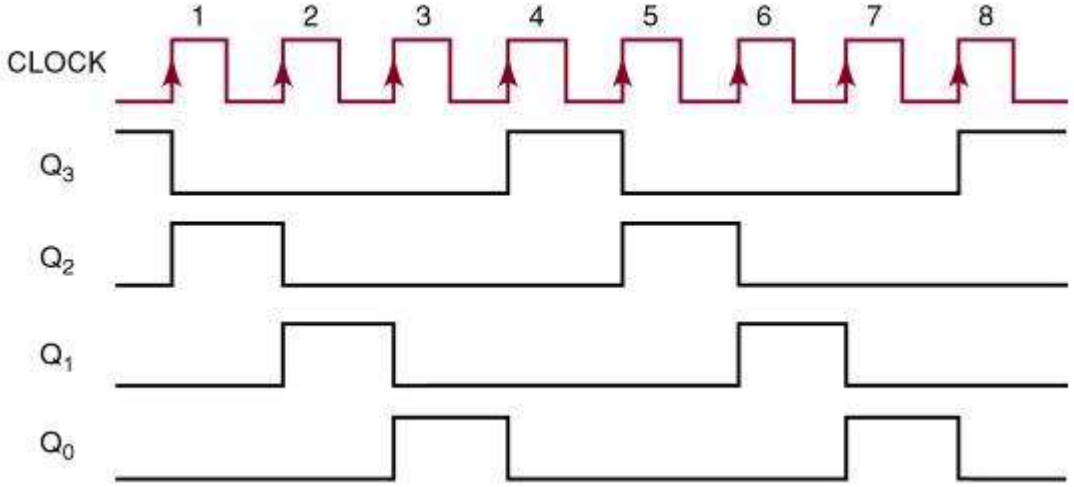
# Thanh ghi

- Mạch đếm vòng 4 bit



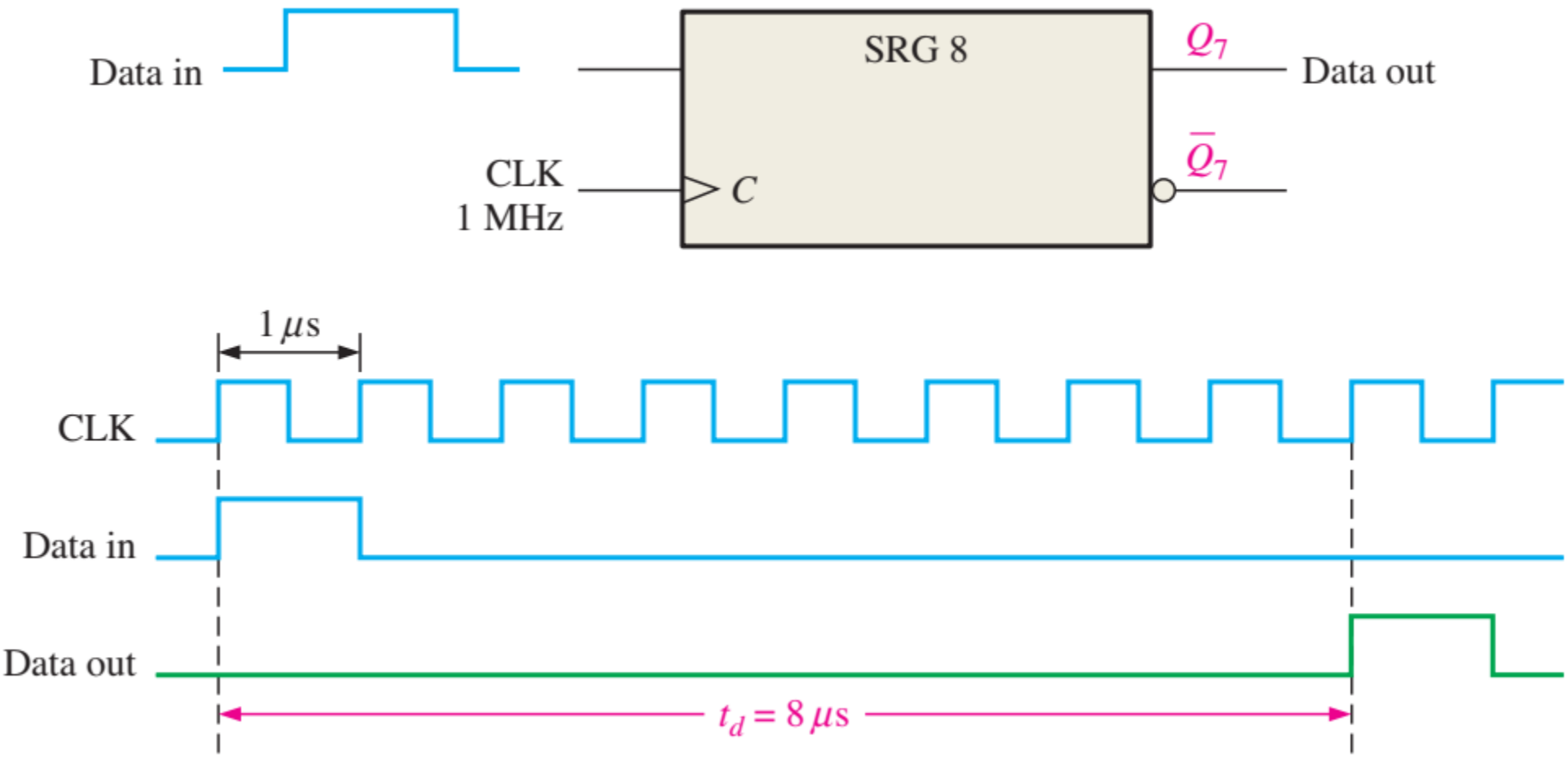



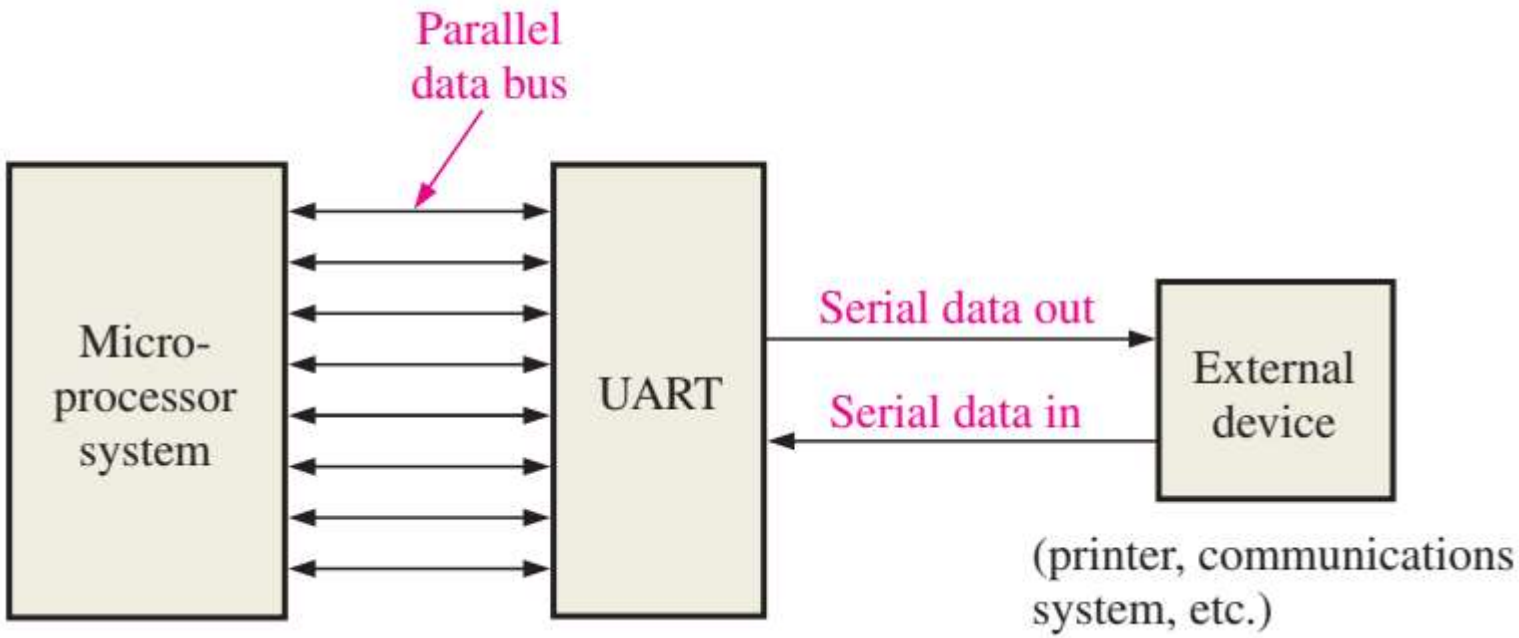

| $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ | CLOCK pulse |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 2 |
| 0 | 0 | 0 | 1 | 3 |
| 1 | 0 | 0 | 0 | 4 |
| 0 | 1 | 0 | 0 | 5 |
| 0 | 0 | 1 | 0 | 6 |
| 0 | 0 | 0 | 1 | 7 |
| . | . | . | . | . |
| . | . | . | . | . |



# Thanh ghi

- Mạch Delay



# Thanh ghi

- Mạch giao tiếp